



Deep Learning-Based Object Pose Estimation: A Comprehensive Survey

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Abstract

Object pose estimation is a fundamental computer vision problem with broad applications in augmented reality and robotics. Over the past decade, deep learning models, due to their superior accuracy and robustness, have increasingly supplanted conventional algorithms reliant on engineered point pair features. Nevertheless, several challenges persist in contemporary methods, including their dependency on labeled training data, model compactness, robustness under challenging conditions, and their ability to generalize to novel unseen objects. A recent survey discussing the progress made on different aspects of this area, outstanding challenges, and promising future directions, is missing. To fill this gap, we discuss the recent advances in deep learning-based object pose estimation, covering all three formulations of the problem, *i.e.*, instance-level, category-level, and unseen (including both instance-unseen and category-unseen cases) object pose estimation. Our survey also covers multiple input data modalities, degrees-of-freedom of output poses, object properties, and downstream tasks, providing the readers with a holistic understanding of this field. Additionally, it discusses training paradigms of different domains, inference modes, application areas, evaluation metrics, and benchmark datasets, as well as reports the performance of current state-of-the-art methods on these benchmarks, thereby facilitating the readers in selecting the most suitable method for their application. Finally, the survey identifies key challenges, reviews the prevailing trends along with their pros and cons, and identifies promising directions for future research. We cover the literature up to our submission date and will continue to follow the latest works at <https://github.com/CNJianLiu/Awesome-Object-Pose-Estimation>.

Keywords Object pose estimation · Deep learning · Comprehensive survey · 3D computer vision

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基于深度学习的目标姿态估计：综述

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摘要

物体姿态估计是计算机视觉领域的基础性问题，在增强现实和机器人技术中具有广泛应用价值。过去十年间，深度学习模型凭借其卓越的精度与鲁棒性，已逐步取代依赖人工设计点对特征的传统算法。然而当前方法仍面临诸多挑战：对标注训练数据的依赖性、模型结构紧凑性、恶劣环境下的鲁棒性表现，以及对新异物体的泛化能力等。目前尚缺乏系统综述该领域研究进展、突出挑战及未来发展方向的权威文献。为填补这一空白，本文深入探讨基于深度学习的物体姿态估计最新进展，涵盖问题的三种建模框架——实例级、类别级及未见物体（包括实例未见与类别未见场景）姿态估计。研究还整合了多模态输入数据、姿态输出自由度、物体属性特征及下游任务等维度，为读者构建该领域的全景式认知框架。此外，本文还探讨了不同领域的训练范式、推理模式、应用领域、评估指标及基准数据集，并报告了当前最先进方法在这些基准测试中的表现，从而帮助读者选择最适合其应用场景的方法。最后，本综述指出了关键挑战，梳理了主流趋势及其优缺点，并明确了未来研究的潜在方向。我们涵盖截至投稿日期的文献，并将持续关注<https://github.com/CNJianLiu/Awesome-Object-Pose-Estimation>上的最新研究成果。

关键词 物体姿态估计 · 深度学习 · 综合测量 · 三维计算机视觉

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1 Introduction

Object pose estimation is a fundamental computer vision problem that aims to estimate the pose of an object in a given image relative to the camera that captured the image. Object pose estimation is a crucial technology for augmented reality (Liu et al., 2022f; He et al., 2022a; Wen et al., 2024), robotic manipulation (Liu et al., 2023d, 2024a), hand-object interaction (Lin et al., 2023b; Rezazadeh et al., 2023), etc. Depending on the application needs, the object pose is estimated up to varying degrees of freedom (DoF) such as 3DoF that only includes 3D rotation, 6DoF that additionally includes 3D translation, or 9DoF which includes estimating the 3D size of the object besides the 3D rotation and 3D translation.

In the pre-deep learning era, many hand-crafted feature-based approaches such as SIFT (Lowe, 2004), FPFH (Rusu et al., 2009), VFH (Rusu et al., 2010), and Point Pair Features (PPF) (Drost et al., 2010; Choi & Christensen, 2012; Choi et al., 2013; Birdal & Ilic, 2015) were designed for object pose estimation. However, these methods exhibit deficiencies in accuracy and robustness when confronted with complex scenes (Xiang et al., 2017; Wang et al., 2019a). These traditional methods have now been supplanted by data driven deep learning-based approaches that harness the power of deep neural networks to learn high-dimensional feature representations from data, leading to improved accuracy and robustness to handle complex environments.

Deep learning-based methods can be divided into instance-level, category-level, and unseen (including both instance-unseen and category-unseen cases) object pose estimation according to the problem formulation. Fig. 1 shows a comparison of the three methods. Early methods were mainly instance-level (Wang et al., 2019a; Peng et al., 2019; He et al., 2020; Li et al., 2019; Zakharov et al., 2019), trained to estimate the pose of specific object instances. Instance-level methods can be further divided into correspondence-based, template-based, voting-based, and regression-based methods. Since instance-level methods are trained on instance-specific data, they can estimate pose with high precision for the given object instances. However, their generalization performance is poor because they are meant to be applied only to the instances on which they are trained. Moreover, many instance-level methods (He et al., 2020, 2021) require CAD models of the objects. Recognizing these limitations, Wang et al. (2019b) proposed the first category-level object pose and size estimation method. They generalize to intra-class unseen objects without necessitating retraining and employing CAD models during inference. Subsequent category-level methods (Lin et al., 2022c; Di et al., 2022; Zheng et al., 2023; Liu et al., 2023c, 2025b) can be divided into shape prior-based and shape prior-free methods. While improving the generalization ability within a category, these category-level

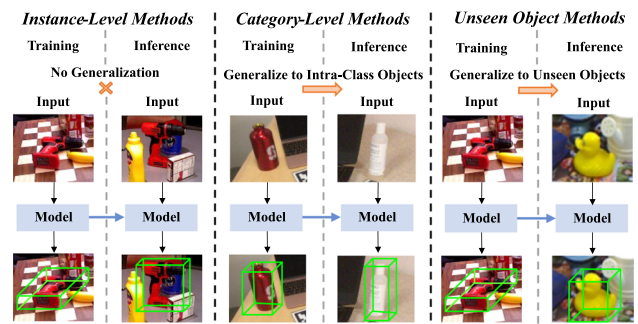


Fig. 1 Comparison of instance-level, category-level, and unseen (including both instance-unseen and category-unseen cases) object pose estimation methods. Instance-level methods can only estimate the pose of specific object instances on which they are trained. Category-level methods can generalize to unseen instances within a known category, rather than being limited to the specific training instances. In contrast, unseen object pose estimation methods aim for stronger generalization, handling both novel instances and entirely new object categories not encountered during training

methods still need to collect and label extensive training data for each object category. Moreover, these methods cannot generalize to unseen object categories. To this end, some unseen object pose estimation methods have been recently proposed (Liu et al., 2022f; Labbé et al., 2022; Nguyen et al., 2024b; Lin et al., 2024a; Wen et al., 2024), which can be further classified into CAD model-based and manual reference view-based methods. These methods further enhance the generalization of object pose estimation, *i.e.*, they can be generalized to unseen objects without retraining. Nevertheless, they still need to obtain the object CAD model or annotate a few reference images of the object.

Although significant progress has been made in the area of object pose estimation, several challenges persist in current methods, such as the reliance on labeled training data, difficulty in generalizing to novel unseen objects, model compactness, and robustness in challenging scenarios. To enable readers to swiftly grasp the current state-of-the-art (SOTA) in object pose estimation and facilitate further research in this direction, it is crucial to provide a thorough review of all the relevant problem formulations. A close examination of the existing academic literature reveals a significant gap when reviewing the various problem formulations in object pose estimation. Current prevailing reviews (Hoque et al., 2021; Marullo et al., 2023; Fan et al., 2022b; Du & Wang, 2021; Guan et al., 2024) tend to exhibit a narrow focus, either confined to particular input modalities (Marullo et al., 2023; Fan et al., 2022b) or tethered to specific application domains (Du & Wang, 2021; Guan et al., 2024). Furthermore, these reviews predominantly scrutinize instance-level and category-level methods, thus neglecting the exploration of the most practical problem formulation in the domain which is unseen object pose estimation. This hinders readers

1 引言

物体姿态估计是计算机视觉中的一个基础问题, 其目标是估算给定图像中物体相对于拍摄相机的姿态。该技术对增强现实 (Liu 等人, 2022f; He 等人, 2022a; Wen 等人, 2024)、机器人操作 (Liu 等人, 2023d、2024a)、人机交互 (Lin 等人, 2023b; Rezazadeh 等人, 2023) 等领域至关重要。根据具体应用需求, 物体姿态估计可支持不同自由度 (DoF): 例如仅包含三维旋转的3DoF、额外包含三维平移的6DoF, 或在三维旋转和平移基础上还需估算物体三维尺寸的9DoF。

在深度学习时代之前, 许多基于特征的手工设计方法——例如 SIFT (Lowe 等人 2004), FPFH (Rusu; 2009), VFH (Rusu; 2010) 和 Point Pair Features (PPF) (Drost 等人 2010; Choi & Christensen, <; 2012; Choi 等人 2013; Birdal & Ilic, <; 2015)——均被用于物体姿态估计。然而, 当面对复杂场景时, 这些方法在准确性和鲁棒性方面存在明显不足 (Xiang 等人 2017; Wang 等人 2019a)。如今, 这些传统方法已被数据驱动的深度学习方法所取代; 后者利用深度神经网络的强大能力从数据中学习高维特征表示, 从而显著提升了处理复杂环境时的准确性和鲁棒性。

基于深度学习的方法可根据问题表述分为实例级、类别级以及未见过物体姿态估计 (包括实例未见过和类别未见过两种情况)。图1展示了三种方法的对比。早期方法主要集中在实例级 (Wang 等人 2019a; Peng 等人 2019; He 等人 2020; Li 等人 2019; Zakharov 等人 2019), 其训练目标是估计特定物体实例的姿态。实例级方法可进一步划分为基于对应关系、基于模板、基于投票和基于回归的方法。由于实例级方法基于实例特定数据训练, 因此能对给定物体实例进行高精度姿态估计。但其泛化性能较差, 因为这些方法仅适用于训练时所涉及的实例。此外, 许多实例级方法 (He 等人 2020、2021) 需要物体的CAD模型。针对这些局限性, Wang 等人 (2019b) 提出了首个类别级物体姿态与尺寸估计方法。该方法无需重新训练且在推理过程中不依赖CAD模型, 即可推广至类内未见过物体。后续类别级方法 (Lin 等人, 2022c; Di 等人, 2022; Zheng 等人, 2023; Liu 等人, 2023c、2025b) 可分为基于形状先验与无形状先验方法。这些类别级方法在提升类别内泛化能力的同时,

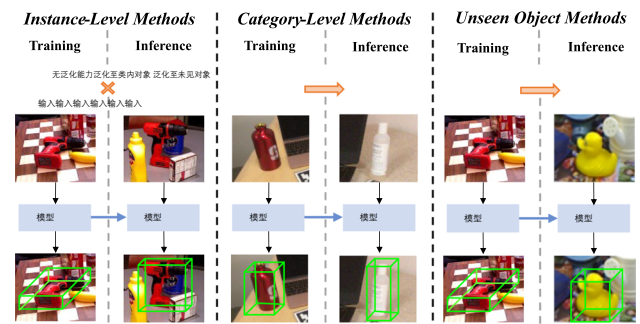


图1实例级、类别级与未见样本 (包含实例未见和类别未见两种情况) 物体姿态估计方法对比。实例级方法仅能对训练时涉及的具体物体实例进行姿态估计; 类别级方法可对已知类别中的未见实例进行泛化, 突破了特定训练样本的局限性; 而未见样本物体姿态估计方法则致力于更强的泛化能力, 既能处理新出现的实例, 也能应对训练过程中未曾接触过的全新物体类别。

这些方法仍需为每个物体类别收集并标注大量训练数据。此外, 这些方法无法泛化到未见过的物体类别。为此, 近期已提出若干未见过物体姿态估计方法 (Liu 等人, 2022f; Labbe 等人, 2022; Nguyen 等人, 2024b; Lin 等人, 2024a; Wen 等人, 2024), 这些方法可进一步分为基于CAD模型的方法和基于人工参考视图的方法。这些方法进一步提升了物体姿态估计的泛化能力, 即无需重新训练即可推广到未见过的物体。然而, 它们仍需获取物体CAD模型或标注少量物体参考图像。

尽管在物体姿态估计领域已取得显著进展, 当前方法仍存在若干挑战, 例如依赖标注训练数据、难以泛化到新未见物体、模型紧凑性以及复杂场景中的鲁棒性。为使读者快速掌握物体姿态估计领域的最新研究进展 (SOTA), 并促进该方向的进一步研究, 对所有相关问题表述进行全面综述至关重要。对现有学术文献的深入分析表明, 在物体姿态估计的各种问题表述方面存在显著空白。当前主流综述 (Hoque 等人, 2021; Marullo 等人, 2023; Fan 等人, 2022b; Du & Wang, 2021; Guan 等人, 2024) 往往呈现狭窄聚焦, 要么局限于特定输入模态 (Marullo 等人, 2023; Fan 等人, 2022b), 要么受限于特定应用领域 (Du & Wang, 2021; Guan 等人, 2024)。此外, 这些综述主要聚焦于实例级和类别级方法, 从而忽略了对该领域中最实用的问题表述——即未见过物体的姿态估计——的探讨。这阻碍了读者的理解。

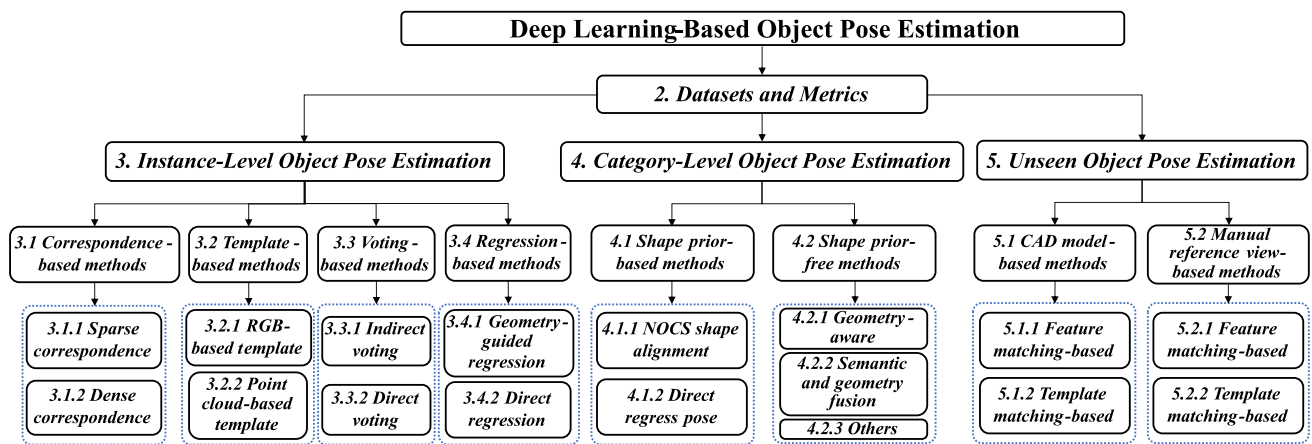


Fig. 2 A taxonomy of this survey. Firstly, we review the datasets and evaluation metrics used to evaluate object pose estimation. Next, we review the deep learning-based methods by dividing them into three categories: instance-level, category-level, and unseen object pose estimation. Instance-level methods can be further classified into

correspondence-based, template-based, voting-based, and regression-based methods. Category-level methods can be further divided into shape prior-based and shape prior-free methods. Unseen object pose estimation methods can be further classified into CAD model-based and manual reference view-based methods

from gaining a comprehensive understanding of the area. For instance, Fan et al. (2022b) provided valuable insights into RGB image-based object pose estimation. However, their focus is limited to a singular modality, hindering readers from comprehensively understanding methods across various input modalities. Conversely, Du and Wang (2021) exclusively examined object pose estimation within the context of the robotic grasping task, which limits the readers to understand object pose estimation only from the perspective of a single specific application.

To address the above problems, we present here a comprehensive survey of recent advancements in deep learning-based methods for object pose estimation. Our survey encompasses all problem formulations, including instance-level, category-level, and unseen object pose estimation, aiming to provide readers with a holistic understanding of this field. Additionally, we discuss different domain training paradigms, application areas, evaluation metrics, and benchmark datasets, as well as report the performance of state-of-the-art methods on these benchmarks, aiding readers in selecting suitable methods for their applications. Furthermore, we also highlight prevailing trends and discuss their strengths and weaknesses, as well as identify key challenges and promising avenues for future research. The taxonomy of this survey is shown in Fig. 2. Our main contributions and highlights are as follows:

- We present a *comprehensive survey* of deep learning-based object pose estimation methods. This is the *first* survey that covers all three problem formulations in the domain, including instance-level, category-level, and unseen object pose estimation.

- Our survey covers popular input data modalities (RGB images, depth images, RGBD images), the different degrees of freedom (3DoF, 6DoF, 9DoF) in output poses, object properties (rigid, articulated) for the task of pose estimation as well as tracking. It is crucial to cover all these aspects in a single survey to give a complete picture to readers, an aspect overlooked by existing surveys which only cover a few of these aspects.
- We discuss different domain training paradigms, inference modes, application areas, evaluation metrics, and benchmark datasets as well as report the performance of existing SOTA methods on these benchmarks to help readers choose the most appropriate ones for deployment in their application.
- We highlight popular trends in the evolution of object pose estimation techniques over the past decade and discuss their strengths and weaknesses. We also identify key challenges that are still outstanding in object pose estimation along with promising research directions to guide future efforts.

The rest of this article is organized as follows. Sec. 2 reviews the datasets and metrics used to evaluate the three categories of object pose estimation methods. We then review instance-level methods in Sec. 3, category-level methods in Sec. 4, and unseen object pose estimation methods in Sec. 5. In the aforementioned three sections, we also discuss the training paradigms, inference modes, challenges, and popular trends associated with representative methods in the particular category. Next, Sec. 6 reviews the common applications of object pose estimation. Finally, Sec. 7 provides an

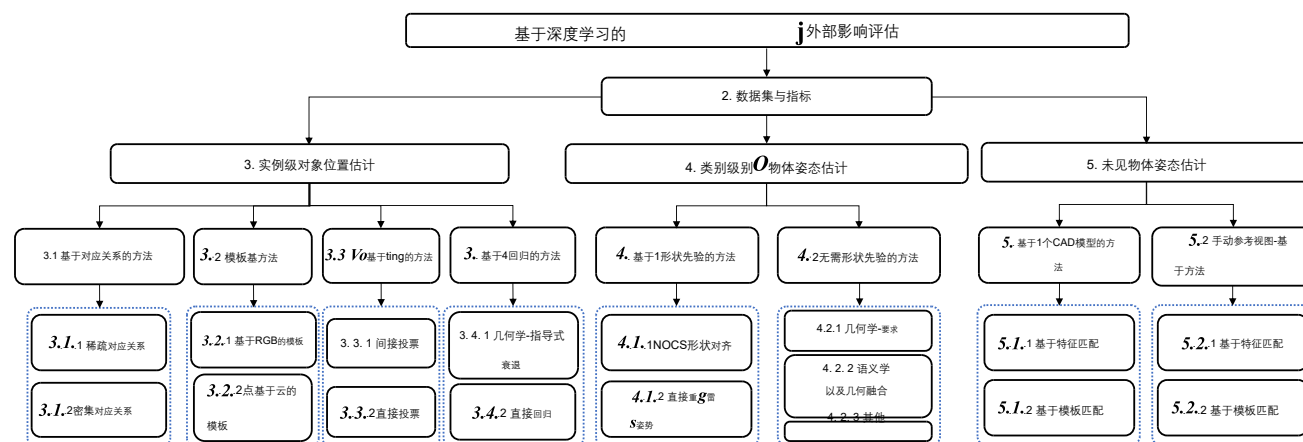


图 2 本调查的分类体系。首先，我们回顾了用于评估物体姿态估计的数据集与评估指标；接着，将基于深度学习的方法分为三类进行综述：实例级、类别级及未见物体姿态估计方法。实例级方法可进一步细分为基于对应关系的方法、基于模板的方法、基于投票的方法以及基于回归的方法；类别级方法可分为基于形状

先验的方法与无需形状先验的方法；未见物体姿态估计方法则可进一步划分为基于CAD模型的方法与基于手动参考视图的方法。

这有助于全面理解该领域。例如，Fan 等人（2022b）在基于RGB图像的对象姿态估计方面提供了宝贵见解。然而，他们的研究仅限于单一输入模态，导致读者难以全面掌握跨多种输入模态的方法。相反，Du 和 Wang（2021）仅针对机器人抓取任务背景下的对象姿态估计展开研究，这使得读者只能从单一具体应用的角度来理解对象姿态估计。

为解决上述问题，本文对基于深度学习的物体姿态估计方法最新进展进行了全面综述。我们的研究涵盖实例级、类别级及未见物体姿态估计等所有问题建模方式，旨在为读者提供该领域的整体认知框架。此外，我们系统探讨了不同领域训练范式、应用场景、评估指标及基准数据集，并展示了前沿方法在这些基准上的性能表现，助力读者选择适合实际应用的解决方案。同时，我们重点分析了当前研究趋势及其优缺点，明确了关键挑战与未来研究方向。本综述的分类体系如图2所示。主要研究成果与亮点如下：

- 我们对基于深度学习的物体姿态估计方法进行了全面综述。这是首个涵盖该领域所有三种问题表述（包括实例级、类别级及未见物体姿态估计）的综述研究。

- 我们的调查涵盖主流输入数据形式（RGB图像、深度图像、RGBD 图像）、输出位姿的不同自由度（3DoF、6DoF、9DoF）、物体属性（刚性、关节式）等位姿估计与跟踪任务所需要素。在单一调查中全面覆盖这些要素对读者而言至关重要，这正是现有调查所忽略的——现有研究往往仅涉及其中部分维度。
- 我们探讨了不同的领域训练范式、推理模式、应用领域、评估指标及基准数据集，并报告了现有 SOTA 方法在这些基准上的性能表现，以帮助读者选择最适合其应用场景的方法。
- 本文重点梳理了过去十年间物体姿态估计技术演进中的主流趋势，并对其优缺点进行系统分析。同时，我们指出了该领域仍存在的挑战，以及具有发展前景的研究方向，为未来研究提供指导方向。

本文其余部分结构如下：第2节综述用于评估三类物体姿态估计方法的数据集与评估指标；随后在第3节中回顾实例级方法，第4节探讨类别级方法，第5节则研究未见物体的姿态估计方法。

5. 在上述三个章节中，我们还讨论了特定类别中代表性方法的训练范式、推理模式、挑战及流行趋势。接下来，第6节回顾了物体姿态估计的常见应用。最后，第7节提供了

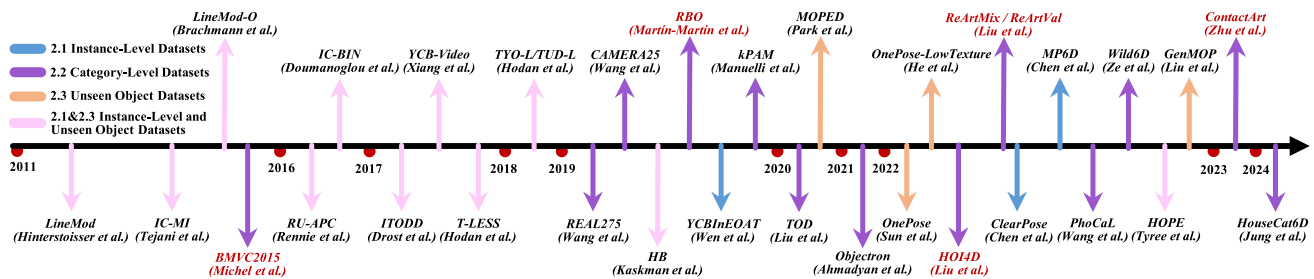


Fig. 3 Chronological overview of the datasets for object pose estimation evaluation. Notably, the pink arrows represent the BOP Challenge datasets, which can be used to evaluate both instance-level and unseen object pose estimation methods. The red references represent the

datasets of articulated objects. From this, we can also see the development trend in the field of object pose estimation, *i.e.*, from instance-level methods to category-level and unseen object pose estimation methods

outlook on emerging trends and future research directions based on the challenges in the field.

2 Datasets and Metrics

The advancement of deep learning-based object pose estimation is closely linked to the creation and utilization of challenging and trustworthy large-scale datasets. This section introduces commonly used mainstream object pose estimation datasets, categorized into instance-level, category-level, and unseen object pose estimation methods based on problem formulation. The chronological overview is shown in Fig. 3. In addition, we also conduct an overview of the related evaluation metrics.

2.1 Datasets for Instance-Level Methods

Since the BOP Challenge datasets (Hodan et al., 2024) are currently the most popular datasets for the evaluation of instance-level methods, we divide the instance-level datasets into BOP Challenge and other datasets for overview.

2.1.1 BOP Challenge Datasets

Linemod Dataset (LM) (Hinterstoisser et al., 2012) comprises 15 RGBD sequences containing annotated RGBD images with ground-truth 6DoF object poses, object CAD models, 2D bounding boxes, and binary masks. Typically following (Brachmann et al., 2016), approximately 15% of images from each sequence are allocated for training, with the remaining 85% reserved for testing. These sequences present challenging scenarios with cluttered scenes, texture-less objects, and varying lighting conditions, making accurate object pose estimation difficult.

Linemod Occlusion Dataset (LM-O) (Brachmann et al., 2014) is an extension of the LM dataset (Hinterstoisser et al., 2012) specifically designed to evaluate the performance in

occlusion scenarios. This dataset consists of 1214 RGBD images from the basic sequence in the LM dataset for 8 heavily occluded objects. It is critical for evaluating and improving pose estimation algorithms in complex environments characterized by occlusion.

IC-MI (Tejani et al., 2014) / **IC-BIN Dataset** (Doumanoglou et al., 2016) contribute to texture-less object pose estimation. IC-MI comprises six objects: 2 texture-less and 4 textured household item models. IC-BIN dataset is specifically designed to address challenges posed by clutter and occlusion in robot garbage bin picking scenarios. IC-BIN includes 2 objects from the IC-MI.

RU-APC Dataset (Rennie et al., 2016) aims to tackle challenges in warehouse picking tasks and provides rich data for evaluating and improving the perception capabilities of robots in a warehouse automation context. The dataset comprises 10,368 registered depth and RGB images, covering 24 types of objects, which are placed in various poses within different boxes on warehouse shelves to simulate diverse experimental conditions.

YCB-Video Dataset (YCB-V) (Xiang et al., 2017) comprises 21 objects distributed across 92 RGBD videos, each video containing 3 to 9 objects from the YCB object dataset (Calli et al., 2015) (totaling 50 objects). It includes 133,827 frames with a resolution of 640×480 , making it well-suited for both object pose estimation and tracking tasks.

T-LESS Dataset (Hodan et al., 2017) is an RGBD dataset designed for texture-less objects commonly found in industrial settings. It includes 30 electrical objects with no obvious texture or distinguishable color properties. In addition, it includes images of varying resolutions. In the training set, images predominantly feature black backgrounds, while the test set showcases diverse backgrounds with varying lighting conditions and occlusions. T-LESS is challenging because of the absence of texture on objects and the intricate environmental settings.

ITODD Dataset (Drost et al., 2017) includes 28 real-world industrial objects distributed across over 800 scenes with

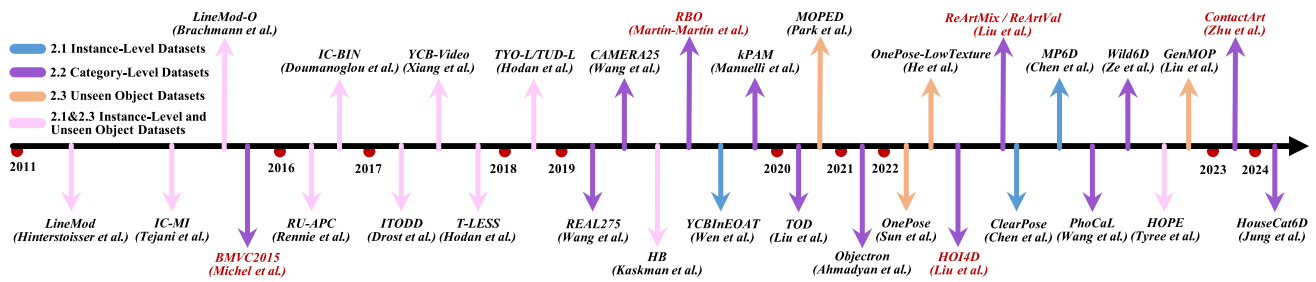


图 3 对象姿态估计评估数据集的时间线概览。值得注意的是, 粉色箭头代表 BOP Challenge 数据集, 可用于评估实例级及未见过物体的姿态估计方法; 红色参考数据集则代表关节运动物体的数据集。由此还可看出物体姿态估计领域的演进趋势: 即从实例

级方法发展到类别级及未见过物体的姿态估计方法。

基于该领域面临的挑战, 对新兴趋势及未来研究方向的展望。

2 数据集与指标

基于深度学习的物体姿态估计技术发展, 与具有挑战性且可信度高的大规模数据集的创建与应用密切相关。本节将按问题建模方式, 从实例级、类别级及未见物体姿态估计方法三个维度, 系统介绍主流物体姿态估计数据集。时间线梳理详见图3, 同时我们还对相关评估指标进行了系统性梳理。

2.1 用于实例级方法的数据集

由于BOP Challenge数据集 (Hodan 等人, 2024) 目前是评估实例级方法最主流的数据集, 为便于概览, 我们将实例级数据集划分为BOP Challenge数据集及其他数据集。

2.1.1 BOP挑战数据集

Linemod数据集 (LM) (Hinterstoisser 等人, 2012) 包含15个 RGBD 序列, 每个序列均配有标注完整的 RGBD 图像, 其中包含真实六自由度物体姿态、物体 CAD模型、二维边界框及二值掩码。通常遵循 (Brachmann 等人, 2016) 的做法, 每个序列中约15%的图像用于训练, 其余85%用于测试。这些序列呈现了具有复杂场景、无纹理物体及多变光照条件的挑战性场景, 导致物体姿态的精确估计难度较高。

Linemod遮挡数据集 (LM-O) (Brachmann 等人, 2014) 是LM数据集 (Hinterstoisser 等人, 2012) 的扩展版本, 专门用于评估遮挡场景下的性能表现。该数

据集包含来自LM数据集基础序列的1214张 RGBD 图像, 涉及8个高度遮挡物体。对于评估和改进复杂遮挡环境中的姿态估计算法具有关键意义。

IC-MI (Tejani et al., 2014) 与 **IC-BIN Dataset** (Doumanoglou et al., 2016) 均致力于无纹理物体姿态估计任务。IC-MI包含六种物体模型: 2种无纹理模型和4种带纹理的家用物品模型; IC-BIN数据集专为解决机器人垃圾箱拾取场景中由物体密集分布和遮挡造成的挑战而设计, 其中包含IC-MI中的两种物体模型。

RU-APC数据集 (Rennie 等人, 2016) 旨在解决仓库拣选任务中的挑战, 并为评估和提升机器人在仓库自动化环境中的感知能力提供丰富数据。该数据集包含10,368张已标注的深度图像和RGB图像, 涵盖24种物体类型, 这些物体以不同姿态放置在仓库货架的不同箱体中, 以模拟多样化的实验条件。

YCB 视频数据集 (YCB-V) (Xiang 等人, 2017) 包含21个目标, 分布在92个 RGBD 视频中, 每个视频包含来自 YCB 目标数据集 (Calli 等人, 2015) 的3至9个目标 (总计50个)。该数据集包含133,827帧图像, 分辨率为640×480, 非常适合用于目标姿态估计和跟踪任务。

T-LESS 数据集 (Hodan 等人, 2017) 是一个专为工业环境中常见的无纹理物体设计的 RGBD 数据集。该数据集包含 30 个缺乏明显纹理或可辨识颜色特征的电气设备图像, 同时涵盖不同分辨率的图像。训练集中的图像主要以黑色背景为主, 而测试集则展示了具有多种背景、不同光照条件及遮挡效果的图像。T-LESS 数据集的挑战性源于物体表面缺乏纹理以及复杂的环境场景设置。

ITODD 数据集 (Drost 等人, 2017) 包含28个真实工业物体, 分布于800多个场景中。

around 3,500 images. This dataset leverages two industrial 3D sensors and three high-resolution grayscale cameras to enable multi-angle observation of the scenes, providing comprehensive and detailed data for industrial object analysis and evaluation.

TYO-L / TUD-L Dataset (Hodan et al., 2024) focus on different lighting conditions. Specifically, TYO-L provides observation of 3 objects under 8 lighting conditions. These scenes are designed to evaluate the robustness of pose estimation algorithms to lighting variations. Unlike TYO-L, the data collection method of TUD-L involves fixing the camera and manually moving the object, providing a more realistic representation of the object's physical movement.

HB Dataset (Kaskman et al., 2019) covers various scenes with changes in occlusion and lighting conditions. It comprises 33 objects, including 17 toys, 8 household items, and 8 industry-related objects, distributed across 13 diverse scenes.

HOPE Dataset (Tyree et al., 2022) is specifically designed for household objects, containing 28 toy grocery objects. The HOPE-Image dataset includes objects from 50 scenes across 10 home/office environments. Each scene includes up to 5 lighting variations, such as backlit and obliquely directed lighting, with shadow-casting effects. Additionally, the HOPE-Video dataset comprises 10 video sequences totaling 2,038 frames, with each scene showcasing between 5 to 20 objects.

2.1.2 Other Datasets

YCBInEOAT Dataset (Wen et al., 2020) is designed for RGBD-based object pose tracking in robotic manipulation. It contains the ego-centric RGBD videos of a dual-arm robot manipulating the YCB objects (Calli et al., 2015). There are 3 types of manipulation: single-arm pick-and-place, within-arm manipulation, and pick-and-place between arms. This dataset comprises annotations of ground-truth poses across 7449 frames, encompassing 5 distinct objects depicted in 9 videos.

ClearPose Dataset (Chen et al., 2022d) is designed for transparent objects, which are widely prevalent in daily life, presenting significant challenges to visual perception and sensing systems due to their indistinct texture features and unreliable depth information. It encompasses over 350K real-world RGBD images and 5M instance annotations across 63 household objects.

MP6D Dataset (Chen et al., 2022c) is an RGBD dataset designed for object pose estimation of metal parts, featuring 20 texture-less metal components. It consists of 20,100 real-world images with object pose labels collected from various scenarios as well as 50K synthetic images, encompassing cluttered and occluded scenes.

2.2 Datasets for Category-Level Methods

In this part, we divide the category-level datasets into rigid and articulated object datasets for elaboration.

2.2.1 Rigid Objects Datasets

CAMERA25 Dataset (Wang et al., 2019b) incorporates 1085 instances across 6 object categories: bowl, bottle, can, camera, mug, and laptop. Notably, the object CAD models in CAMERA25 are sourced from the synthetic ShapeNet dataset (Chang et al., 2015). Each image within this dataset contains multiple instances, accompanied by segmentation masks and 9DoF pose labels.

REAL275 Dataset (Wang et al., 2019b) is a real-world dataset comprising 18 videos and approximately 8K RGBD images. The dataset is divided into three subsets: a training set (7 videos), a validation set (5 videos), and a testing set (6 videos). It includes 42 object instances across 6 categories, consistent with those in the CAMERA25 dataset. REAL275 is a prominent real-world dataset extensively used for category-level object pose estimation in academic research.

kPAM Dataset (Manuelli et al., 2019) is tailored specifically for robotic applications, emphasizing the use of keypoints. Notably, it adopts a methodology involving 3D reconstruction followed by manual keypoint annotation on these reconstructions. With a total of 117 training sequences and 245 testing sequences, kPAM offers a substantial collection of data for training and evaluating algorithms related to robotic perception and manipulation.

TOD Dataset (Liu et al., 2020) consists of 15 transparent objects categorized into 6 classes, each annotated with pertinent 3D keypoints. It encompasses a vast collection of 48K stereo and RGBD images capturing both transparent and opaque depth variations. The primary focus of the TOD dataset is on transparent 3D object applications, providing essential resources for tasks such as object detection and pose estimation in challenging scenarios involving transparency.

Objectron Dataset (Ahmadyan et al., 2021) contains 15K annotated video clips with over 4M labeled images belonging to categories of bottles, books, bikes, cameras, chairs, cereal boxes, cups, laptops, and shoes. This dataset is sourced from 10 countries spanning 5 continents, ensuring diverse geographic representation. Due to its extensive content, it is highly advantageous for evaluating the RGB-based category-level object pose estimation and tracking methods.

Wild6D Dataset (Ze & Wang, 2022) is a substantial real-world dataset used to assess self-supervised category-level object pose estimation methods. It offers annotations exclusively for 486 test videos with diverse backgrounds, showcasing 162 objects across 5 categories (excluding the "can" category found in CAMERA25 and REAL275).

该数据集包含约3,500张图像。通过采用两台工业级3D传感器与三台高分辨率灰度相机,实现了场景的多角度观测,为工业物体分析与评估提供了全面详实的数据支持。

TYO -L/ TUD -L数据集 (Hodan 等人, 2024) 专注于不同光照条件下的场景分析。具体而言, TYO -L数据集包含8种光照条件下对3个物体的观测数据,这些场景设计用于评估姿态估计算法对光照变化的鲁棒性。与 TYO -L不同, TUD-L的数据采集方法采用固定摄像机并手动移动物体的方式,能更真实地呈现物体的物理运动状态。

HB数据集 (Kaskman 等人, 2019) 涵盖多种场景,包含遮挡变化和光照条件变化。该数据集包含33个物体,包括17件玩具、8件家居用品和8件工业相关物品,分布在13个不同场景中。**HOPE数据集** (Tyree 等人, 2022) 专为家居物品设计,包含28件玩具类杂货物品。HOPE-Image数据集包含来自10个家庭/办公环境的50个场景中的物体。每个场景包含多达5种光照变化(如背光和斜射照明),并具有阴影投射效果。此外, HOPE-Video数据集包含10个视频序列,总计2,038帧,每个场景展示5至20个物体。

2.1.2 其他数据集

YCBInEOAT数据集 (Wen 等人, 2020) 专为机器人操作中的基于 RGBD 的物体姿态跟踪设计。该数据集包含双臂机器人操作 YCB 物体时的自我中心 RGBD 视频 (Calli 等人, 2015), 涵盖三种操作类型: 单臂取放、臂内操作及双臂间取放。数据集包含7449帧视频的真实姿态标注, 涵盖9个视频中呈现的5种不同物体。

ClearPose数据集 (Chen 等人, 2022d) 专为透明物体设计——这类物体在日常生活中极为常见,但由于其纹理特征模糊且深度信息不可靠,给视觉感知与传感系统带来了重大挑战。该数据集包含超过35万张真实世界的 RGBD 图像,以及涵盖63种家用物品的500万条实例标注数据。

MP6D数据集 (Chen 等人, 2022c) 是一个专为金属零件物体姿态估计设计的 RGBD 数据集,包含20个无纹理金属部件。该数据集由20,100张来自不同场景并带有物体姿态标签的真实世界图像,以及50K张涵盖杂乱场景和遮挡场景的合成图像组成。

2.2 类别水平方法的数据集

在本部分中,我们将类别级别的数据集划分为刚性物体数据集和关节式物体数据集以供详细阐述。

2.2.1 刚性物体数据集

CAMERA25数据集 (Wang 等人, 2019b) 包含6个物体类别的1085个实例:碗、瓶子、罐子、相机、马克杯和笔记本电脑。值得注意的是, CAMERA25中的物体CAD模型源自合成ShapeNet数据集 (Chang 等人, 2015)。该数据集中的每张图像均包含多个实例,并附有分割掩码和9自由度姿态标签。

REAL275数据集 (Wang 等人, 2019b) 是一个包含18个视频和约8K张 RGBD 图像的真实世界数据集。该数据集分为三个子集:训练集(7个视频)、验证集(5个视频)和测试集(6个视频)。数据集包含6个类别中的42个物体实例,与CAMERA25数据集中的分类标准一致。REAL275是学术研究中广泛用于类别级物体姿态估计的重要真实世界数据集。

kPAM数据集 (Manuellet等人, 2019) 专为机器人应用场景设计,重点强调关键点的使用。该数据集采用三维重建技术,随后对重建结果进行人工关键点标注,整体方法论具有显著优势。包含117个训练序列和245个测试序列, kPAM为机器人感知与操作算法的训练与评估提供了海量数据支持。

TOD数据集 (Liu 等人, 2020) 包含15个透明物体,分为6个类别,每个类别均标注了相关的3D关键点。该数据集涵盖4.8万张立体和 RGBD 图像,捕捉了透明与不透明物体的深度变化特征。TOD数据集的核心应用方向是透明3D物体识别,为涉及透明度的复杂场景中的物体检测与姿态估计等任务提供了重要资源。**Objectron数据集** (Ahmadyan 等人, 2021) 包含1.5万段标注视频片段及超过400万张标注图像,涵盖瓶子、书籍、自行车、相机、椅子、麦片盒、杯子、笔记本电脑和鞋子等类别。该数据集源自五大洲10个国家,确保了地理分布的多样性。凭借其丰富的内容资源,该数据集在评估基于RGB的类别级物体姿态估计与跟踪方法方面具有显著优势。

Wild6D数据集 (Ze & Wang, 2022) 是一个规模庞大的真实世界数据集,用于评估自监督类别的物体姿态估计方法。该数据集仅包含486段测试视频的标注信息,视频背景多样,涵盖5个类别中的162个物体(不包括CAMERA25和REAL275中包含的“罐子”类别)。

PhoCaL Dataset (Wang et al., 2022b) incorporates both RGBD and RGB-P (Polarisation) modalities. It consists of 60 meticulously crafted 3D models representing household objects, including symmetric, transparent, and reflective items. PhoCaL focuses on 8 specific object categories across 24 sequences, deliberately introducing challenges such as occlusion and clutter.

HouseCat6D Dataset (Jung et al., 2024) is a comprehensive dataset designed for multi-modal category-level object pose estimation and grasping tasks. The dataset encompasses a wide range of household object categories, featuring 194 high-quality 3D models. It includes objects of varying photometric complexity, such as transparent and reflective items, and spans 41 scenes with diverse viewpoints. The dataset is specifically curated to address challenges in object pose estimation, including occlusions and the absence of markers, making it suitable for evaluating algorithms under real-world conditions.

2.2.2 Articulated Objects Datasets

BMVC Dataset (Michel et al., 2015) includes 4 articulated objects: laptop, cabinet, cupboard, and toy train. Each object is modeled as a motion chain comprising components and interconnected heads. Joints are constrained to one rotational and one translational DoF. This dataset provides CAD models and accompanying text files detailing the topology of the underlying motion chain structure for each object.

RBO Dataset (Martín-Martín et al., 2019) contains 14 commonly found articulated objects in human environments, with 358 interaction sequences, resulting in a total of 67 minutes of manual manipulation under different experimental conditions, including changes in interaction type, lighting, viewpoint, and background settings.

HOI4D Dataset (Liu et al., 2022e) is pivotal for advancing research in category-level human-object interactions. It comprises 2.4M RGBD self-centered video frames depicting interactions between over 9 participants and 800 object instances. These instances are divided into 16 categories, including 7 rigid and 9 articulated objects.

ReArtMix / ReArtVal Datasets (Liu et al., 2022c) are formulated to tackle the challenge of partial-level multiple articulated objects pose estimation featuring unknown kinematic structures. The ReArtMix dataset encompasses over 100,000 RGBD images rendered against diverse background scenes. The ReArtVal dataset consists of 6 real-world desk-top scenes comprising over 6,000 RGBD frames.

ContactArt Dataset (Zhu et al., 2024) is generated using a remote operating system (Qin et al., 2022a) to manipulate articulated objects in a simulation environment. This system utilizes smartphones and laptops to precisely annotate poses and contact information. This dataset contains 5 prevalent categories of articulated objects: laptops, drawers, safes,

microwaves, and trash cans, for a total of 80 instances. All object models are sourced from the PartNet dataset (Mo et al., 2019), thus promoting scalability.

2.3 Datasets for Unseen Object Pose Estimation

The current mainstream datasets for evaluating unseen objects are the BOP Challenge datasets, as discussed in Sec. 2.1.1. Besides these BOP Challenge datasets, there are also some datasets designed for evaluating manual reference view-based methods as follows.

MOPED Dataset (Park et al., 2020a) is a model-free object pose estimation dataset featuring 11 household objects. It includes reference and test images that encompass all views of the objects. Each object in the test sequences is depicted in five distinct environments, with approximately 300 test images per object.

GenMOP Dataset (Liu et al., 2022f) includes 10 objects ranging from flat objects to thin structure objects. For each object, there are two video sequences collected from various backgrounds and lighting situations. Each video sequence consists of approximately 200 images.

OnePose Dataset (Sun et al., 2022b) comprises over 450 real-world video sequences of 150 objects. These sequences are collected in a variety of background conditions and capture all angles of the objects. Each environment has an average duration of 30 seconds. The dataset is randomly partitioned into training and validation sets.

OnePose-LowTexture Dataset (He et al., 2022a) is introduced as a complement to the testing set of the existing OnePose dataset (Sun et al., 2022b), which predominantly features textured objects. This dataset comprises 40 household objects with low texture. For each object, there are two video sequences: one serving as the reference video and the other for testing. Each video is captured at a resolution of 1920×1440, 30 Frames Per Second (FPS), and approximately 30 seconds in duration.

2.4 Metrics

In this part, we divide the metrics into 3DoF, 6DoF, 9DoF, and other evaluation metrics for a comprehensive overview.

2.4.1 3DoF Evaluation Metrics

The geodesic distance (Wohlhart and Lepetit, 2015) between the ground-truth and predicted 3D rotations is a commonly used 3DoF pose estimation metric. Calculating the angle error between two rotation matrices can visually evaluate their relative deviation. It can be formulated as follows:

$$d(R_{gt}, R) = \arccos\left(\frac{\text{tr}(R_{gt}^T R) - 1}{2}\right) / \pi, \quad (1)$$

PhoCaL数据集 (Wang 等人, 2022b) 同时包含 RGBD 和 RGB-P (偏振) 两种模态。该数据集包含 60 个精心制作的 3D 模型, 用于表征家用物品, 涵盖对称、透明及反射性物体。PhoCaL 聚焦于 24 个序列中的 8 个特定物体类别, 并刻意引入遮挡与杂乱等挑战。

HouseCat6D数据集 (Jung 等人, 2024) 是一个专为多模态类别级物体姿态估计与抓取任务设计的综合性数据集。该数据集涵盖广泛的家庭用品类别, 包含 194 个高质量 3D 模型, 涉及透明物品、反光物品等不同光度复杂度的物体, 并包含 41 个具有多样化视角的场景。该数据集经过专门整理, 旨在解决物体姿态估计中的遮挡问题及标记缺失等挑战, 因此非常适合用于评估算法在真实场景下的表现。

2.2.2 可动物体数据集

BMVC数据集 (Michel 等人, 2015) 包含四个可活动物体: 笔记本电脑、柜子、橱柜和玩具火车。每个物体均被建模为由多个组件及相互连接的头部组成的运动链。关节仅允许一个旋转自由度和一个平移自由度。该数据集提供了 CAD 模型及配套文本文件, 详细描述了每个物体底层运动链结构的拓扑关系。

RBO数据集 (Martin-Martin 等人, 2019) 包含 14 种常见于非人类环境中的可活动物体, 共包含 358 条交互序列, 对应在不同实验条件下 (包括交互类型、光照、视角和背景设置的变化) 累计 67 分钟的人工操作数据。

HOI4D数据集 (Liu 等人, 2022e) 对推动类别级人机交互研究具有关键意义。该数据集包含 240 万帧 RGBD 自中心视频帧, 记录了超过 9 名参与者与 800 个物体实例之间的交互行为。这些物体实例被划分为 16 个类别, 其中包括 7 种刚性物体和 9 种可动关节物体。

ReArtMix / ReArtVal数据集 (Liu 等, 2022c) 旨在解决具有未知运动学结构的部分层级多关节物体姿态估计问题。ReArtMix 数据集包含超过 100,000 张在多样化背景场景下渲染的 RGBD 图像。ReArtVal 数据集由 6 个真实桌面场景组成, 包含超过 6,000 RGBD 帧。

ContactArt数据集 (Zhu 等, 2024) 通过远程操作系统 (Qin 等, 2022a) 在仿真环境中操控关节物体生成, 该系统利用智能手机和笔记本电脑精确标注姿态与接触信息。该数据集包含 5 种常见关节物体类别: 笔记本电脑、抽屉、保险柜、微波炉和垃圾桶, 共计

80 个实例。所有物体模型均源自 PartNet 数据集 (Mo 等, 2019), 从而提升可扩展性。

2.3 未见物体姿态估计数据集

目前用于评估未见过物体的主流数据集是 BOP Challenge 数据集, 具体详见第 2.1.1 节。除这些 BOP Challenge 数据集外, 还存在一些专门用于评估基于人工参考视图方法的数据集, 具体如下。

moped数据集 (Park 等人, 2020a) 是一个无模型的物体姿态估计数据集, 包含 11 种家用物品。该数据集包含参考图像和测试图像, 涵盖了这些物品的所有视角。测试序列中的每个物品均出现在五个不同的环境中, 每个物品约有 300 张测试图像。

GenMOP数据集 (Liu 等人, 2022f) 包含 10 种物体, 涵盖从平面物体到细结构物体的多种类型。每个物体均采集了来自不同背景和光照条件下的两组视频序列, 每组视频序列包含约 200 张图像。

OnePose数据集 (Sun 等人, 2022b) 包含超过 450 个包含 150 个物体的真实世界视频序列。这些序列采集于多种背景条件下, 完整捕捉了物体的各个角度视角。每个场景的平均时长为 30 秒。该数据集被随机划分为训练集和验证集。

OnePose低纹理数据集 (He 等人, 2022a) 作为现有 OnePose 数据集 (Sun 等人, 2022b) 测试集的补充数据集被引入, 该数据集主要包含纹理丰富的物体。该数据集包含 40 个低纹理家居物品, 每个物品对应两个视频序列: 一个作为参考视频, 另一个用于测试。所有视频均以 1920×1440 分辨率、每秒 30 帧 (FPS) 的帧率拍摄, 时长约 30 秒。

2.4 指标

在本部分中, 我们将评估指标分为 3 自由度 (3DoF)、6 自由度 (6DoF)、9 自由度 (9DoF) 及其他评估指标, 以提供全面概述。

2.4.1 三维自由度评估指标

真实三维旋转与预测三维旋转之间的测地距离 (Wohlfarth 和 Lepetit, 2015) 是常用的三自由度姿态估计指标。通过计算两个旋转矩阵之间的角度误差, 可直观评估它们的相对偏差。其计算公式如下:

$$d(R_{gt}, R) = \arccos\left(\frac{\text{tr}(R_{gt}^T R) - 1}{2}\right) / \pi, \quad (1)$$

where R_{gt} and R denote the ground-truth and predicted 3D rotations, respectively. $\top$ represents matrix transpose. tr denotes the trace of a matrix, which refers to the sum of the elements on the main diagonal. Typically, 3D rotation estimation accuracy is defined as the percentage of objects whose angle error is below a specific threshold and whose predicted class is correct. It can be expressed as follows:

$$Acc. = \begin{cases} 1, & \text{if } d(R_{gt}, R) < \lambda \text{ and } c = c_{gt} \\ 0, & \text{otherwise} \end{cases}, \quad (2)$$

where c , c_{gt} , and λ denote the predicted class, ground-truth class and predefined threshold, respectively.

2.4.2 6DoF Evaluation Metrics

Currently, the BOP metric (*BOP-M*) (Hodan et al., 2024) is the most popular metric, which is the Average Recall (*AR*) of the Visible Surface Discrepancy (*VSD*), Maximum Symmetry-Aware Surface Distance (*MSSD*), and Maximum Symmetry-Aware Projection Distance (*MSPD*) metrics. Specifically, the *VSD* (Hodan et al., 2024) metric treats poses that are indistinguishable in shape as equivalent by only measuring the misalignment of the visible object surface. It can be expressed as follows:

$$e_{VSD}(\hat{D}, \bar{D}, \hat{V}, \bar{V}, \tau) = avg_{p \in \hat{V} \cup \bar{V}} \begin{cases} 0, & \text{if } p \in \hat{V} \cap \bar{V} \wedge |\hat{D}(p) - \bar{D}(p)| < \tau \\ 1, & \text{otherwise} \end{cases}, \quad (3)$$

where the symbols $\hat{D}$ and $\bar{D}$ represent distance maps generated by rendering the object model M in two different poses: $\hat{P}$ (an estimated pose) and $\bar{P}$ (the ground-truth pose), respectively. In these maps, each pixel p stores the distance from the camera center to a 3D point x_p that projects onto p . These distance values are derived from depth maps, which are typical outputs of sensors like Kinect, containing the Z coordinate of x_p . These distance maps are compared with the distance map D_I of the test image I to derive visibility masks $\hat{V}$ and $\bar{V}$. These masks identify pixels where the model M is visible in the image I . The parameter τ represents the tolerance for misalignment. In addition, the *MSSD* (Hodan et al., 2024) metric is a suitable factor for determining the likelihood of successful robotic manipulation and is not significantly affected by object geometry or surface sampling density. It can be formulated as follows:

$$e_{MSSD}(\hat{P}, \bar{P}, S_M, V_M) = \min_{S \in S_M} \max_{x \in V_M} \|\hat{P}x - \bar{P}Sx\|_2, \quad (4)$$

where the set S_M comprises global symmetry transformations for the object model M , while V_M represents the

vertices of the model. Furthermore, the *MSPD* (Hodan et al., 2024) metric is ideal for evaluating RGB-only methods in augmented reality, focusing on perceivable discrepancies and excluding alignment along the optical (Z) axis, which can be represented as follows:

$$e_{MSPD}(\hat{P}, \bar{P}, S_M, V_M) = \min_{S \in S_M} \max_{x \in V_M} \|proj(\hat{P}x) - proj(\bar{P}Sx)\|_2, \quad (5)$$

where the function $proj()$ represents the 2D projection (pixel-level), and the other symbols have the same meanings as in *MSSD*.

Besides the *BOP-M*, the average point distance (*ADD*) (Hinterstoisser et al., 2012) and average closest point distance (*ADD-S*) (Hinterstoisser et al., 2012) are also commonly leveraged to evaluate the performance of 6DoF object pose estimation. They can intuitively quantify the geometric error between the estimated and the ground-truth poses by computing the average distance between corresponding points on the object CAD model. Specifically, the *ADD* metric is designed for asymmetric objects, while the *ADD-S* metric is designed explicitly for symmetric objects. Given the ground-truth rotation R_{gt} and translation t_{gt} , as well as the estimated rotation R and translation t , *ADD* calculates the average pairwise distance between the 3D model points $x \in O$ corresponding to the transformation between the ground truth and estimated pose:

$$ADD = avg_{x \in O} \|(R_{gt}x + t_{gt}) - (Rx + t)\|. \quad (6)$$

For symmetric objects, the matching between points in certain views is inherently ambiguous. Therefore, the average distance is calculated using the nearest point distance as follows:

$$ADD-S = avg \min_{x_1 \in O} \min_{x_2 \in O} \|(R_{gt}x_1 + t_{gt}) - (Rx_2 + t)\|. \quad (7)$$

Meanwhile, the area under the *ADD* and the *ADD-S* curve (*AUC*) are often leveraged for evaluation. Specifically, if the *ADD* and *ADD-S* are smaller than a given threshold, the predicted pose will be considered correct. Moreover, there are many methods (Wang et al., 2020b; Jiang et al., 2022; Li et al., 2022) that evaluate asymmetric and symmetric objects using *ADD* and *ADD-S*, respectively. This metric is termed *ADD(S)*.

In addition, $n^\circ mcm$ (Shotton et al., 2013) is also currently a prevalent evaluation metric (especially in category-level object pose estimation). It directly quantifies the errors in predicted 3D rotation and 3D translation. An object pose prediction is deemed correct if its rotation error is below threshold n° and its translation error is below threshold mcm . It can be defined as an indicator function as follows:

其中 R_{gt} 和 R 分别表示真实三维旋转角度和预测三维旋转角度。 T 表示矩阵转置运算。 tr 表示矩阵的迹运算,即主对角线元素之和。通常,三维旋转估计精度定义为角度误差低于特定阈值且预测类别正确的物体所占百分比,其计算公式如下:

$$Acc. = \begin{cases} 1, & \text{if } d(R_{gt}, R) < \lambda \text{ and } c = c_{gt} \\ 0, & \text{otherwise} \end{cases}, \quad (2)$$

其中 c 、 c_{gt} 和 λ 分别表示预测类别、真实类别和预定义阈值。

2.4.2 6自由度评估指标

目前,BOP指标($BOP-M$) (Hodan 等人, 2024)是最受欢迎的指标,它由可见表面差异(VSD)、最大对称感知表面距离(MSSD)和最大对称感知投影距离(MSPD)指标的平均召回率(AR)组成。具体而言,VSD (Hodan 等人, 2024)指标通过仅测量可见物体表面的错位程度,将形状无法区分的姿态视为等效。其表达式如下:

$$e_{VSD}(\hat{D}, \bar{D}, \hat{V}, \bar{V}, \tau) = \text{avg}_{p \in \hat{V} \cup \bar{V}} \begin{cases} 0, & \text{if } p \in \hat{V} \cap \bar{V} \wedge |\hat{D}(p) - \bar{D}(p)| < \tau \\ 1, & \text{otherwise} \end{cases}, \quad (3)$$

其中符号 $\hat{D}$ 和 $\bar{D}$ 表示通过以两种不同姿态渲染对象模型 M 生成的距离图:

$\hat{P}$ (估计姿态)和 $\bar{P}$ (真实姿态)。在这些地图中,每个像素 p 存储了从相机中心到投影到 p 上的三维点 x_p 的距离。这些距离值源自深度图,深度图是Kinect等传感器的典型输出,包含 x_p 的 Z 坐标。这些距离图与

distance 进行比较map DI: 测试图像 I 的可见性参数

掩模 V 和 $\bar{V}$ 这些掩模用于识别图像 I 中模型 M 可见的像素位置。参数 τ 表示对错位的容差。此外,MSSD (Hodan 等人, 2024)度量指标是评估机器人操作成功率的合适因素,且不受物体几何形状或表面采样密度的显著影响。其可表述如下:

$$e_{MSSD}(\hat{P}, \bar{P}, S_M, V_M) = \min_{S \in S_M} \max_{x \in V_M} \|\hat{P}x - \bar{P}Sx\|_2, \quad (4)$$

其中集合 S_M 包含对象模型 M 的全局对称变换,而 V_M 表示模型的顶点。此外,MSPD (Hodan 等人, 2024)

度量指标非常适合评估增强现实中的纯RGB方法,其重点关注可感知的差异,并排除沿光学(Z)轴的对齐问题,具体表示如下:

$$e_{MSPD}(\hat{P}, \bar{P}, S_M, V_M) = \min_{S \in S_M} \max_{x \in V_M} \left\| \text{proj}(\hat{P}x) - \text{proj}(\bar{P}Sx) \right\|_2, \quad (5)$$

其中函数 $\text{proj}()$ 表示二维投影(像素级),其余符号含义与MSSD相同。

除了BOP-M之外,平均点距离(ADD) (Hinterstoisser等人, 2012)和平均最近点距离(ADD-S) (Hinterstoisser等人, 2012)也被广泛用于评估6DoF物体姿态估计的性能。它们通过计算物体CAD模型上对应点之间的平均距离,直观量化估计姿态与真实姿态之间的几何误差。具体而言,ADD指标专为非对称物体设计,而ADD-S指标则专门针对对称物体设计。给定真实旋转 R_{gt} 和平移 t_{gt} ,以及估计旋转 R 和平移 t ,ADD计算真实姿态与估计姿态变换对应的三维模型点 $x \in O$ 之间的平均成对距离:

$$ADD = \text{avg}_{x \in O} \|(R_{gt}x + t_{gt}) - (Rx + t)\|. \quad (6)$$

对于对称物体而言,特定视图中各点之间的匹配本质上具有模糊性。因此,平均距离是通过计算最近点距离得出的,具体公式如下:

$$ADD-S = \text{avg}_{x_1 \in O} \min_{x_2 \in O} \|(R_{gt}x_1 + t_{gt}) - (Rx_2 + t)\|. \quad (7)$$

与此同时,ADD曲线与ADD-S曲线下的面积(AUC)常被用于评估。具体而言,若ADD与ADD-S值低于给定阈值,则预测姿态将被视为正确。此外,已有多种方法(Wang 等人, 2020b; Jiang 等人, 2022; Li 等人, 2022)分别利用ADD与ADD-S评估非对称与对称物体。该指标被称为ADD(S)。

此外, mcm (Shotton 等人, 2013)目前也是一种普遍采用的评估指标(尤其在类别级物体姿态估计中)。该指标直接量化预测的三维旋转与三维平移误差。若物体姿态预测的旋转误差低于阈值 n° 且平移误差低于阈值 mcm ,则判定其预测结果正确。其可定义为如下指标函数:

$$I_{n^{\circ}mcm}(e_R, e_t) = \begin{cases} 1, & \text{if } e_R < n^{\circ} \text{ and } e_t < mcm \\ 0, & \text{otherwise} \end{cases}, \quad (8)$$

where e_R and e_t represent the rotation and translation errors between the estimated and ground-truth values, respectively.

Furthermore, compared to directly comparing 6DoF pose in 3D space, the simplicity and practicality of the 2D Projection metric (Brachmann et al., 2016) make it suitable for evaluation as well, which quantifies the average distance between CAD model points when projected under the estimated object pose and the ground-truth pose. A pose is considered correct if the projected distances are less than 5 pixels.

2.4.3 9DoF Evaluation Metric

IoU_{3D} denotes the Intersection-over-Union (IoU) (Wang et al., 2019b) percentage between the ground-truth and predicted 3D bounding boxes, which can evaluate the 6DoF pose estimation as well as the 3DoF size estimation. It can be expressed as:

$$IoU_{3D} = \frac{P_B \cap G_B}{P_B \cup G_B}, \quad (9)$$

where G_B and P_B represent the ground-truth and the predicted 3D bounding boxes, respectively. The symbols $\cap$ and $\cup$ represent the intersection and union, respectively. The correctness of the predicted object pose is determined based on whether the IoU_{3D} value exceeds a predefined threshold.

2.4.4 Other Metric

Since some Normalized Object Coordinate Space (NOCS) shape alignment-based category-level methods reconstruct the 3D object shape before estimating the object pose, the Chamfer Distance (CD) metric (Tian et al., 2020a), which not only captures the global shape deviation but is also sensitive to local shape differences, is commonly leveraged to evaluate the NOCS shape reconstruction accuracy of these methods

as follows:

$$D_{cd} = \sum_{x \in N} \min_{y \in M_{gt}} \|x - y\|_2^2 + \sum_{y \in M_{gt}} \min_{x \in N} \|x - y\|_2^2, \quad (10)$$

where N and M_{gt} represent the reconstructed and ground-truth NOCS shape, respectively.

3 Instance-Level Object Pose Estimation

Instance-level object pose estimation describes the task of estimating the pose of the objects that have been seen during the training of the model. We classify existing instance-level methods into four categories: correspondence-based (Sec. 3.1), template-based (Sec. 3.2), voting-based (Sec. 3.3), and regression-based (Sec. 3.4) methods.

3.1 Correspondence-Based Methods

Correspondence-based object pose estimation refers to techniques that involve identifying correspondences between the input data and the given complete object CAD model. Correspondence-based methods can be divided into sparse (Sec. 3.1.1) and dense (Sec. 3.1.2) correspondences. The illustration of these two types of methods is shown in Fig. 4. The attributes and performance of some representative methods are shown in Table 1.

3.1.1 Sparse Correspondence Methods

Sparse correspondence-based methods involve detecting object keypoints in the input image or point cloud to establish 2D-3D or 3D-3D correspondences between the input data and the object CAD model, followed by the Perspective-n-Point (PnP) algorithm (Fischler & Bolles, 1981) or least square method to determine the object pose. As a representative method, Rad and Lepetit (2017) first used a segmentation method to detect the object of interest in an RGB image. Then, they predicted the 2D projections of the object's 3D

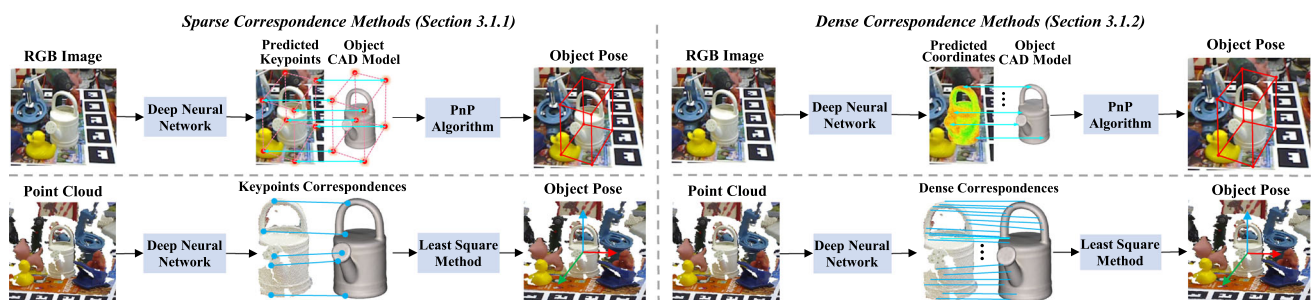


Fig. 4 Illustration of the correspondence-based methods (Sec. 3.1), which involve establishing correspondences between input data and a provided object CAD model

$$I_{n^\circ mcm}(e_R, e_t) = \begin{cases} 1, & \text{if } e_R < n^\circ \text{ and } e_t < mcm \\ 0, & \text{otherwise} \end{cases}, \quad (8)$$

其中 e_R 和 e_t 分别表示估计值与真实值之间的旋转误差和平移误差。

此外, 相较于直接比较三维空间中的六自由度姿态, 二维投影度量指标 (Brachmann 等人, 2016) 的简洁性和实用性也使其适用于评估。该指标通过量化 CAD 模型点在估计物体姿态与真实姿态投影下的平均距离来实现。若投影距离小于 5 像素, 则判定姿态正确。

2.4.3 9自由度评估指标

IoU_{3D} 表示真实值与预测三维边界框之间的交并比 (IoU) (Wang 等人, 2019b), 可用于评估六自由度姿态估计及三维尺寸估计。其表达式为:

$$IoU_{3D} = \frac{P_B \cap G_B}{P_B \cup G_B}, \quad (9)$$

其中 G_B 和 P_B 分别表示真实三维边界框和预测三维边界框。符号 $\cap$ 和 $\cup$ 分别表示交集和并集。预测物体姿态的正确性是通过 IoU_{3D} 值是否超过预设阈值来判定的。

2.4.4 其他指标

由于基于归一化对象坐标空间 (NOCS) 形状对齐的类别级方法在估计物体姿态前会先重建三维物体形状, 因此常采用锥度距离 (CD) 度量指标 (Tian 等人, 2020a) 来评估这些方法的 NOCS 形状重建精度。该指标不仅能捕捉整体形状偏差, 还对局部形状差异具有敏感性, 具体评估方式如下:

$$D_{cd} = \sum_{x \in N} \min_{y \in M_{gt}} \|x - y\|_2^2 + \sum_{y \in M_{gt}} \min_{x \in N} \|x - y\|_2^2, \quad (10)$$

其中 N 和 M_{gt} 分别表示重构 NOCS 形状与真实形状。

3 实例级对象位姿估计

实例级对象姿态估计描述了在模型训练过程中对已观测到对象的姿态进行估计的任务。我们将现有的实例级方法分为四类: 基于对应关系的方法 (第3.1节)、基于模板的方法 (第3.2节)、基于投票的方法 (第3.3节) 以及基于回归的方法 (第3.4节)。

3.1 基于通信量的方法

基于对应关系的对象姿态估计是指通过识别输入数据与给定完整对象 CAD 模型之间的对应关系来实现的技术。基于对应关系的方法可分为稀疏对应关系 (第3.1.1节) 和密集对应关系 (第3.1.2节)。这两种方法的示意图如图4所示。部分代表性方法的属性与性能数据见表1。

3.1.1 稀疏对应方法

基于稀疏对应关系的方法包括: 在输入图像或点云中检测物体的关键点, 以建立输入数据与物体 CAD 模型之间的二维-三维或三维-三维对应关系; 随后采用透视 n 点 (PnP) 算法 (Fischler & Bolles, 1981) 或最小二乘法来确定物体的姿态。作为代表性方法, Rad 和 Lepetit (2017) 首先采用分割算法在 RGB 图像中检测目标物体, 进而预测该物体三维空间的二维投影。

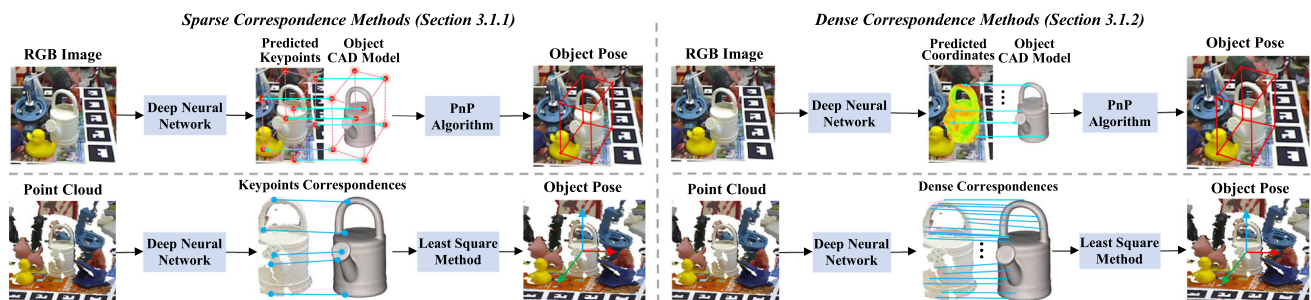


图 4 基于对应关系的方法示意图 (第3.1节), 该方法涉及在输入数据与提供的对象 CAD 模型之间建立对应关系

Table 1 Representative instance-level methods. For each method, we report its 10 properties: published year, training input, inference input, pose DoF (3DoF, 6DoF, and 9DoF), object property (rigid, articulated), task (estimation, tracking, and refinement), domain training paradigm (source domain, domain adaptation, and domain generalization), inference mode, application area, and its performance of key metrics on key datasets. Notably, for the input of training and inference, we only focus on the input of the pose estimation model, not the input of the front-end segmentation method (because it can be obtained through RGB as well as through depth or RGBD). D, S, C, T, V, P, and R denote object detection, instance segmentation, correspondence prediction, template matching, voting, pose solution/regression, and pose refinement, respectively. We report the average recall of ADD(S) within 10% of the object diameter (termed ADD(S)-0.1d) of LM-O and LM datasets, and the AUC of ADD-S (< 0.1m) of YCB-V dataset (Sec. 2)

Methods	Published Year	Training Input	Inference Input	Pose DoF	Object Property	Task	Domain Training Paradigm	Inference Mode	Application Area	LM-O LM ADD(S)-0.1d	YCB-V ADD-S (< 0.1m)
Correspondence-Based Methods											
	Rad and Lepetit (2017)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	four-stage, S+C+P+R	symmetrical objects	-	62.7
	Tekin et al. (2018)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	two-stage, C+P	general	-	56.0
	Hu et al. (2019)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	two-stage, C+P	occlusion	27.0	-
	Song et al. (2020)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	three-stage, C+P+R	symmetrical objects, occlusion	47.5	91.3
	Hu et al. (2021)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	two-stage, C+P	large scale variations	48.6	-
	Chang et al. (2021)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	two-stage, C+P	transparent, symmetrical objects	-	-
	Guo et al. (2023)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	three-stage, D+C+P	general	44.5	-
	Li et al. (2019)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	three-stage, D+C+P	general	-	89.9
	Hodan et al. (2020)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	two-stage, C+P	symmetrical objects	-	-
	Shugurov et al. (2021)	RGB/RGBD, CAD Model	RGB/RGBD, CAD Model	6DoF	rigid	estimation	source	four-stage, D+C+P+R	general	-	99.9
Template-Based Methods	Chen et al. (2022a)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	three-stage, D+C+P	general	-	95.8
	Haugaard and Buch (2022)	RGB/RGBD, CAD Model	RGB/RGBD, CAD Model	6DoF	rigid	estimation	generalization	four-stage, D+C+P+R	general	-	-
	Li et al. (2023a)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	three-stage, D+C+P	general	51.4	97.8
	Xu et al. (2024b)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	refinement	source	two-stage, P+R	occlusion	60.7	97.4
	Sundermeyer et al. (2018)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	generalization	three-stage, D+T+P	general	-	31.4
	Papadimitris et al. (2020)	RGB, CAD Model	RGB, CAD Model	3DoF	rigid	estimation	source	two-stage, D+T	general	-	-
	Li and Ji (2020)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	three-stage, D+T+R	general	-	88.6
	Deng et al. (2021)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	tracking	generalization	two-stage, D+T	symmetrical objects	-	-
	Li et al. (2022)	RGBD, CAD Model	RGBD, CAD Model	6DoF	rigid	estimation	source	three-stage, S+P+R	general	70.6	99.5
	Jiang et al. (2024)	Depth, CAD Model	Depth, CAD Model	6DoF	rigid	estimation	source	two-stage, S+P	general	-	-
	Dang et al. (2024)	Depth, CAD Model	Depth, CAD Model	6DoF	rigid	estimation	source	two-stage, S+P	general	52.0	69.0

表1代表性实例级方法。针对每种方法, 我们报告其10项属性: 发表年份、训练输入、推理输入、姿态自由度 (3DoF、6DoF和9DoF)、物体属性 (刚性、分体式)、任务类型 (估计、跟踪与精化)、领域训练范式 (源域、领域适应与领域泛化)、推理模式、应用领域, 以及关键数据集上主要指标的性能表现。值得注意的是, 对于训练与推理的输入, 我们仅关注姿态估计模型的输入, 而非前端分割方法的输入 (因其可通过RGB、深度或RGBD获取)。D、S、C、T、V、P和R分别表示物体检测、实例分割、对应关系预测、模板匹配、投票决策、姿态解/回归及姿态精化。我们报告了LM-O与LM数据集上ADD(S)在物体直径10%范围内平均召回率 (称为ADD(S-0.1m)的AUC值 (详见第2节))。

方法	稀疏对应关系	Rad 与 Lepetit (2017)	发表年份	训练输入	推理输入	姿态自由度	对象属性	任务	领域训练范式	推理模式	应用区域	LM-O LM ADD(S)-0.1d	YCB -V (<0.1米)
基于通信量的方法			2017	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	根源	四级 S+C+P+R	对称物体	-	62.7
		Tekin等 (2018)	2018	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	根源	两阶段C+P 法	一般的	-	56.0
		Hu 等 人 (2019)	2019	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	根源	两阶段C+P 法	闭塞	-	-
		Song等人 (2020)	2020	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	根源	三阶段: C+ P+R	对 称 物 体 , 遮挡	47.5	91.3
		Hu 等 人 (2021)	2021	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	根源	两阶段C+P 法	大规模变异	48.6	-
		Chang等 (2021)	2021	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	根源	两阶段, C+ P	透明的	-	-
		Gao 等 人 (2023)	2023	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	根源	三阶段D+C+ P	一般的	44.5	-
	致密对应关系	Li 等人 (2019)	2019	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	根源	三阶段D+C+ P	一般的	-	89.9
		Hodan等人 (2020)	2020	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	根源	两阶段C+P 法	对称物体	-	-
		Shugurov 等人 (2021)	2021	RGB/ RGBD, CAD模型	RGB/ RGBD, CAD模型	6自由度	严格的	评价	根源	四级 D+C+P+R	一般的	-	99.9
		Chen 等人 (2022)	2022	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	根源	三阶段D+C+ P	一般的	-	95.8
基于模板的方法		Haugard 与 Bush (2022)	2022	RGB/ RGBD, CAD模型	RGB/ RGBD, CAD模型	6自由度	严格的	评价	一般化	四级 D+C+P+R	一般的	-	-
		Li 等 人 (2023)	2023	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	根源	三阶段D+C+ P	一般的	51.4	97.8
		Xu 等 人 (2024b)	2024	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	精炼	根源	两阶段P+R	闭塞	60.7	97.4
	基于RGB的	松德迈尔 等人 (2018)	2018	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	一般化	三阶段的 D+T+P两阶段	一般的	-	31.4
		Papaioannidis 等人 (2020)	2020	RGB, CAD模型	RGB, CAD模型	3自由度	严格的	评价	根源	段: D+T	一般的	-	-
		Li 和 Ji (2020)	2020	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	评价	根源	三阶段: D+ T+R	一般的	-	88.6
		Deng等 (2021)	2021	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	跟踪	一般化	两阶段, D+ T	对称物体	-	-
	点云	Li 等人 (2022)	2022	RGBD, CAD模型	RGBD, CAD模型	6自由度	严格的	评价	根源	三阶段S+P+ R	一般的	70.6	99.5
		Jiang等 (2024)	2023	深度, CAD模型	深度, CAD模型	6自由度	严格的	评价	根源	两阶段S+P法	一般的	-	-
		Dang等 (2024)	2024	深度, CAD模型	深度, CAD模型	6自由度	严格的	评价	根源	两阶段S+P法	一般的	52.0	69.0

Table 1 continued

Methods	Published Year	Training Input	Inference Input	Pose DoF	Object Property	Task	Domain Training Paradigm	Inference Mode	Application Area	LM-O LM ADD(S)-0.1d	YCB-V ADD-S (<0.1m)
Voting-Based Methods											
	Peng et al. (2019)	RGB	RGB	6DoF	rigid	estimation	source	two-stage, V+P	occlusion	40.8	86.3
	He et al. (2020)	RGBD, CAD Model	RGBD, CAD Model	6DoF	rigid	estimation	source	three-stage, S+V+P	general	-	99.4
	He et al. (2021)	RGBD, CAD Model	RGBD, CAD Model	6DoF	rigid	estimation	source	three-stage, S+V+P	general	66.2	99.7
	Cao et al. (2022)	RGB, CAD Model	RGB	6DoF	rigid	estimation	source	end to end	general	58.7	-
	Wu et al. (2022)	RGBD, CAD Model	RGBD, CAD Model	6DoF	rigid	estimation	source	three-stage, S+V+P	general	70.2	99.4
	Zhou et al. (2023a)	RGBD, CAD Model	RGBD, CAD Model	6DoF	rigid	estimation	source	three-stage, S+V+P	general	77.7	99.8
	Wang et al. (2019a)	RGBD	RGBD	6DoF	rigid	estimation	source	two-stage, P+R	general	-	94.3
	Tian et al. (2020b)	RGBD, CAD Model	RGBD, CAD Model	6DoF	rigid	estimation	source	three-stage, S+V+P	general	-	92.9
	Zhou et al. (2021b)	RGBD, CAD Model	RGBD, CAD Model	6DoF	rigid	estimation	source	two-stage, S+P	general	65.0	99.6
	Mo et al. (2022)	RGBD	RGBD	6DoF	rigid	estimation	source	two-stage, S+P	general	-	-
	Hong et al. (2024)	RGBD	RGBD	6DoF	rigid	estimation	source	two-stage, S+P	general	71.1	96.7
Regression-Based Methods											
	Chen et al. (2020b)	RGBD	RGBD	6DoF	rigid	estimation	source	end to end	general	-	98.7
	Hu et al. (2020)	RGBD, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	end to end	general	43.3	-
	Labbé et al. (2020)	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	three-stage, D+P+R	general	-	-
	Wang et al. (2021c)	RGBD, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	two-stage, D+P	general	62.2	-
	Di et al. (2021)	RGBD, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	source	two-stage, D+P	occlusion	62.32	96.0
	Wang et al. (2021b)	RGBD, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	adaptation	two-stage, D+P	occlusion	59.8	85.6
	Xiang et al. (2017)	RGB	RGB	6DoF	rigid	estimation	source	three-stage, S+V+P	cluttered	24.9	-
	Li et al. (2018a)	RGBD	RGBD	6DoF	rigid	estimation	source	two-stage, P+R	general	-	-

表 1继续

方法	发布年份	训练输入	推理输入	姿势自由度	对象属性	任务	领域培训范式	推理模式	应用区域	LM-O LM ADD(S),Id	YCB -V (<0.1米)	ADDS
基于投票的方法	Peng等 (2019)	RGB	RGB	6自由度	严格的	评价	根源	二阶段 V+P	闭塞	40.8 86.3	-	
	He等 (2020)	RGBD, CAD 模型	RGBD, CAD 模型	6自由度	严格的	评价	根源	三阶段的 S+V+P	一般的	- 99.4	95.5	
	He等人 (2021)	RGBD, CAD 模型	RGBD, CAD 模型	6自由度	严格的	评价	根源	三阶段的; S+V+P	一般的	66.2 99.7	96.6	
	Cao 等人 (2022)	RGB, CAD模型	RGB	6自由度	严格的	评价	根源	首尾相连地	一般的	58.7 -	90.9	
	Wu等 (2022)	RGBD, CAD 模型	RGBD, CAD 模型	6自由度	严格的	评价	根源	三阶段的; S+V+P	一般的	70.2 99.4	96.6	
	周等人 (2023a)	RGBD, CAD 模型	RGBD, CAD 模型	6自由度	严格的	评价	根源	三阶段的 S+V+P	一般的	77.7 99.8	96.7	
	Wang等 (2019a)	RGBD	RGBD	6自由度	严格的	评价	根源	二阶段 P+R	一般的	- 94.3	93.1	
	Tan等 (2020b)	RGBD, CAD 模型	CAD RGBD	6自由度	严格的	评价	根源	三阶段的 S+V+P	一般的	- 92.9	91.8	
	周等人 (2021b)	RGBD, CAD 模型	CAD RGBD	6自由度	严格的	评价	根源	两阶段S+P法	一般的	65.0 99.6	95.8	
	Mo 等人 (2022)	RGBD	RGBD	6自由度	严格的	评价	根源	两阶段S+P法	一般的	- -	93.6	
	Hong等 (2024)	RGBD	RGBD	6自由度	严格的	评价	根源	两阶段S+P法	一般的	71.1 96.7	92.7	
	Chen 等人 (2020b)	RGBD	RGBD	6自由度	严格的	评价	根源	首尾相连地	一般的	- 98.7	92.4	
基于回归的方法	胡等人 (2020)	RGBD, CAD 模型	CAD RGB	6自由度	严格的	评价	根源	首尾相连地	一般的	43.3 -	-	
	Labbe等 (2020)	RGB, CAD模型	RGB, CAD 模型	6自由度	严格的	评价	根源	三阶段的 D+P+R	一般的	- -	93.4	
	Wang等 (2021c)	RGBD, CAD 模型	CAD RGB	6自由度	严格的	评价	根源	二阶段 D+P	一般的	62.2 -	91.6	
	Di 等人 (2021)	RGBD, CAD 模型	CAD RGB	6自由度	严格的	评价	根源	两阶段, D+P	闭塞	62.32 96.0	90.9	
	Wang等 (2021b)	RGBD, CAD 模型	CAD RGB	6自由度	严格的	评价	改编本	二阶段 D+P	闭塞	59.8 85.6	90.5	
	Xiang等 (2017)	RGB	RGB	6自由度	严格的	评价	根源	三阶段的 S+V+P	杂乱的	24.9 -	75.9	
	李等人 (2018a)	RGBD	RGBD	6自由度	严格的	评价	根源	二阶段 P+R	一般的	- -	94.3	

Table 1 continued

Methods	Published Year	Training Input	Inference Input	Pose DoF	Object Property	Task	Domain Training Paradigm	Inference Mode	Application Area	LM-O LM ADD(S)-0.1d	YCB-V ADD-S (<0.1m)
Li et al. (2018c)	2018	RGB, CAD Model	RGB, Model	CAD 6DoF	rigid	refinement, tracking	source	end to end	general	55.5 88.6	81.9
Manhardt et al. (2018)	2018	RGB, CAD Model	RGB, Model	CAD 6DoF	rigid	refinement, tracking	generalization	end to end	general	- -	-
Manhardt et al. (2019)	2019	RGB	RGB	6DoF	rigid	estimation	source	end to end	symmetrical objects	- -	-
Papaioannidis and Pitas (2019)	2019	RGB	RGB	3DoF	rigid	estimation	source	two-stage, D+P	general	- -	-
Liu et al. (2019b)	2019	RGB	RGB	3DoF	rigid	estimation	source	two-stage, D+P	texture-less	- -	-
Wen et al. (2020)	2020	RGBD, Model	CAD RGBD	6DoF	rigid	tracking	generalization	end to end	general	- -	93.9
Wang et al. (2020b)	2020	RGBD, Model	CAD RGBD	6DoF	rigid	estimation	generalization	end to end	general	32.1 58.9	-
Jiang et al. (2022)	2022	RGBD	RGBD	6DoF	rigid	estimation	source	end to end	general	30.8 97.0	95.2
Hai et al. (2023b)	2023	RGB, CAD Model	RGB	6DoF	rigid	refinement	source	end to end	general	66.4 99.3	-
Li et al. (2024a)	2024	RGB, CAD Model	RGB, Model	CAD 6DoF	rigid	refinement	source	two-stage, S+P	general	- -	97.0

表 1 继续

方法	发布年份	训练输入	推理输入	推理	姿势自由度	对象属性	任务	领域训练范式	推理模式	应用区域	LM-O LM ADD(S)-0.1d	VCB -V (<0.1米)	ADDS
李等人 (2018c)	2018	RGB, CAD模型	RGB, CAD模型	RGB, CAD模型	CAD 6自由度	严格的	细化, 跟踪	根源	首尾相连地	一般的	55.5 88.6	81.9	
曼哈特等 al. (2018)	2018	RGB, CAD模型	RGB, CAD模型	RGB, CAD模型	CAD 6自由度	严格的	细化跟踪	泛化	首尾相连地	一般的	-	-	
曼哈特等 al. (2019)	2019	RGB	RGB	RGB	6自由度	严格的	评价	根源	首尾相连地	对称的 目标	-	-	
帕帕伊奥安尼迪斯 以及 Pitas (2019)	2019	RGB	RGB	RGB	3自由度	严格的	评价	根源	二阶段 D+P	一般的	-	-	
Liu等 (2019b)	2019	RGB	RGB	RGB	3自由度	严格的	评价	根源	二阶段 D+P	无纹理	-	-	
Wen等 (2020)	2020	RGBD, CAD模型	RGBD, CAD模型	RGBD, CAD模型	6自由度	严格的	跟踪	泛化	首尾相连地	一般的	-	93.9	
Wang等 (2020b)	2020	RGBD, CAD模型	RGBD, CAD模型	RGBD, CAD模型	6自由度	严格的	评价	泛化	首尾相连地	一般的	32.1 58.9	-	
Jiang等 (2022)	2022	RGBD	RGBD	RGBD	6自由度	严格的	评价	根源	首尾相连地	一般的	30.8 97.0	95.2	
Hai 等人 (2023b)	2023	RGB, CAD模型	RGB, CAD模型	RGB, CAD模型	6自由度	严格的	精炼	根源	首尾相连地	一般的	66.4 99.3	-	
李等人 (2024a)	2024	RGB, CAD模型	RGB, CAD模型	RGB, CAD模型	CAD 6自由度	严格的	精炼	根源	两阶段, S+P	一般的	-	97.0	

bounding box corners. Finally, they used the PnP algorithm (Fischler & Bolles, 1981) to estimate the object pose. Additionally, they employed a classifier to determine the pose range in real-time, addressing the issue of ambiguity in symmetric objects. Tekin et al. (2018) proposed a CNN network inspired by YOLO (Redmon et al., 2016) to integrate object detection and pose estimation, directly predicting the locations of the projected vertices of the 3D object bounding box. Unlike Rad and Lepetit (2017) and Tekin et al. (2018), Pavlakos et al. (2017) predicted the 2D projections of predefined semantic keypoints. Doosti et al. (2020) introduced a compact model comprising two adaptive graph convolutional neural networks (GCNNs) (Kipf & Welling, 2016), collaborating to estimate object and hand poses. To further enhance the robustness of object pose estimation, Song et al. (2020) employed a hybrid intermediate representation to convey geometric details in the input image, encompassing keypoints, edge vectors, and symmetry correspondences. Liu et al. (2021a) proposed a multi-directional feature pyramid network along with a method that calculates object pose estimation confidence by incorporating spatial and plane information. Hu et al. (2021) introduced a single-stage hierarchical end-to-end trainable network to address pose estimation challenges associated with scale variations in aerospace objects. In a recent development, Lian and Ling (2023) increased the number of predefined 3D keypoints to enhance the establishment of correspondences. Moreover, they devised a hierarchical binary encoding approach for localizing keypoints, enabling gradual refinement of correspondences and transforming correspondence regression into a more efficient classification task. To estimate transparent object pose, Chang et al. (2021) used a 3D bounding box prediction network and multi-view geometry. Their method first detects 2D projections of 3D bounding box vertices, and then reconstructs 3D points based on the multi-view detected 2D projections incorporating camera motion data. Additionally, they introduced a generalized pose definition to address pose ambiguity for symmetric objects. To enhance the efficiency of pose estimation networks, Guo et al. (2023) integrated knowledge distillation into object pose estimation by distilling the teacher's distribution of local predictions into the student network. Liu et al. (2023b) argued that differentiable PnP strategies conflict with the averaging nature of the PnP problem, resulting in gradients that may encourage the network to degrade the accuracy of individual correspondences. To mitigate this, they introduced a linear covariance loss, which can be used for both sparse and dense correspondence-based methods.

To mitigate vulnerability caused by large occlusions, Crivellaro et al. (2017) used several control points to represent each object part. Then, they predicted the 2D projections of these control points to calculate the object pose. Some researchers solved the occlusion problem by predicting

keypoints using small patches. Oberweger et al. (2018) processed each patch separately to generate heatmaps and then aggregated the results to achieve precise and reliable predictions. Additionally, they offered a straightforward but efficient strategy to resolve ambiguities between patches and heatmaps during training. Hu et al. (2019) unveiled a segmentation-driven pose estimation framework in which every visible object part offers a local pose prediction through 2D keypoint locations. Furthermore, Huang et al. (2021) conceptualized 2D keypoint locations as probabilistic distributions within the loss function and designed a confidence-based network.

Reducing the reliance on annotated real-world data is also an important task. Some methods exploit geometric consistency as additional information to alleviate the need for annotation. Zhao et al. (2020) employed image pairs with object annotations and relative transformation between viewpoints to automatically identify objects' 3D keypoints that are geometrically and visually consistent. In addition, Yang et al. (2021) used a keypoint consistency regularization for dual-scale images with a labeled 2D bounding box. Using semi-supervised learning, Liu et al. (2021b) developed a unified framework for estimating 3D hand and object poses. They constructed a joint learning framework that conducts explicit contextual reasoning between hand and object representations. To generate pseudo labels in semi-supervised learning, they utilized the spatial-temporal consistency found in large-scale hand-object videos as a constraint. Synthetic data is also a way to solve the annotation problem. Georgakis et al. (2019) reduced the need for expensive 3DoF pose annotations by selecting keypoints and maintaining viewpoint and modality invariance in RGB images and CAD model renderings. Sock et al. (2020) utilized self-supervision to minimize the gap between synthetic and real data and enforced photometric consistency across different object views to fine-tune the model. Further, Zhang et al. (2021) utilized the invariance of geometry relations between keypoints across real and synthetic domains to accomplish domain adaptation. Thalhammer et al. (2021) introduced a specialized feature pyramid network to compute multi-scale features, enabling the simultaneous generation of pose hypotheses across various feature map resolutions.

Overall, sparse correspondence-based methods can estimate object pose efficiently. However, relying on only a few control points can lead to sub-optimal accuracy.

3.1.2 Dense Correspondence Methods

Dense correspondence-based methods aim to establish dense 2D-3D or 3D-3D correspondences. They utilize a significantly larger number of correspondences compared to sparse correspondence-based methods. This enables them to achieve higher accuracy and handle occlusions more

边界框角点。最后,他们采用PnP算法(Fischler & Bolles, 1981)来估计物体姿态。此外,他们运用分类器实时确定姿态范围,解决了对称物体中的模糊性问题。Tekin等人(2018)提出了一种受YOLO(Redmon等人, 2016)启发的CNN网络,将物体检测与姿态估计相结合,直接预测三维物体边界框投影顶点的位置。与Rad和Lepetit(2017)及Tekin等人(2018)不同,Pavlakos等人(2017)预测了预定义语义关键点的二维投影。Doosti等人(2020)引入了一个由两个自适应图卷积神经网络(GCNNs)(Kipf & Welling, 2016)组成的紧凑模型,协同估计物体与手部姿态。为进一步提升物体姿态估计的鲁棒性,Song等人(2020)采用混合中间表征来传递输入图像中的几何细节,涵盖关键点、边缘向量及对称对应关系。Liu等人(2021a)提出了一种多方向特征金字塔网络,并结合了一种通过整合空间与平面信息来计算物体姿态估计置信度的方法。Hu等人(2021)提出了一种单阶段分层端到端可训练网络,以解决航空航天物体尺度变化带来的姿态估计难题。在最新研究中,Lian和Ling(2023)增加了预定义的3D关键点数量以提升对应关系的建立精度,并设计了一种分层二进制编码方法用于关键点定位,从而实现对应关系的逐步优化,并将对应关系回归转化为更高效的分类任务。为估计透明物体的姿态,Chang等人(2021)采用了3D边界框预测网络与多视角几何方法:首先检测3D边界框顶点的2D投影,随后结合相机运动数据基于多视角检测到的2D投影重建3D点;此外,他们还提出了广义姿态定义以解决对称物体的姿态不确定性问题。为提升姿态估计网络的效率,Guo等人(2023)通过将教师网络的局部预测分布蒸馏到学生网络中,将知识蒸馏技术整合到物体姿态估计过程中。Liu等人(2023b)指出,可微分的点对点(PnP)策略与PnP问题的平均化特性存在冲突,可能导致梯度优化反而会降低单个对应点的精度。为缓解这一问题,他们引入了线性协方差损失函数,该方法既适用于稀疏对应点方法,也适用于密集对应点方法。

为缓解大范围遮挡导致的模型脆弱性,Crivellaro等人(2017)采用多个控制点来表征每个物体部件,通过预测这些控制点的二维投影来计算物体姿态。部分研究者通过使用小块图像预测关键点来解决遮挡问

题,Oberweger团队(2018)采用分块处理方式生成热力图并进行结果聚合,从而实现精准可靠的预测。此外,他们还提出了一种简单高效的策略,用于在训练过程中消除图像块与热力图之间的不确定性。Hu团队(2019)提出了一种基于分割的姿态估计框架,通过二维关键点位置为每个可见物体部件提供局部姿态预测。Huang团队(2021)将二维关键点位置视为损失函数中的概率分布,并设计了基于置信度的网络模型。

降低对标注真实世界数据的依赖同样是重要课题。部分方法通过利用几何一致性作为辅助信息来减少标注需求。Zhao等人(2020)采用带有物体标注及视角间相对变换的图像对,自动识别几何与视觉特征一致的物体三维关键点。此外,Yang团队(2021)针对带有标注二维边界框的双尺度图像,引入了关键点一致性正则化方法。Liu等人(2021b)通过半监督学习构建了三维手部与物体姿态估计统一框架,其联合学习架构实现了手部与物体表征间的显式上下文推理。在半监督学习中生成伪标签时,他们以大规模手部-物体视频中的时空一致性作为约束条件。合成数据同样是解决标注难题的有效途径,Georgakis等人(2019)通过选择关键点并在RGB图像和CAD模型渲染中保持视角与模态不变性,减少了对昂贵的3DoF姿态标注的需求。Sock等人(2020)采用自监督学习方法缩小合成数据与真实数据之间的差异,并确保不同物体视角下的光度一致性以优化模型性能。此外,Zhang等人(2021)利用关键点在真实与合成领域间几何关系的不变性实现了领域适应。Thalhammer等人(2021)提出了一种专用特征金字塔网络用于计算多尺度特征,从而能够同时生成不同特征图分辨率下的姿态假设。

总体而言,基于稀疏对应关系的方法能够高效地估计物体姿态。然而,仅依赖少量控制点可能导致精度欠佳。

3.1.2 密集对应方法

基于密集对应关系的方法旨在建立密集的二维-三维或三维-三维对应关系。与基于稀疏对应关系的方法相比,这类方法使用了显著更多的对应点。这使得它们能够实现更高的精度,并更好地处理遮挡问题。

effectively. Specifically, for the RGB image, they leverage every pixel or multiple patches to generate pixel-wise correspondences, while for the point cloud, they use the entire point cloud to find point-wise correspondences. Li et al. (2019) argued for the differentiation between rotation and translation, proposing the coordinates-based disentangled pose network. This network separates pose estimation into distinct predictions for rotation and translation. Zakharov et al. (2019) introduced the dense multi-class 2D-3D correspondence-based object pose detector and a tailored deep learning-based refinement process. In addition, Cai and Reid (2020) proposed a technique to automatically identify and match image landmarks consistently across different views, aiming to enhance the process of learning 2D-3D mapping. Wang et al. (2021a) developed a pose estimation pipeline guided by reconstruction, capitalizing on geometric consistency. Further, Shugurov et al. (2021) built upon Zakharov et al. (2019) by developing a unified deep network capable of accommodating multiple image modalities (such as RGB and Depth) and integrating a differentiable rendering-based pose refinement method. Su et al. (2022) introduced a discrete descriptor realized by hierarchical binary grouping, capable of densely representing the object surface. As a result, this method can predict fine-grained correspondences. Chen et al. (2022a) introduced a probabilistic PnP (Fischler & Bolles, 1981) layer designed for general end-to-end pose estimation. This layer generates a pose distribution on the $SE(3)$ manifold. On the other hand, Xu et al. (2022) argued that encoding pose-sensitive local features and modeling the statistical distribution of inlier poses are crucial for accurate and robust 6DoF pose estimation. Inspired by PPF (Drost et al., 2010), they exploited pose-sensitive information carried by each pair of oriented points and an ensemble of redundant pose predictions to achieve robust performance on severe inter-object occlusion and systematic noises in scene point clouds.

Some methods recover object poses by establishing 3D-3D correspondences. Huang et al. (2022) used an RGB image to predict 3D object coordinates in the camera frustum, thus establishing 3D-3D correspondences. Further, Jiang et al. (2023) introduced a center-based decoupled framework, leveraging bird's eye and front views for object center voting. They utilized feature similarity between the center-aligned object and the object CAD model to establish correspondences for Singular Value Decomposition (SVD)-based (Besl & McKay, 1992) rotation estimation. More recently, Lin et al. (2024d) utilized an RGBD image as input and employed point-to-surface matching to estimate the object surface correspondence. They establish 3D-3D correspondences by iteratively constricting the surface, transitioning it into a correspondence point while progressively eliminating outliers.

Some methods put more effort into handling challenging cases, such as symmetric objects (Park et al., 2019b; Hodan

et al., 2020; Wu et al., 2023) and texture-less objects (Wu et al., 2021a). Park et al. (2019b) utilized generative adversarial training to reconstruct occluded parts to alleviate the impact of occlusion. They handle symmetric objects by guiding predictions towards the nearest symmetric pose. In addition, Hodan et al. (2020) modeled an object using compact surface fragments to handle symmetries in object modeling effectively. For each pixel, the network predicts: the likelihood of each object's presence, the probability of the fragments conditional on the object's presence, and the exact 3D translation of each fragment. Finally, the object pose is determined using a robust and efficient version of the PnP-RANSAC algorithm (Fischler & Bolles, 1981). Further, Wu et al. (2023) employed a geometric-aware dense matching network to acquire visible dense correspondences. Additionally, they utilized the distance consistency of these correspondences to mitigate ambiguity in symmetrical objects. For texture-less objects, Wu et al. (2021a) leveraged information from the object CAD model and established 2D-3D correspondences using a pseudo-Siamese neural network.

With research development, domain adaptation, weak supervision, and self-supervision techniques have been introduced into pose estimation. Li et al. (2021b) noticed that images with varying levels of realism and semantics exhibit different transferability between synthetic and real domains. Consequently, they decomposed the input image into multi-level semantic representations and merged the strengths of these representations to mitigate the domain gap. Further, Hu et al. (2022) introduced a method exclusively trained on synthetic images, which infers the necessary pose correction for refining rough poses. Haugaard and Buch (2022) utilized learned distributions to sample, score, and refine pose hypotheses. Correspondence distributions are learned using a contrastive loss. This method is unsupervised regarding visual ambiguities. More recently, Li et al. (2023a) introduced a weakly-supervised reconstruction-based pipeline. Initially, they reconstructed the objects from various viewpoints using an implicit neural representation. Subsequently, they trained a network to predict pixel-wise 2D-3D correspondences. Hai et al. (2023a) proposed a refinement strategy that uses the geometry constraint in synthetic-to-real image pairs captured from multiple viewpoints.

There are also methods that focus on pose refinement. Lipson et al. (2022) iteratively refined pose and correspondences in a tightly coupled manner. They incorporated a differentiable layer to refine the pose by solving the bidirectional depth-augmented PnP problem. In addition, Xu et al. (2024b) formulated object pose refinement as a non-linear least squares problem using the estimated correspondence field, *i.e.*, the correspondence between the RGB image and the rendered image using the initial pose. The non-linear least squares problem is then solved by a differentiable

具体而言, 对于RGB图像, 他们利用每个像素或多个像素块来生成像素级对应关系; 而对于点云, 则利用整个点云来寻找点级对应关系。Li 等人 (2019) 主张区分旋转和平移, 提出了基于坐标的解耦姿态网络, 该网络将姿态估计分解为旋转和平移的独立预测任务。Zakharov 等人 (2019) 提出了基于密集多类别二维-三维对应关系的对象姿态检测器及定制化的深度学习精炼流程。此外, Cai 和 Reid (2020) 提出了一种能够在不同视角下一致识别和匹配图像地标的技术, 旨在优化二维-三维映射的学习过程。Wang 等人 (2021a) 开发了基于几何一致性、以重建为导向的姿态估计流程; 而 Shugurov 等人 (2021) 则在 Zakharov 等人工作的基础上进行了进一步改进。(2019) 通过开发能够兼容多种图像模态 (如RGB和深度) 的统一深度网络, 并集成基于可微分渲染的姿态优化方法。Su 等人 (2022) 提出了一种通过分层二值分组实现的离散描述符, 能够密集表征物体表面, 从而实现细粒度对应关系预测。Chen 等人 (2022a) 设计了一种面向通用端到端姿态估计的概率性 PnP (Fischler & Bolles, 1981) 层, 该层在SE(3)流形上生成姿态分布。另一方面, Xu 等人 (2022) 指出, 编码姿态敏感局部特征并建模内点姿态的统计分布对实现精确且鲁棒的6DoF姿态估计至关重要。受 PPF (Drost 等人, 2010) 启发, 他们利用每对定向点携带的姿态敏感信息及冗余姿态预测集成, 实现了对严重物体遮挡和场景点云系统性噪声的鲁棒性处理。

某些方法通过建立三维-三维对应关系来恢复物体姿态。Huang 等人 (2022) 利用RGB图像预测相机视锥中的三维物体坐标, 从而建立三维-三维对应关系。此外, Jiang 等人 (2023) 提出了一种基于中心点的解耦框架, 利用鸟瞰视角和正视视角进行物体中心投票。他们通过中心对齐物体与物体CAD模型之间的特征相似性, 为基于奇异值分解 (SVD) (Besl & McKay, 1992) 的旋转估计建立对应关系。最近, Lin 等人 (2024d) 以 RGBD 图像作为输入, 采用点到面匹配法估计物体表面对应关系。他们通过迭代收缩表面、将其转化为对应点并逐步剔除异常值来建立三维-三维对应关系。部分方法更侧重于处理复杂案例, 例如对称物体 (Park 等人, 2019b; Hodan 等人, 2020; Wu 等

人, 2023) 和无纹理物体 (Wu 等人, 2021a)。Park 等人 (2019b) 采用生成对抗训练技术重建被遮挡部分, 以减轻遮挡效应的影响。他们通过引导预测结果朝向最近的对称姿态来处理对称物体。此外, Hodan 等人 (2020) 采用紧凑的表面片段建模方法, 有效处理物体建模中的对称性问题。对于每个像素, 网络会预测: 物体存在的可能性、物体存在条件下各片段的概率, 以及各片段的精确三维平移量。最终, 采用鲁棒高效的PnP- RANSAC 算法 (Fischler & Bolles, 1981) 确定物体姿态。Wu 等人 (2023) 则运用几何感知密集匹配网络获取可见区域的密集对应关系, 并利用这些对应关系的距离一致性来消除对称物体的歧义。对于无纹理物体, Wu 等人 (2021a) 利用物体CAD模型信息, 通过伪孪生神经网络建立了二维-三维对应关系。

随着研究发展, 领域适应、弱监督和自监督技术已被引入姿态估计领域。Li 等人 (2021b) 注意到, 具有不同真实度和语义特征的图像在合成域与真实域间表现出差异化的迁移能力。为此, 他们将输入图像分解为多层级语义表征, 并融合这些表征的优势以缓解领域差异。此外, Hu 等人 (2022) 提出了一种仅基于合成图像训练的方法, 通过推断必要姿态校正来优化粗略姿态。Haugaard 与 Buch (2022) 利用学习到的分布进行姿态假设的采样、评分与优化, 对应分布通过对比损失函数进行学习。该方法对视觉模糊性具有无监督特性。近期, Li 等人 (2023a) 提出了一种弱监督重构流程: 首先利用隐式神经表征从多视角重建物体, 随后训练网络预测像素级二维-三维对应关系。Hai 等人 (2023a) 提出了一种优化策略, 该策略利用从多个视角捕获的合成图像与真实图像对中的几何约束。

此外, 还存在专注于姿态优化的方法。Lipson 等人 (2022) 采用紧密耦合的方式迭代优化姿态与对应关系, 通过引入可微分层来求解双向深度增强点对点 (PnP) 问题以实现姿态优化。此外, Xu 等人 (2024b) 将物体姿态优化建模为非线性最小二乘问题, 利用基于初始姿态估计的RGB图像与渲染图像之间的对应关系场进行求解。该非线性最小二乘问题通过可微分算法进行求解。

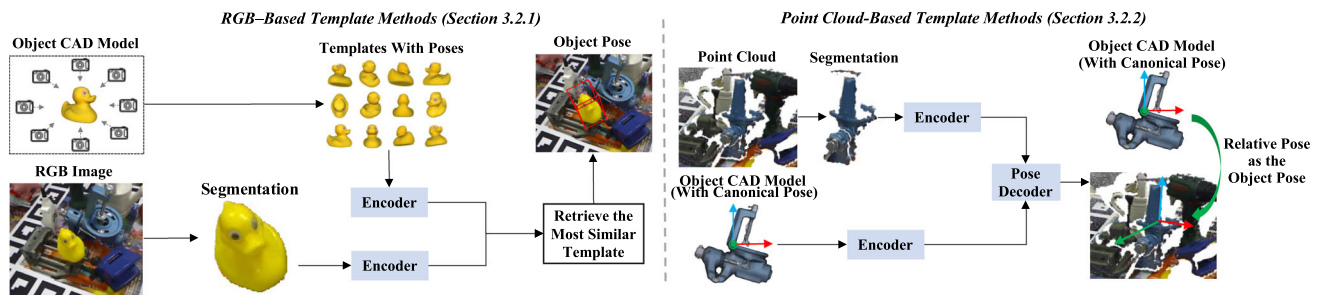


Fig. 5 Illustration of the template-based methods (Sec. 3.2), which involve identifying the most similar template from a set of templates labeled with ground-truth object poses

levenberg-marquardt algorithm (Moré, 2006), enabling end-to-end training.

Brief Discussion: In general, the aforementioned correspondence-based methods exhibit robustness to occlusion since they use local correspondences to predict object pose. However, these methods may encounter challenges when handling objects that lack salient shape features or texture.

3.2 Template-Based Methods

By leveraging global information from the image, template-based methods can effectively address the challenges posed by texture-less objects. Template-based methods involve identifying the most similar template from a set of templates labeled with ground-truth object poses. They can be categorized into RGB-based template (Sec. 3.2.1) and point cloud-based template (Sec. 3.2.2) methods. These two methods are illustrated in Fig. 5. The characteristics and performance of some representative methods are shown in Table 1.

3.2.1 RGB-Based Template Methods

When the input is an RGB image, the templates comprise 2D projections extracted from object CAD models, with annotations of ground-truth poses. This process reformulates object pose estimation as an image retrieval task. As a seminal contribution, Sundermeyer et al. (2018) achieved 3D rotation estimation through a variant of denoising autoencoder, which learns an implicit representation of object rotation. If depth is available, it can be used for pose refinement. Liu et al. (2019a) developed a CNN akin to an autoencoder to reconstruct arbitrary scenes featuring the target object and extract the object area. In addition, Zhang et al. (2020b) utilized an object detector and a keypoint extractor to simplify the template search process. Papaioannidis et al. (2020) suggested that estimating object poses in synthetic images is more straightforward. Therefore, they employed a generative adversarial network to convert real images into synthetic ones

while preserving the object pose. Li and Ji (2020) utilized a new pose representation (*i.e.*, 3D location field) to guide an auto-encoder to distill pose-related features, thereby enhancing the handling of pose ambiguity. Stevšič and Hilliges (2020) proposed a spatial attention mechanism to identify and utilize spatial details for pose refinement. Different from the above methods, Deng et al. (2021) addressed the 6DoF object pose tracking problem within the Rao-Blackwellized particle filtering (Doucet et al., 2001) framework. They finely discretized the rotation space and trained an autoencoder network to build a codebook of feature embeddings for these discretized rotations. This method efficiently estimates the 3D translation along with the full distribution over the 3D rotation.

RGB cameras are widely used as visual sensors, yet they struggle to capture sufficient information under poor lighting conditions. This results in poor pose estimation performance.

3.2.2 Point Cloud-Based Template Methods

With the popularity of consumer-grade 3D cameras, point cloud-based methods take full advantage of their ability to adapt to poor illumination and capture geometric information. When dealing with a point cloud, the template comprises the object CAD model in canonical pose. Notably, we classify the methods that directly regress the relative pose between the object CAD model and the observed point cloud as template-based methods. This is because these methods can be interpreted as seeking the optimal relative pose that aligns the observed point cloud with the template. Consequently, the determined relative pose serves as the object pose. Li et al. (2022) adopted a feature disentanglement and alignment module to establish part-to-part correspondences between the partial point cloud and object CAD model, enhancing geometric constraint. Jiang et al. (2024) proposed a point cloud registration framework based on the SE(3) diffusion model, which gradually perturbs the optimal rigid transformation of a pair of point clouds by continuously injecting perturbation transformation through the SE(3) forward diffusion process. Then, the SE(3) reverse denoising process is used to gradu-

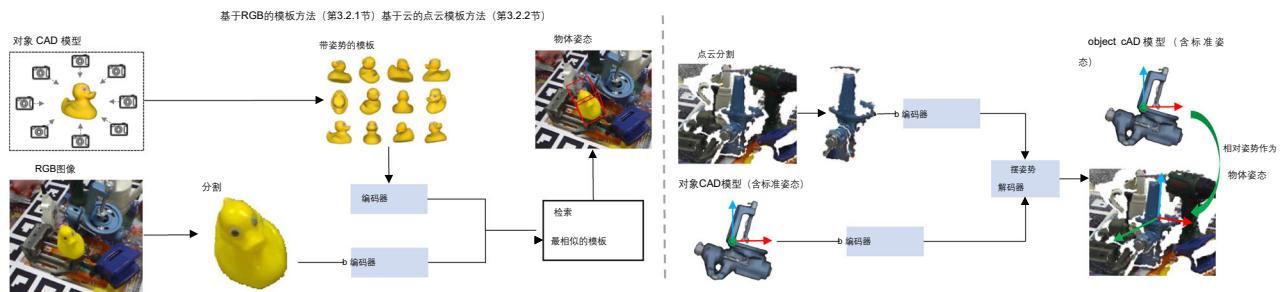


图5基于模板的方法示意图（第3.2节），该方法涉及从一组标注有真实物体姿态的模板中识别最相似的模板

Levenberg-Marquardt算法（More，2006），支持端到端训练。

简要讨论：通常而言，上述基于对应关系的方法对遮挡具有较强的鲁棒性，因其通过局部对应关系来预测物体姿态。然而，当处理缺乏显著形状特征或纹理特征的物体时，这些方法可能面临挑战。

3.2 基于模板的方法

通过利用图像中的全局信息，基于模板的方法能够有效应对无纹理物体带来的挑战。这类方法的核心是从一组标注了真实物体姿态的模板中识别出最相似的模板。它们可分为基于RGB的模板方法（第3.2.1节）和基于点云的模板方法（第3.2.2节），这两种方法如图5所示。部分代表性方法的特征与性能详见表1。

3.2.1 基于RGB的模板方法

当输入为RGB图像时，模板包含从物体CAD模型中提取的二维投影，并附有真实姿态标注。该过程将物体姿态估计重新定义为图像检索任务。作为开创性贡献，Sundermeyer等人（2018）通过去噪自编码器的变体实现了三维旋转估计，该方法学习了物体旋转的隐式表征。若具备深度信息，可将其用于姿态精炼。Liu等人（2019a）开发了类似自编码器的CNN来重建包含目标物体的任意场景并提取物体区域。此外，Zhang等人（2020b）采用物体检测器与关键点提取器以简化模板搜索流程。Papaioannidis等人（2020）提出在合成图像中估计物体姿态更为直接，因此他们采用生成对抗网络将真实图像转换为合成图像同时保留物体姿态。Li与Ji（2020）利用新型姿态表征（即三维位置

场）引导自编码器提炼与姿态相关的特征，从而增强了对姿态模糊性的处理能力。Stevsić Hilliges等人

（2020）提出了一种空间注意力机制，用于识别并利用空间细节以实现姿态优化。与上述方法不同，Deng等人（2021）在Rao-Blackwell化粒子滤波框架（Doucet等人，2001）中解决了6自由度物体姿态跟踪问题。他们对旋转空间进行精细离散化处理，并训练自编码器网络构建特征嵌入码本，用于表征离散化旋转状态。该方法能高效估计三维平移运动，并同时获得三维旋转的完整分布信息。

RGB相机作为视觉传感器被广泛应用，但在光照条件较差的情况下难以捕获足够的信息，这导致其姿态估计性能较差。

3.2.2 基于点云的模板方法

随着消费级3D相机的普及，基于点云的方法充分利用了其适应弱光照环境和捕捉几何信息的能力。在处理点云时，模板包含处于标准姿态下的物体CAD模型。值得注意的是，我们将直接回归物体CAD模型与观测点云之间相对姿态的方法归类为基于模板的方法，因为这些方法本质上是在寻找使观测点云与模板对齐的最佳相对姿态。由此确定的相对姿态即作为物体姿态。Li等人（2022）采用特征解耦与对齐模块，在部分点云与物体CAD模型之间建立部件对应关系，从而增强几何约束。Jiang等人（2024）提出基于SE(3)扩散模型的点云配准框架，通过SE(3)前向扩散过程持续注入扰动变换，逐步扰动一对点云的最优刚性变换。随后，采用SE(3)反向去噪过程进行逐步处理。

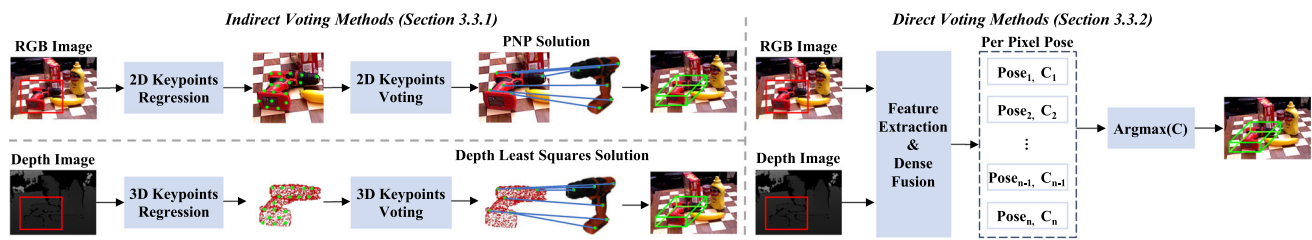


Fig. 6 Illustration of the voting-based methods (Sec. 3.3), which determine object pose through a pixel-level or point-level voting scheme

ally denoise, making it closer to the optimal transformation for accurate pose estimation. Dang et al. (2024) proposed two key contributions to enhance pose estimation performance on real-world data. First, they introduced a directly supervised loss function that bypasses the SVD (Besl & McKay, 1992) operation, mitigating the sensitivity of SVD-based loss functions to the rotation range between the input partial point cloud and the object CAD model. Second, they devised a match normalization strategy to address disparities in feature distributions between the partial point cloud and the CAD model.

Brief Discussion: Overall, template-based methods leverage global information from the image, enabling them to effectively handle texture-less objects. Moreover, they are advantageous in that they can naturally handle the one-to-many mapping from an image to multiple possible poses, which often arises due to pose ambiguity. However, the high pose estimation accuracy comes at the cost of increased memory usage by the templates and a rapid rise in computational complexity. Additionally, they may also exhibit poor performance when confronted with occluded objects.

3.3 Voting-Based Methods

Voting-based methods determine object pose through a pixel-level or point-level voting scheme, which can be categorized into two main types: indirect voting (Sec. 3.3.1) and direct voting (Sec. 3.3.2). The illustration of these types of methods is shown in Fig. 6. The attributes and performance of some representative methods are shown in Table 1.

3.3.1 Indirect Voting Methods

Indirect voting methods first estimate a set of 2D keypoint hypotheses from an RGB image via pixel-level voting, or a set of 3D keypoint hypotheses from a point cloud via point-level voting, and then estimate the keypoint distributions based on these hypotheses. Subsequently, the object pose is determined through 2D-3D or 3D-3D keypoint correspondences as described in Sec. 3.1. For instance, some researchers predicted 2D keypoints and then derived the

object pose through 2D-3D keypoints correspondence. Liu et al. (2021c) introduced a continuous representation method called keypoint distance field (KDF), which extracts 2D keypoints by voting on each KDF. Meanwhile, Cao et al. (2022) proposed a method called dynamic graph PnP (Fischler & Bolles, 1981) to learn the object pose from 2D-3D correspondence, enabling end-to-end training. Moreover, Liu et al. (2023f) introduced a bidirectional depth residual fusion network to fuse RGBD information, thereby estimating 2D keypoints precisely. Inspired by the diffusion model, Xu et al. (2024a) proposed a diffusion-based framework to formulate 2D keypoint detection as a denoising process to establish more accurate 2D-3D correspondences.

Unlike the aforementioned methods that predict 2D keypoints, He et al. (2020) proposed a depth hough voting network to predict 3D keypoints. Subsequently, they estimated the object pose through levenberg-marquardt algorithm (Moré, 2006). Furthermore, He et al. (2021) introduced a bidirectional fusion network to complement RGB and depth heterogeneous data, thereby better predicting the 3D keypoints. To better capture features among object points in 3D space, Mei et al. (2022) utilized graph convolutional networks to facilitate feature exchange among points in 3D space, aiming to improve the accuracy of predicting 3D keypoints. Wu et al. (2022) proposed a 3D keypoint voting scheme based on cross-spherical surfaces, allowing for generating smaller and more dispersed 3D keypoint sets, thus improving estimation efficiency. To obtain more accurate 3D keypoints, Wang et al. (2023a) presented an iterative 3D keypoint voting network to refine the initial localization of 3D keypoints. Most recently, Zhou et al. (2023a) introduced a novel weighted vector 3D keypoints voting algorithm, which adopts a non-iterative global optimization strategy to precisely localize 3D keypoints, while also achieving near real-time inference speed.

In response to challenging scenarios such as cluttered or occlusion, Peng et al. (2019) introduced a pixel-wise voting network to regress pixel-level vectors pointing to 3D keypoints. These vectors create a flexible representation for locating occluded or truncated 3D keypoints. Since most industrial parts are parameterized, Zeng et al. (2021b) defined

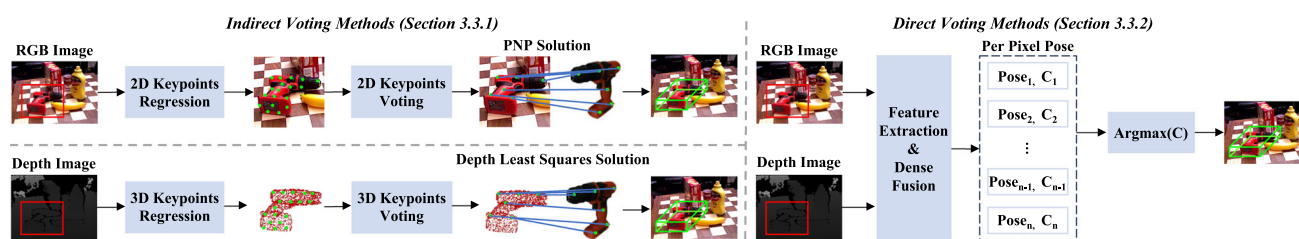


图6基于投票机制的方法示意图（第3.3节），该方法通过像素级或点级投票方案确定物体姿态

通过降噪处理，使变换结果更接近最优解，从而实现更精准的姿态估计。Danget等人（2024）针对真实数据场景提出了两项关键改进方案以提升姿态估计性能：首先，他们创新性地引入直接监督损失函数，绕过了奇异值分解（SVD）（Besl & McKay, 1992）操作步骤，有效缓解了基于SVD的损失函数对输入部分点云与物体CAD模型旋转范围的敏感性；其次，他们设计了匹配归一化策略，有效解决了部分点云与CAD模型间特征分布差异带来的挑战。

简要讨论：总体而言，基于模板的方法能够利用图像中的全局信息，从而有效处理无纹理物体。此外，这类方法具有天然优势——可自然处理图像到多种可能姿态之间的一一对应关系，这种现象通常由姿态模糊性导致。然而，高姿态估计精度的实现需要以模板占用更多内存和计算复杂度急剧上升为代价。另外，在处理被遮挡物体时，这类方法也可能表现出较差的性能表现。

3.3 基于投票的方法

基于投票的方法通过像素级或点级投票方案确定物体姿态，可分为两大主要类型：间接投票（第3.3.1节）和直接投票（第3.3.2节）。这些方法的示意图如图6所示。部分代表性方法的属性与性能数据见表1。

3.3.1 间接投票方法

间接投票方法首先通过像素级投票从RGB图像中估计一组二维关键点假设，或通过点云中点级投票估计一组三维关键点假设，然后基于这些假设估计关键点分布。随后，如第3.1节所述，通过二维-三维或三维-三维关键点对应关系确定物体姿态。例如，部分研究者先预测二维关键点，再通过二维-三维关键点对应关系推导物体姿态。Liu等人（2021c）提出了一种称为关

键点距离场（KDF）的连续表示方法，通过对每个KDF进行投票来提取二维关键点。与此同时，Cao等人（2022）提出了一种称为动态图点对点（Fischler & Bolles, 1981）的方法，通过二维-三维对应关系学习物体姿态，实现端到端训练。此外，Liu等人（2023f）引入双向深度残差融合网络以融合RGBD信息，从而精确估计二维关键点。受扩散模型启发，Xu等人（2024a）提出了一种基于扩散的框架，将二维关键点检测建模为去噪过程，从而建立更精确的二维-三维对应关系。

与前述预测二维关键点的方法不同，He等人（2020）提出了一种深度Hough投票网络用于预测三维关键点。随后，他们通过Levenberg-Marquardt算法（More, 2006）估算了物体的姿态。此外，He等人（2021）引入了双向融合网络以整合RGB和深度异构数据，从而更准确地预测三维关键点。为了更好地捕捉三维空间中物体点间的特征，Mei等人（2022）采用图卷积网络促进三维空间内点间的特征交换，旨在提高三维关键点预测的准确性。Wu等人（2022）提出了一种基于交叉球面的三维关键点投票方案，能够生成更小且分布更分散的三维关键点集，从而提升估计效率。为获得更精确的三维关键点，Wang等人（2023a）提出了迭代式三维关键点投票网络以优化三维关键点的初始定位。最近，Zhou等人.....（2023a）提出了一种新颖的加权向量三维关键点投票算法，该算法采用非迭代全局优化策略，能够精确定位三维关键点，同时实现接近实时的推理速度。

针对杂乱或遮挡等复杂场景，Peng等人（2019）提出了一种像素级投票网络，用于回归指向三维关键点的像素级向量。这些向量为定位被遮挡或截断的三维关键点提供了灵活的表征方式。由于大多数工业零件具有参数化特征，Zenget等人（2021b）进一步定义了

3D keypoints linked to parameters through driven parameters and symmetries. This approach effectively addresses the pose estimation of objects in stacking scenes.

Rather than utilizing a single-view RGBD image as input, Duffhauss et al. (2022) employed multi-view RGBD images as input. They extracted visual features from each RGB image, while geometric features were extracted from the object point cloud (generated by fusing all depth images). This multi-view RGBD feature fusion-based method can accurately predict object pose in cluttered scenes.

Some researchers have proposed new training strategies to improve pose estimation performance. Yu et al. (2020) developed a differentiable proxy voting loss that simulates hypothesis selection during the voting process, enabling end-to-end training. In addition, Lin et al. (2022b) proposed a novel learning framework, which utilizes the accurate result of the RGBD-based pose refinement method to supervise the RGB-based pose estimator. To bridge the domain gap between synthetic and real data, Ikeda et al. (2022) introduced a method to transfer object style transfer from synthetic to realistic without manual intervention.

In general, indirect voting-based methods provide an excellent solution for instance-level object pose estimation. However, the accuracy of pose estimation heavily relies on the quality of the keypoints, which can result in lower robustness.

3.3.2 Direct Voting Methods

Since the performance of the indirect voting methods heavily depends on the selection of keypoints, direct voting methods have been proposed as an alternative solution, which directly predict the pose and confidence at the pixel-level or point-level, then select the pose with the highest confidence as the object pose. Tian et al. (2020b) uniformly sampled rotation anchors in $SO(3)$. Subsequently, they predicted constraint deviations for each anchor towards the target, using the uncertainty score to select the best prediction. Then, they detected the 3D translation by aggregating point-to-center vectors towards the object center to recover the 6DoF pose. Wang et al. (2019a) fused RGB and depth features on a per-pixel basis and utilized a pose predictor to generate 6DoF pose and confidence for each pixel. Subsequently, they selected the pose of the pixel with the highest confidence as the final pose. Zhou et al. (2020) employed CNNs (LeCun et al., 1989) to extract RGB features, which are then integrated into the point cloud to obtain fused features. Unlike (Wang et al., 2019a), the fused features take the form of point sets rather than feature mappings.

However, the aforementioned RGBD fusion methods merely concatenate RGB and depth features without delving into their intrinsic relationship. Therefore, Zhou et al. (2021b) proposed a new multi-modal fusion graph convo-

lutional network to enhance the fusion of RGB and depth images, capturing the inter-modality correlations through local information propagation. Liu et al. (2024b) decoupled scale-related and scale-invariant information in the depth image to guide the network in perceiving the scene's 3D structure and provide scene texture for the RGB image feature extraction. Unlike the aforementioned approaches that use still images, Mu et al. (2021) proposed a time fusion model integrating temporal motion information from RGBD images for 6DoF object pose estimation. This method effectively captures object motion and changes, thereby enhancing pose estimation accuracy and stability.

Symmetric objects may have multiple true poses, leading to ambiguity in pose estimation. To address this issue, Mo et al. (2022) designed a symmetric-invariant pose distance metric, which enables the network to estimate symmetric objects accurately. Cai et al. (2022b) introduced a 3D rotation representation to learn the object implicit symmetry, eliminating the need for additional prior knowledge about object symmetry.

To reduce dependency on annotated real data, Zeng et al. (2021a) trained their model solely on synthetic dataset. Then, they utilized a sim-to-real learning network to improve their generalization ability. During pose estimation, they transformed scene points into centroid space and obtained object pose through clustering and voting.

Brief Discussion: Overall, voting-based methods have demonstrated superior performance in pose estimation tasks. However, the voting process is time-consuming and increases computational complexity (Buchschat & Vetter, 2020).

3.4 Regression-Based Methods

Regression-based methods aim to directly obtain the object pose from the learned features. They can be divided into two main types: geometry-guided regression (Sec. 3.4.1) and direct regression (Sec. 3.4.2). The illustration of these two types of methods is shown in Fig. 7. The attributes and performance of some representative methods are shown in Table 1.

3.4.1 Geometry-Guided Regression Methods

Geometry-guided regression methods leverage geometric information from RGBD images (such as object 3D structural features or 2D-3D geometric constraints) to assist in object pose estimation. Gao et al. (2020) employed decoupled networks for rotation and translation regression from the object point cloud. Meanwhile, Chen et al. (2020b) introduced a rotation residual estimator to estimate the residual between the predicted rotation and the ground truth, enhancing the accuracy of rotation prediction. Lin et al. (2021c) used a network to extract the geometric features of the object point

通过驱动参数与对称性将三维关键点与参数关联。该方法有效解决了堆叠场景中物体的姿态估计问题。

Duffhauss 等人 (2022) 并未采用单视图 RGBD 图像作为输入, 而是采用了多视图 RGBD 图像。他们从每张 RGB 图像中提取视觉特征, 同时从物体点云 (通过融合所有深度图像生成) 中提取几何特征。这种基于多视图 RGBD 特征融合的方法能够在复杂场景中准确预测物体姿态。

部分研究人员提出了新的训练策略以提升姿态估计性能。Yu 等人 (2020) 开发了一种可微分的代理投票损失函数, 该函数模拟了投票过程中的假设选择机制, 从而实现了端到端训练。此外, Lin 等人 (2022b) 提出了一种创新的学习框架, 该框架利用基于 RGBD 的姿态精炼方法的精确结果来监督基于 RGB 的姿态估计器。为弥合合成数据与真实数据之间的领域差异, Ikeda 等人 (2022) 提出了一种无需人工干预即可将对象风格迁移从合成数据转移到真实数据的方法。

总体而言, 基于间接投票的方法为实例级物体姿态估计提供了极佳的解决方案。然而, 姿态估计的准确性高度依赖于关键点的质量, 这可能导致系统鲁棒性降低。

3.3.2 直接投票方法

由于间接投票方法的性能在很大程度上取决于关键点的选择, 因此提出了直接投票方法作为替代方案: 该方法直接在像素级或点级预测姿态和置信度, 然后选择置信度最高的姿态作为物体姿态。Tian 等人 (2020b) 在 $SO(3)$ 空间中均匀采样旋转锚点, 随后利用不确定性评分预测每个锚点相对于目标的约束偏差, 并选择最佳预测结果; 接着通过聚合指向物体中心的点到中心向量来检测三维平移, 从而恢复 6 自由度姿态。Wang 等人 (2019a) 在逐像素基础上融合 RGB 和深度特征, 并采用姿态预测器为每个像素生成 6 自由度姿态及置信度, 最终选择置信度最高的像素姿态作为最终姿态。Zhou 等人 (2020) 采用 CNN (LeCun 等人, 1989) 提取 RGB 特征, 并将其整合到点云中以获得融合特征。与 Wang 等人提出的方案不同.....2019a) 中, 融合特征以点集形式呈现, 而非特征映射形式。

然而, 上述 RGBD 融合方法仅简单拼接 RGB 和深度特征, 而未深入探究二者之间的内在关联。为此, Zhou 等人 (2021b) 提出了一种新型多模态融合图卷积网络, 通过局部信息传播捕捉模态间相关性, 从而提

升 RGB 与深度图像的融合效果。Liu 等人 (2024b) 将深度图像中的尺度相关信息与尺度不变信息分离, 引导网络感知场景的三维结构, 并为 RGB 图像特征提取提供场景纹理信息。与上述基于静态图像的方法不同, Mu 等人 (2021) 提出了时间融合模型, 整合 RGBD 图像中的时序运动信息用于六自由度物体姿态估计。该方法能有效捕捉物体运动与变化, 从而显著提升姿态估计的精度与稳定性。

对称物体可能存在多种真实姿态, 导致姿态估计存在不确定性。为解决这一问题, Mo 等人 (2022) 设计了一种对称不变姿态距离度量方法, 使网络能够准确估计对称物体的姿态。Cai 等人 (2022b) 提出三维旋转表征技术来学习物体的隐式对称性, 无需依赖额外的物体对称性先验知识。

为降低对标注真实数据的依赖, Zeng 等人 (2021a) 仅使用合成数据集训练其模型。随后, 他们采用模拟到真实的学习网络以提升模型的泛化能力。在姿态估计过程中, 他们将场景点转换至质心空间, 并通过聚类与投票机制获得物体姿态。

简要讨论: 总体而言, 基于投票的方法在姿态估计任务中表现出更优的性能。然而, 投票过程耗时较长且会增加计算复杂度 (Bukchat & Vetter, 2020)。

3.4 基于回归的方法

基于回归的方法旨在直接从学习到的特征中获取物体姿态。它们可分为两大主要类型: 几何引导回归 (第 3.4.1 节) 和直接回归 (第 3.4.2 节)。这两种方法的示意图如图 7 所示。部分代表性方法的属性与性能数据见表 1。

3.4.1 几何引导回归方法

几何引导回归方法通过利用 RGBD 图像中的几何信息 (如物体三维结构特征或二维-三维几何约束条件) 来辅助物体姿态估计。Gao 等人 (2020) 采用解耦网络从物体点云中进行旋转和平移回归分析。与此同时, Chen 等人 (2020b) 引入旋转残差估计器来计算预测旋转与真实值之间的残差, 从而提升旋转预测精度。Lin 等人 (2021c) 则通过网络提取物体的几何特征进行建模分析。

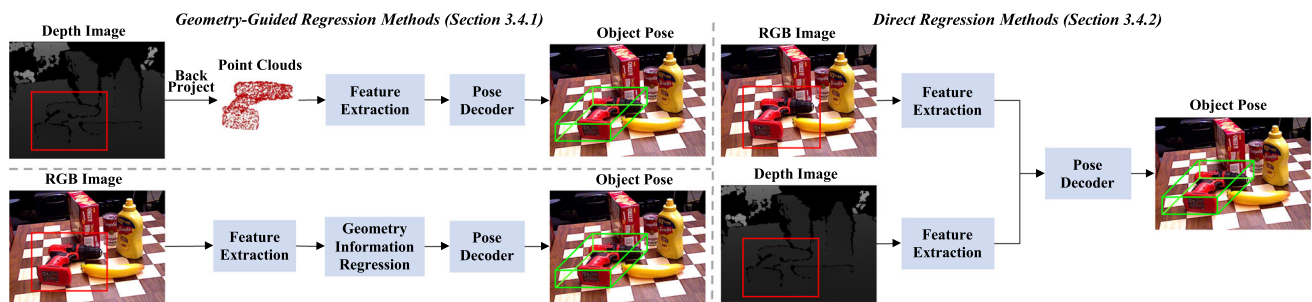


Fig. 7 Illustration of the regression-based methods (Sec. 3.4), which aim to obtain the object pose directly from the learned features

cloud. Then, they enhanced the pairwise consistency of geometric features by applying spectral convolution on pairwise compatibility graphs. Additionally, Shi et al. (2021) learned geometric and contextual features within point cloud blocks. Then, they trained a sub-block network to predict the pose of each point cloud block. Finally, the most reliable block pose is selected as the object pose. To address the challenge of point cloud-based object pose tracking, Liu et al. (2024c) proposed a shifted point convolution operation between the point clouds of adjacent frames to facilitate the local context interaction.

Approaches solely relying on object point cloud often overlook the object texture details. Therefore, Wen et al. (2021) and An et al. (2024) leveraged the complementary nature of RGB and depth information. They improved cross-modal fusion strategies by employing attention mechanisms to effectively align and integrate these two heterogeneous data sources, resulting in enhanced performance.

In contrast to the aforementioned methods that directly derive geometric information from the depth image or the object CAD model, many researchers focused more on generating geometric constraints from the RGB image. Hu et al. (2020) learned the 2D offset from the CAD model center to the 3D bounding box corners from the RGB image, and then directly regressed the object pose from the 2D-3D correspondence. Di et al. (2021) used a shared encoder and two independent decoders to generate 2D-3D correspondence and self-occlusion information, improving the robustness of object pose estimation under occlusion. Further, Wang et al. (2021c) proposed a Geometry-Guided Direct Regression Network (GDR-Net) to learn object pose from dense 2D-3D correspondence in an end-to-end manner. Wang et al. (2021b) introduced noise-augmented student training and differentiable rendering based on GDR-Net (Wang et al., 2021c), enabling robustness to occlusion scenes through self-supervised learning with multiple geometric constraints. Zhang et al. (2022d) proposed a transformer-based pose estimation approach that consists of a patch-aware feature fusion module and a transformer-based pose refinement module to address the limitation of CNN-based networks in capturing global dependencies. Most recently, Feng et al. (2024) decou-

pled rotation into two sets of corresponding 3D normals. This decoupling strategy significantly improves the rotation accuracy.

Given the labor-intensive nature of real-world data annotation, some methods leverage synthetic data training to generalize to the real world. Gao et al. (2021) constructed a lightweight synthetic point cloud generation pipeline and leveraged an enhanced point cloud-based autoencoder to learn latent object pose information to regress object pose. To improve generalization to real-world scenes, Zhou et al. (2021a) utilized annotated synthetic data to supervise the network convergence. They proposed a self-supervised pipeline for unannotated real data by minimizing the distance between the CAD model transformed from the predicted pose and the input point cloud. Tan and Dong (2023) proposed a self-supervised monocular object pose estimation network consisting of teacher and student modules. The teacher module is trained on synthetic data for initial object pose estimation, and the student model predicts camera pose from the unannotated real image. The student module acquires knowledge of object pose estimation from the teacher module by imposing geometric constraints derived from the camera pose.

Geometry-guided regression methods typically require additional processing steps to extract and handle geometric information, which increases computational costs and complexity.

3.4.2 Direct Regression Methods

Direct regression methods aim to directly recover the object pose from the RGBD image without additional transformation steps, thus reducing complexity. These methods encompass various strategies, including coupled pose output, decoupled pose output, and 3D rotation (3DoF pose) output. Coupled pose involves predicting object rotation and translation together, while decoupled pose involves predicting them separately. Moreover, the 3DoF pose output focuses solely on predicting object rotation without considering translation. These strategies are discussed in detail below.

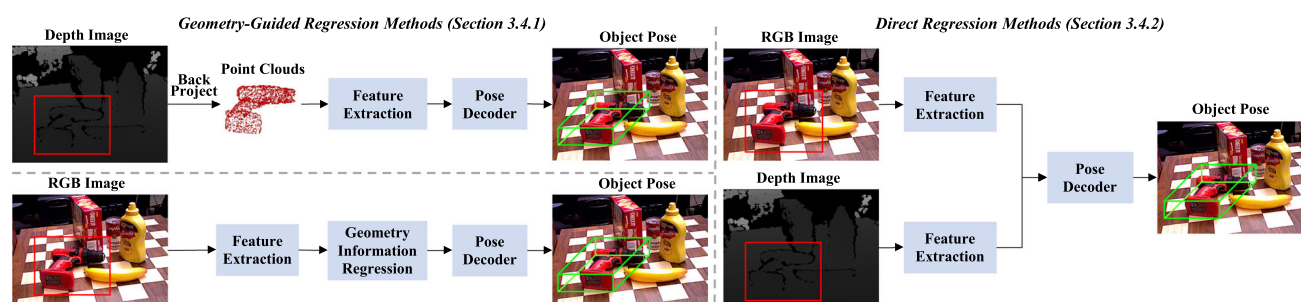


图7基于回归方法的示意图（第3.4节），该方法旨在直接从学习到的特征中获取物体姿态

随后，他们通过对成对兼容性图应用谱卷积运算，提升了几何特征的成对一致性。此外，Shi 等人（2021）在点云块内学习了几何与上下文特征，进而训练了一个子块网络来预测每个点云块的姿态。最终，选择最可靠的块姿态作为物体姿态。为应对基于点云的物体姿态跟踪挑战，Liu 等人（2024c）提出在相邻帧的点云之间进行位移点卷积运算，以促进局部上下文交互。

单纯依赖物体点云的方法往往忽视物体纹理细节。为此，Wen 等人（2021）与 An 等人（2024）充分利用了RGB颜色信息与深度信息的互补特性。他们通过引入注意力机制优化跨模态融合策略，有效对齐并整合这两种异构数据源，从而显著提升了模型性能。

与前述直接从深度图像或物体CAD模型中提取几何信息的方法不同，许多研究者更侧重于从RGB图像中生成几何约束。Hu 等人（2020）通过RGB图像学习了从CAD模型中心到三维边界框角点的二维偏移量，随后直接从二维-三维对应关系中回归物体姿态。Di 等人（2021）采用共享编码器与两个独立解码器生成二维-三维对应关系及自遮挡信息，提升了遮挡场景下物体姿态估计的鲁棒性。此外，Wanget等人（2021c）提出几何引导直接回归网络（GDR-Net），以端到端方式从密集二维-三维对应关系中学习物体姿态。Wang 等人（2021b）基于GDR-Net引入噪声增强学生训练与可微分渲染技术（Wang 等人，2021c），通过多几何约束的自监督学习实现对遮挡场景的鲁棒性。Zhang 等人（2022d）提出了一种基于Transformer的姿态估计方法，该方法包含一个基于块感知的特征融合模块和一个基于Transformer的姿态精炼模块，旨在解决基于CNN的网络在捕捉全局依赖关系方面的局限性。最近，Feng 等人（2024）将旋转分解为两组对应的三维

法线向量，这一分解策略显著提升了旋转精度。

鉴于真实世界数据标注具有劳动密集型特征，部分方法通过合成数据训练来实现对现实场景的泛化能力。Gao 等人（2021）构建了轻量级合成点云生成流程，并采用增强型基于点云的自编码器学习潜在物体姿态信息以回归物体姿态。为提升对真实场景的泛化能力，Zhou 等人（2021a）利用标注合成数据监督网络收敛过程，提出通过最小化预测姿态转换后的CAD模型与输入点云之间的距离来处理未标注真实数据的自监督流程。Tan 与 Dong（2023）提出由教师模块与学生模块组成的自监督单目物体姿态估计网络：教师模块在合成数据上进行初始物体姿态估计训练，学生模型则根据未标注真实图像预测相机姿态。学生模块通过

通过教师模块施加源自相机姿态的几何约束，实现物体姿态估计知识的获取。

几何引导回归方法通常需要额外的处理步骤来提取和处理几何信息，这会增加计算成本和复杂度。

3.4.2 直接回归方法

直接回归方法旨在无需额外变换步骤即可直接从RGBD图像中恢复物体的姿态，从而降低计算复杂度。这些方法涵盖多种策略，包括耦合姿态输出、解耦姿态输出以及三维旋转（3DoF姿态）输出。耦合姿态输出需同时预测物体的旋转和平移，而解耦姿态输出则分别预测两者；3DoF姿态输出则仅关注物体旋转的预测，不考虑平移。下文将对这些策略进行详细阐述。

Coupled Pose: To overcome lighting variations in the environment, Rambach et al. (2018) used a pencil filter to normalize the input image into light-invariant representations, and then directly regressed the object coupled pose using a CNN network. Additionally, Kleeberger and Huber (2020) introduced a robust framework to handle occlusions between objects and estimate the multiple objects pose in the image. This framework is capable of running in real-time at 65 FPS. Sarode et al. (2019) introduced a PointNet-based (Qi et al., 2017) framework to align point clouds for pose estimation, aiming to reduce sensitivity to pose misalignment. Estimating object pose from a single RGB image introduces an inherent ambiguity problem. Manhardt et al. (2019) suggested explicitly addressing these ambiguities. They predicted multiple 6DoF poses for each object to estimate specific pose distributions caused by symmetry and repetitive textures. Inspired by the visible surface difference metric, Bengtson et al. (2021) relied on a differentiable renderer and the CAD model to generate multiple weighted poses, avoiding falling into local minima. Moreover, Park and Cho (2022) proposed a method for pose estimation based on the local grid in object space. The method locates the grid region of interest on a ray in camera space and transforms the grid into object space via the estimated pose. The transformed grid is a new standard for sampling mesh and estimating pose.

For object pose tracking, Garon and Lalonde (2017) proposed a real-time tracking method that learns transformation relationships from consecutive frames during training, and used FCN (Long et al., 2015) to obtain the relative pose between two frames for training and inference.

Coupled pose may lead to information coupling between rotation and translation, making it difficult to distinguish their relationship during the optimization process, thus affecting estimation accuracy.

Decoupled Pose: Decoupling the 6DoF object pose enables explicit modeling of the dependencies and independencies between object rotation and translation (Xiang et al., 2017).

In object pose estimation, Xiang et al. (2017) estimated the 3D translation by locating the object center in the image and predicting the distance from the object center to the camera. They further estimated the 3D rotation by regressing to a quaternion representation, and introduced a novel loss function to handle symmetric objects better. Meanwhile, Kehl et al. (2017) extended the SSD framework (Liu et al., 2016) to generate 2D bounding boxes, as well as confidence scores for each viewpoint and in-plane rotation. Then, they chose the 2D bounding box through non-maximum suppression, along with the highest confidence viewpoint and in-plane rotation to infer the 3D translation, resulting in the full 6DoF object pose. Wu et al. (2018), Do et al. (2018) and Bukschat and Vetter (2020) used two parallel FCN (Long et al., 2015) branches to regress the object rotation and translation independently. To eliminate the dependence on annotations of

real data, Wang et al. (2020b) used synthetic RGB data for fully supervised training, and then leveraged neural rendering for self-supervised learning on unannotated real RGBD data. Moreover, Jiang et al. (2022) fused RGBD, built-in 2D-pixel coordinate encoding, and depth normal vector features to better estimate the object rotation and translation. Single-view methods suffer from ambiguity, therefore, Li et al. (2018a) proposed a multi-view fusion framework to reduce the ambiguity inherent in single-view frameworks. Further, Labbé et al. (2020) proposed a unified approach for multi-view, multi-object pose estimation. Initially, they utilized a single-view, single-object pose estimation technique to derive pose hypotheses for individual objects. Then, they aligned these object pose hypotheses across multiple input images to collectively infer both the camera viewpoints and object pose within a unified scene. Most recently, Hsiao et al. (2024) introduced a score-based diffusion method to solve the pose ambiguity problem in RGB-based object pose estimation.

For object pose tracking, Wen et al. (2020) proposed a data-driven optimization strategy to stabilize the 6DoF object pose tracking. Specifically, they predicted the 6DoF pose by predicting the relative pose between the adjacent frames. Liu et al. (2022b) proposed a new subtraction feature fusion module based on (Wen et al., 2020) to establish sufficient spatiotemporal information interaction between adjacent frames, improving the robustness of object pose tracking in complex scenes. Different from the method based on RGBD input, Ge and Loianno (2021) designed a novel deep neural network architecture that integrates visual and inertial features to predict the relative object pose between consecutive image frames.

In object pose refinement, Li et al. (2018c) iteratively refined pose through aligning RGB image with rendered image of object CAD model. Additionally, they predicted optical flow and foreground masks to stabilize the training procedure. Manhardt et al. (2018) refined the 6DoF pose by aligning object contour between the RGB image and rendered contour. The rendered contour is obtained from the object CAD model using the initial pose. Hai et al. (2023b) proposed a shape-constraint recursive matching framework to refine the initial pose. They first computed a pose-induced flow based on the initial and currently estimated pose, and then directly decoupled the 6DoF pose from the pose-induced flow. To address the low running efficiency of the pose refinement methods, Iwase et al. (2021) introduced a deep texture rendering-based pose refinement method for fast feature extraction using an object CAD model with a learnable texture. Most recently, Li et al. (2024a) proposed a two-stage method. The first stage performs pose classification and renders the object CAD model in the classified poses. The second stage performs regression to predict fine-grained residual in

耦合姿态: 为克服环境中的光照变化, Rambach 等人 (2018) 采用铅笔滤波器将输入图像归一化为光照不变表示, 随后通过CNN网络直接回归物体耦合姿态。此外, Kleeberger 和 Huber (2020) 提出了一种稳健框架以处理物体间的遮挡并估计图像中多物体姿态, 该框架可实现实时65帧/秒的运行速度。Sarode 等人 (2019) 基于PointNet框架 (Qi 等人, 2017) 进行点云对齐以实现姿态估计, 旨在降低姿态错位敏感性。从单张RGB图像估计物体姿态存在固有模糊性问题, Manhardt 等人 (2019) 提出明确解决这些模糊性的方法, 通过预测每个物体的多个6DoF姿态来估计由对称性和重复纹理导致的特定姿态分布。受可见表面差异度量启发, Bengtson 等人 (2021) 研究采用可微分渲染器与CAD模型生成多重加权姿态, 有效避免陷入局部极小值。此外, Park 与 Cho (2022) 提出基于物体空间局部网格的姿态估计方法: 该方法首先在相机空间中确定光线上的网格感兴趣区域, 再通过估计姿态将网格转换至物体空间。这种转换后的网格结构已成为网格采样与姿态估计的新标准。

在物体姿态跟踪领域, Garon 和 Lalonde (2017) 提出了一种实时跟踪方法, 该方法在训练过程中通过连续帧学习变换关系, 并采用FCN (Long 等人, 2015) 获取两帧间的相对姿态用于训练和推理。

耦合姿态可能导致旋转与平移之间的信息耦合, 使得在优化过程中难以区分二者关系, 从而影响估计精度。

解耦姿态: 通过解耦六自由度 (6DoF) 物体姿态, 能够明确建模物体旋转与平移之间的依赖关系与独立性 (Xiang 等人, 2017)。在物体姿态估计领域, Xiang 等人 (2017) 通过定位物体中心在图像中的位置并预测物体中心到相机的距离来估算三维平移; 随后通过回归四元数表示来估算三维旋转, 并引入了新的损失函数以更好地处理对称物体。与此同时, Kehl 等人 (2017) 扩展了SSD框架 (Liu 等人, 2016), 生成二维边界框以及每个视角和面内旋转的置信度评分; 随后通过非最大值抑制算法选取二维边界框, 并结合置信度最高的视角与面内旋转来推断三维平移, 从而获得完整的六自由度物体姿态。Wu 等人 (2018)、Do 等人 (2018) 以及 Bukschat 与 Vetter (2020) 采用了两个并行FCN (Long 等人, 2015) 分支用于独立回归物体旋转和平移。为消除对真实数据标注的依赖,

Wang 等人 (2020b) 采用合成RGB数据进行全监督训练, 随后利用神经渲染技术对未标注的真实RGBD数据进行自监督学习。此外, Jiang 等人 (2022) 融合了RGBD、内置二维像素坐标编码及深度法向量特征以更准确估计物体旋转和平移。单视图方法存在模糊性问题, 因此Li 等人 (2018a) 提出多视图融合框架以降低单视图框架固有的模糊性。进一步地, Labbe 等人 (2020) 提出了多视图多目标位姿估计统一方法: 首先采用单视图单目标位姿估计技术获取个体物体的位姿假设, 随后将这些位姿假设在多输入图像间进行对齐, 从而在统一场景中同时推断相机视角与物体位姿。最新研究中, Hsiao 等人 (2024) 提出了一种基于分数的扩散方法, 用于解决基于RGB的物体姿态估计中的姿态模糊问题。

在物体姿态跟踪领域, Wen 等人 (2020) 提出了一种数据驱动优化策略, 用于稳定六自由度物体姿态跟踪。具体而言, 他们通过预测相邻帧间的相对姿态来推算六自由度姿态。Liu 等人 (2022b) 基于 (Wen 等人, 2020) 提出了一种新型减法特征融合模块, 通过建立相邻帧间充分的时空信息交互, 显著提升了复杂场景中物体姿态跟踪的鲁棒性。与基于RGBD输入的方法不同, Ge 和 Loiano (2021) 设计了一种创新的深度神经网络架构, 通过整合视觉特征与惯性特征来预测连续图像帧间的相对物体姿态。

在物体姿态优化中, Li 等人 (2018c) 通过将RGB图像与物体CAD模型的渲染图像对齐, 迭代优化姿态。此外, 他们预测光流和前景掩码以稳定训练过程。Manhardt 等人 (2018) 通过在RGB图像与渲染轮廓之间对齐物体轮廓来优化6DoF姿态。渲染轮廓是通过初始姿态从物体CAD模型中获取的。Hai 等人 (2023b) 提出了一种基于形状约束的递归匹配框架来优化初始姿态。他们首先根据初始姿态和当前估计姿态计算姿态诱导流, 然后将6DoF姿态与姿态诱导流解耦。为解决姿态优化方法运行效率低的问题, Iwase 等人 (2021) 引入了一种基于深度纹理渲染的姿态优化方法, 利用具有可学习纹理的物体CAD模型实现快速特征提取。最新研究中, Li 等人 (2024a) 提出两阶段方法: 第一阶段进行姿态分类并在分类后的姿态下渲染物体CAD模型。第二阶段通过回归分析预测细粒度残差

the classified poses. This method improves robustness by guiding residual pose regression through pose classification.

3DoF Pose: Some researchers pursue more efficient and practical pose estimation by solely regressing the 3D rotation. Papaioannidis and Pitas (2019) proposed a novel quaternion-based multi-objective loss function, which integrates manifold learning and regression for learning 3DoF pose descriptors. They obtained the 3DoF pose through the regression of the learned descriptors. Liu et al. (2019b) trained a triple network based on convolutional neural networks to extract discriminative features from binary images. They incorporated pose-guided methods and regression constraints into the constructed triple network to adapt the features for the regression task, enhancing robustness. In addition, Josifovski et al. (2018) estimated the camera viewpoint related to the object coordinate system by constructing a viewpoint estimation model, thereby obtaining the 3DoF pose appearing in the bounding box.

Brief Discussion: Overall, direct regression methods simplify the object pose estimation process and further enhance the performance of instance-level methods.

3.5 Discussion

The methods reviewed above demonstrate diverse strategies for instance-level object pose estimation. To place them in a broader context, we discuss approaches from two complementary perspectives: the input modality (RGB vs. RGBD vs. RGB + Point Cloud) that supplies information and the methodological paradigm (Correspondence vs. Template vs. Voting vs. Regression) that defines how it is exploited.

3.5.1 Input Modalities

RGB-based methods rely on semantic richness and the wide availability of cameras, which makes them efficient and practical for large-scale deployment. Their limitation is sensitivity to illumination changes and surface texture. RGBD pipelines address this by explicitly fusing photometric and geometric information. When the fusion is designed carefully, for example by exploiting inter-modality relations, separating depth-dependent scale signals, or integrating temporal RGBD motion, they consistently achieve higher accuracy and stability. Point cloud-centric methods directly leverage dense geometry. They are robust in low-light conditions and can capture fine structural details, yet they face challenges with partial views, missing regions, and sensor noise. In summary, RGB inputs remain attractive when efficiency or hardware simplicity is the priority, whereas RGBD and point cloud inputs provide significant advantages due to their active and reliable sensing of the geometric data.

3.5.2 Methodological Paradigms

Correspondence-based methods, whether sparse or dense, are effective under occlusion because they rely on local matches and geometric consistency. However, they are less reliable on texture-less or weakly distinctive objects, although recent work shows improvements through stronger feature descriptors and symmetry-aware modeling. Template-based methods exploit global image or shape evidence, which allows them to handle texture-less objects and pose ambiguities, but they remain sensitive to occlusion, background clutter, and illumination changes, and become computationally heavy when matching against a large number of templates. Voting-based methods achieve high accuracy by aggregating pixel or point hypotheses. Additionally, indirect voting depends strongly on keypoint quality, and both indirect and direct variants require considerable computation. Regression-based methods provide a simplified pipeline by predicting pose end-to-end. Geometry-guided regression, in particular, benefits from explicit 2D–3D constraints or point cloud structures to improve stability. Overall, correspondence is suitable for scenarios with partial visibility, template methods are effective on texture-less objects, voting methods are appropriate when accuracy is prioritized over speed, and regression is preferred when efficiency and throughput are critical.

In general, instance-level methods can only estimate specific object instances in the training data, limiting their generalization to unseen objects. Additionally, most instance-level methods require accurate object CAD models, which is a challenge, especially for objects with complex shapes and textures.

4 Category-Level Object Pose Estimation

Research on category-level methods has garnered significant attention due to their potential for generalizing to unseen objects within established categories (Sahin & Kim, 2018). In this section, we review category-level methods by dividing them into shape prior-based (Sec. 4.1) and shape prior-free (Sec. 4.2) methods. The illustration of these two categories is shown in Fig. 8 and Fig. 9, respectively. The characteristics and performance of some representative SOTA methods are shown in Table 2.

4.1 Shape Prior-Based Methods

Shape prior-based methods first learn a neural network using CAD models of intra-class seen objects in offline mode to derive shape priors, and then utilize them as 3D geometry prior information to guide intra-class unseen object pose estimation. In this part, we divide the shape prior-based methods

分类姿态。该方法通过姿态分类引导残差姿态回归, 从而提升鲁棒性。**三自由度姿态:** 部分研究者通过仅回归三维旋转来实现更高效实用的姿态估计。Papai-oannidis 与 Pitas (2019) 提出了一种基于四元数的多目标损失函数, 该方法融合流形学习与回归技术以学习三自由度姿态描述符, 并通过回归学习到的描述符获得三自由度姿态。Liu 等人 (2019b) 基于卷积神经网络训练三重网络以从二值图像中提取判别特征, 将姿态引导方法与回归约束整合至构建的三重网络中, 使特征适应回归任务从而增强鲁棒性。此外, Josifovski 等人 (2018) 通过构建视角估计模型推算与物体坐标系相关的相机视角, 进而获得边界框中呈现的三自由度姿态。

简要讨论: 总体而言, 直接回归方法简化了物体姿态估计过程, 并进一步提升了实例级方法的性能表现。

3.5 讨论

上述综述方法展示了实例级物体姿态估计的多样化策略。为将其置于更广阔的背景中, 我们从两个互补视角探讨相关方法: 一是提供信息的输入模态 (RGB、RGBD 或 RGB+点云), 二是定义信息利用方式的方法论范式 (对应关系、模板匹配、投票决策或回归分析)。

3.5.1 输入模式

基于 RGB 的方法依赖于丰富的语义信息和摄像头的广泛普及, 这使其在大规模部署中具有高效性和实用性。其局限性在于对光照变化和表面纹理较为敏感。RGBD 处理流程通过显式融合光度信息与几何信息来解决这一问题: 当融合过程经过精心设计——例如利用多模态关系、分离深度相关的尺度信号或整合时间 RGBD 运动——它们能持续实现更高的准确性和稳定性。以点云为中心的方法直接利用密集几何数据, 在低光照条件下表现稳健, 并能捕捉精细的结构细节, 但面对部分视角、缺失区域及传感器噪声时仍面临挑战。总之, 当效率或硬件简洁性是首要考量时, RGB 输入仍具吸引力; 而 RGBD 和点云输入则因其能主动且可靠地感知几何数据而具有显著优势。

3.5.2 方法论范式

基于对应关系的方法 (无论是稀疏还是密集型) 在遮挡场景下表现优异, 因其依赖局部匹配与几何一致性。然而对于无纹理或特征模糊的物体, 这类方法可靠性较低——尽管最新研究表明, 通过采用更强特征描述符和对称性感知建模可实现性能提升。基于模板的方法利用全局图像或形状特征, 能够有效处理无纹理物体与姿态模糊问题, 但其仍易受遮挡、背景杂乱及光照变化影响, 且在大规模模板匹配时计算量显著增加。基于投票机制的方法通过聚合像素或点位预测结果实现高精度, 但间接投票机制对关键点质量依赖性强, 且直接与间接变体均需大量计算资源。回归方法通过端到端姿态预测提供简化流程, 其中几何引导回归尤其得益于显式二维-三维约束或点云结构的引入, 从而提升系统稳定性。总体而言, 对应关系算法适用于部分可见性场景, 模板方法对无纹理物体效果显著, 投票法在优先考虑准确率而非速度时适用, 而回归算法则在效率与吞吐量至关重要的情况下更受青睐。

总体而言, 实例级方法仅能估计训练数据中的特定对象实例, 这限制了其对未见过对象的泛化能力。此外, 大多数实例级方法需要精确的对象 CAD 模型, 这对形状和纹理复杂的对象而言尤为困难。

4 类别层级对象位姿估计

类别级方法的研究因其能够将已知类别中的未见对象进行泛化而受到广泛关注 (Sahin & Kim, 2018)。本节通过将类别级方法划分为基于形状先验的方法 (第 4.1 节) 和无形状先验的方法 (第 4.2 节) 进行系统梳理。两类方法的示意图分别如图 8 和图 9 所示。部分代表性 SOTA 方法的特征与性能数据见表 2。

4.1 基于形状先验的方法

基于形状先验的方法首先通过离线模式下类内已知物体的 CAD 模型训练神经网络以获取形状先验, 随后将这些先验作为三维几何先验信息用于指导类内未知物体的姿态估计。本节将对基于形状先验的方法进行系统阐述。

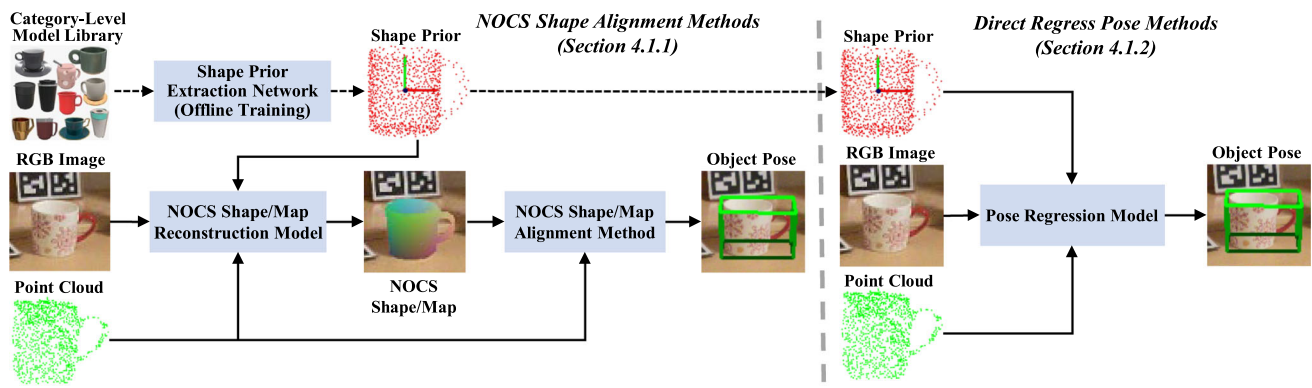


Fig. 8 Illustration of the shape prior-based category-level methods (Sec. 4.1). The dashed arrows indicate offline training, which means that we need to train a model offline using the category-level model library to obtain shape priors. Taking RGBD input as an example, NOCS shape alignment methods (Sec. 4.1.1) first learn a model to predict the

NOCS shape/map of the object, and then align the object point cloud with the NOCS shape/map through a non-differentiable pose solution method such as the Umeyama algorithm (Umeyama, 1991) to solve the object pose. In contrast, direct regress pose methods (Sec. 4.1.2) directly regress the object pose from the extracted input features

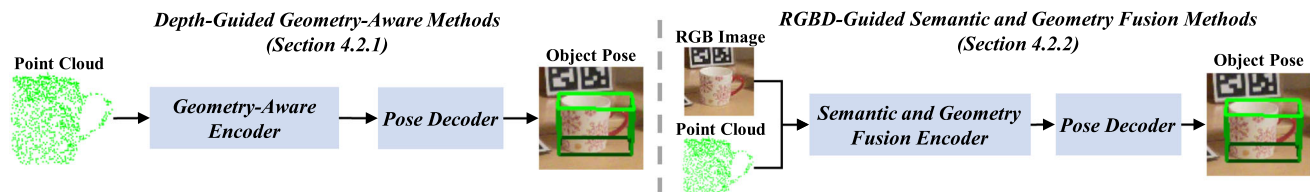


Fig. 9 Illustration of the shape prior-free category-level methods (Sec. 4.2), which do not have the process of shape priors regression: Depth-guided geometry-aware methods (Sec. 4.2.1) focus on perceiving the global and local geometric information of the object and leverage these

3D geometric features to estimate the object pose. Conversely, RGBD-guided semantic and geometry fusion methods (Sec. 4.2.2) regress the object pose by fusing the 2D semantic and 3D geometric information of the object

into two categories based on their approach to addressing object pose estimation. The first category is Normalized Object Coordinate Space (NOCS) shape alignment methods (Sec. 4.1.1), and the other is the direct pose regression methods (Sec. 4.1.2). The illustration of the two categories is shown in Fig. 8.

4.1.1 NOCS Shape Alignment Methods

The NOCS shape alignment methods first predict the NOCS shape/map, and then use an offline pose solution method (such as the Umeyama algorithm (Umeyama, 1991)) to align the object point cloud with the predicted NOCS shape/map to obtain the object pose. As a pioneering work, Tian et al. (2020a) first extracted the shape prior in offline mode, which is used to represent the mean shape of a category of objects. For example, mugs are composed of a cylindrical cup body and an arc-shaped handle. Next, they introduced a shape prior deformation network for the intra-class unseen object to reconstruct its NOCS shape. Finally, the Umeyama algorithm (Umeyama, 1991) is employed to solve the object pose by aligning the NOCS shape and the object point cloud.

Following Tian et al. (2020a), some methods aim to reconstruct the NOCS shape more accurately. Specifically, Wang et al. (2021d) designed a recurrent reconstruction network to iteratively refine the reconstructed NOCS shape. Further, Chen and Dou (2021) adjusted the shape prior dynamically by using the structure similarity between the RGBD image and the shape prior. Zou et al. (2022) proposed two multi-scale transformer-based networks (Pixelformer and Pointformer) for extracting RGB and point cloud features, and subsequently merging them for shape prior deformation. Different from the previous methods, Fan et al. (2024b) introduced an adversarial canonical representation reconstruction framework, which includes a reconstructor and a discriminator of NOCS representation. Specifically, the reconstructor mainly consists of a pose-irrelevant module and a relational reconstruction module to reduce the sensitivity to rotation and translation, as well as to generate high-quality features, respectively. Then, the discriminator is used to guide the reconstructor to generate realistic NOCS representations. Nie et al. (2023) improved the accuracy of pose estimation via geometry-informed instance-specific priors and multi-stage shape reconstruction. More recently,

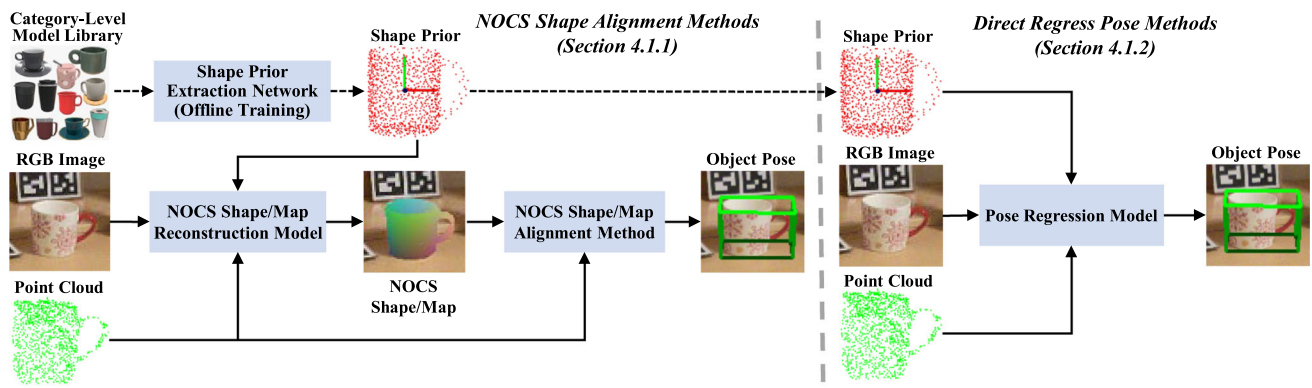


图8基于形状先验的类别级方法示意图(第4.1节)。虚线箭头表示离线训练,即需要使用类别级模型库进行离线模型训练以获取形状先验。以RGBD输入为例,NOCS形状对齐方法(第4.1.1节)首先学习预测物体NOCS形状/地图的模型,随后通过非可微分姿态解算方法(如Umeyama 1991提出的Umeyama算法)将

物体点云与NOCS形状/地图对齐以求解物体姿态。相比之下,直接回归姿态方法(第4.1.2节)直接从提取的输入特征中回归物体姿态。

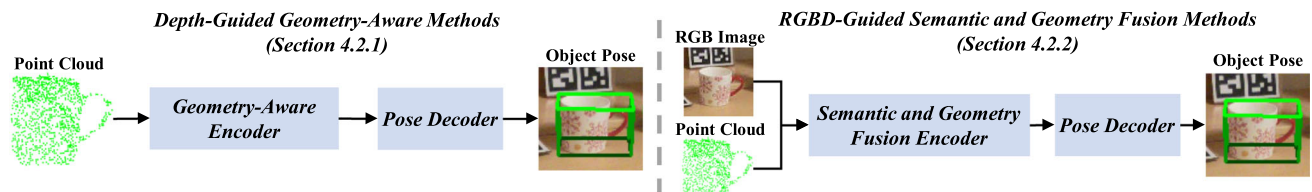


图9无形状先验的类别级方法示意图(第4.2节),该方法无需进行形状先验回归:深度引导的几何感知方法(第4.2.1节)专注于感知物体的全局与局部几何信息,并利用这些三维几何特征来估计物体姿态;相反,RGBD引导的语义与几何融合方法(第

4.2.2节)通过融合物体的二维语义信息与三维几何信息来回归物体姿态。

根据其解决物体姿态估计的方法,可分为两类。第一类是归一化物体坐标空间(NOCS)形状对齐方法(第4.1.1节),另一类是直接姿态回归方法(第4.1.2节)。两类方法的示意图如图8所示。

4.1.1 NOCS 形状配准方法

NOCS形状对齐方法首先预测NOCS形状/地图,然后采用离线姿态求解方法(如上山算法(Umeyama, 1991))将物体点云与预测的NOCS形状/地图对齐以获取物体姿态。作为开创性研究,Tian等人(2020a)首次在离线模式下提取形状先验,该先验用于表征某一类物体的平均形状。例如,马克杯由圆柱形杯体和弧形把手构成。随后,他们为类内未见物体引入形状先验变形网络以重构其NOCS形状。最终采用上山算法(Umeyama, 1991)通过NOCS形状与物体点云的对齐来求解物体姿态。

根据Tian等人(2020a)的研究,部分方法致力于更精确地重建NOCS形状。具体而言,Wang等人(2021d)设计了循环重建网络以迭代优化重建NOCS形状。此外,Chen和Dou(2021)通过利用RGBD图像与形状先验之间的结构相似性,动态调整形状先验。Zou等人(2022)提出两种基于多尺度变换器的网络(Pixelformer与Pointformer)用于提取RGB和点云特征,并将其融合以实现形状先验变形。与先前方法不同,Fan等人(2024b)引入了对抗性典范表示重建框架,该框架包含NOCS表示的重构器与判别器。其中重构器主要由姿态无关模块和关系重建模块组成,分别用于降低对旋转和平移的敏感性及生成高质量特征;判别器则用于引导重构器生成逼真的NOCS表示。Nie等人(2023)通过几何信息驱动的实例特定先验和多阶段形状重建技术,显著提升了姿态估计的准确性。近期研究中,

Table 2 Representative category-level methods. For each method, we report its 10 properties, which have the same meanings as described in Table. 1. D, S, N, K, and P denote object detection, instance segmentation, NOCS shape/map regression, keypoints detection, and pose solution/regression, respectively. Moreover, we report the 5°5cm metric of CAMERA25 and REAL275 datasets (Sec. 2)

Methods	Published Year	Training Input	Inference Input	Pose DoF	Object Property	Prop.	Task	Domain Training Paradigm	Inference Mode	Application Area	CAMERA25 5°5cm (mAP)	REAL275 5°5cm (mAP)
Shape Prior-Based Methods	Tian et al. (2020a)	RGBD, CAD Model	RGBD	9DoF	rigid		estimation	source	three-stage, S+N+P	general	59.0	21.4
	Wang et al. (2021d)	RGBD, CAD Model	RGBD	9DoF	rigid		estimation	source	three-stage, S+N+P	general	76.4	34.3
	Chen and Dou (2021)	RGBD, CAD Model	RGBD	9DoF	rigid		estimation	source	three-stage, S+N+P	general	74.5	39.6
	Zou et al. (2022)	RGBD, CAD Model	RGBD	9DoF	rigid		estimation	source	three-stage, S+N+P	general	76.7	41.9
	Fan et al. (2022a)	RGB, CAD Model	RGB	9DoF	rigid		estimation	source	three-stage, S+N+P	general	-	-
	Wei et al. (2023)	RGB, CAD Model	RGB	9DoF	rigid		estimation	source	three-stage, S+N+P	general	-	-
	Irshad et al. (2022)	RGBD, CAD Model	RGBD	9DoF	rigid		estimation	source	end-to-end	general	66.2	29.1
	Lin et al. (2022a)	Depth	Depth	9DoF	rigid		estimation	generalization	two-stage, S+P	general	70.9	42.3
	Zhang et al. (2022c)	Depth, CAD Model	Depth	9DoF	rigid		estimation	source	two-stage, S+P	general	75.5	44.6
	Zhang et al. (2022b)	Depth, CAD Model	Depth	9DoF	rigid		estimation	source	two-stage, S+P	general	79.6	48.1
	Liu et al. (2022d)	Depth	Depth	9DoF	rigid		refinement, tracking	source	two-stage, S+P	general	80.3	54.4
	Lin et al. (2022c)	RGBD, CAD Model	RGBD	9DoF	rigid		estimation	generalization	two-stage, S+P	general	-	45.0
	Ze and Wang (2022)	RGBD, CAD Model	RGBD	9DoF	rigid		estimation	adaptation	two-stage, S+P	general	-	33.9
	Liu et al. (2025a)	RGBD, CAD Model	RGBD	9DoF	rigid		estimation	generalization	two-stage, S+P	general	-	50.1
	Li et al. (2020)	Depth	Depth	9DoF	articulated		estimation	generalization	two-stage, S+P	general	-	-
Shape Prior-Free Methods	Chen et al. (2021)	Depth	Depth	9DoF	rigid		estimation	source	two-stage, D+P	general	-	28.2
	Weng et al. (2021)	Depth	Depth	9DoF	rigid, articulated		tracking	source	two-stage, N+P	general	-	62.2
	Di et al. (2022)	Depth	Depth	9DoF	rigid		estimation	source	two-stage, S+P	general	79.1	42.9
	You et al. (2022)	Depth	Depth	9DoF	rigid		estimation	generalization	two-stage, S+P	general	-	16.9
	Zheng et al. (2023)	Depth	Depth	9DoF	rigid		estimation	source	two-stage, S+P	general	80.5	55.2
	Zhang et al. (2023)	Depth	Depth	6DoF	rigid		estimation, tracking	source	two-stage, S+P	general	-	60.9

表 2 代表性类别级方法。针对每种方法, 我们列出了其 10 项特性, 这些特性的含义与表 1 中所述一致。D、S、N、K 和 P 分别表示目标检测、实例分割、NOCS 形状地图回归、关键点检测以及姿态解/回归。此外, 我们还报告了 CAMERA25 和 REAL275 数据集上的 $s \circ 5\text{cm}$ 指标 (详见第 2 节)。

方法	发布年份	训练输入	推理输入	姿势自由度	对象属性	任务	领域培训范式	推理模式	应用区域	CAMERA25 $s \circ 5\text{厘米}$ (平均精度)	REAL275 $s \circ 5\text{厘米}$ (平均精度)
形状 基于先验的方 法	田 乙 等 (2020a)	RGBD, CAD 模型	RGBD	9 自由度	严格的	评价	根源	三阶段的 S+N+P	一般的	59.0	21.4
	Wang 等 (2021d)	RGBD, CAD 模型	RGBD	9 自由度	严格的	评价	根源	三阶段的 S+N+P	一般的	76.4	34.3
	陈和宴 (2021)	RGBD, CAD 模型	RGBD	9 自由度	严格的	评价	根源	三阶段的 S+N+P	一般的	74.5	39.6
	邹 乙 等 (2022)	RGBD, CAD 模型	RGBD	9 自由度	严格的	评价	根源	三阶段的 S+N+P	一般的	76.7	41.9
	迷 乙 等 (2022a)	RGB, CAD 模型	RGB	9 自由度	严格的	评价	根源	三阶段的 S+N+P	一般的	-	-
	魏 乙 等 (2023)	RGB, CAD 模型	RGB	9 自由度	严格的	评价	根源	三阶段的 S+N+P	一般的	-	-
	Irshad 等人 (2022)	RGBD, CAD 模型	RGBD	9 自由度	严格的	评价	根源	两阶段 S+P 法 端到端	一般的	66.2	29.1
	Lin 等人 (2022 a)	深	深	9 自由度	严格的	评价	一般化	两阶段 S+P 法	一般的	70.9	42.3
	Zhang 等 (2022c)	深 模型	CAD	9 自由度	严格的	评价	根源	两阶段 S+P 法	一般的	75.5	44.6
	Zhang 等 (2022b)	深 模型	CAD	9 自由度	严格的	评价	根源	两阶段 S+P 法	一般的	79.6	48.1
	刘 乙 等 (2022a)	深	深	9 自由度	严格的	精炼 跟踪	根源	两阶段 S+P 法	一般的	80.3	54.4
	林 乙 等 (2022c)	RGBD, CAD 模型	RGBD	9 自由度	严格的	评价	一般化	两阶段 S+P 法	一般的	-	45.0
	泽与王 (2022)	RGBD, CAD 模型	RGBD	9 自由度	严格的	评价	改编本	两阶段 S+P 法	一般的	-	33.9
	刘 乙 等 (2025a)	RGBD, CAD 模型	RGBD	9 自由度	严格的	评价	一般化	两阶段 S+P 法	一般的	-	50.1
形状优先级 免费方法	Li 等人 (2020)	深	深	9 自由度	清楚地表达	评价	一般化	两阶段 S+P 法	一般的	-	-
	陈等人 (2021)	深	深	9 自由度	严格的	评价	根源	二阶段 D+P	一般的	-	28.2
	Weng 等 (2021)	深	深	9 自由度	严格的 迟来的 关节	跟踪	根源	二阶段 N+P	一般的	-	62.2
	Di 等人 (2022)	深	深	9 自由度	严格的	评价	根源	两阶段 S+P 法	一般的	79.1	42.9
	You 等人 (2022)	深	深	9 自由度	严格的	评价	一般化	两阶段 S+P 法	一般的	-	16.9
	Zheng 等人 (2023)	深	深	9 自由度	严格的	评价	根源	两阶段 S+P 法	一般的	80.5	55.2
	Zhang 等 (2023)	深	深	6 自由度	严格的	评价 跟踪	根源	两阶段 S+P 法	一般的	-	60.9

Table 2 continued

Methods	Published Year	Training Input	Inference Input	Pose DoF	Object- erty	Prop- erty	Task	Domain Train- ing Paradigm	Inference Mode	Application Area	CAMERA25 5°5cm (mAP)	REAL275 5°5cm (mAP)
RGBD-guided semantic & geometry fusion	Wang et al. (2019b)	2019	RGBD	RGBD	9DoF	rigid	estimation	source	two-stage, (S+N)+P	general	40.9	10.0
	Wang et al. (2020a)	2020	RGBD	RGBD	6DoF	rigid	tracking	source	two-stage, K+P	general	-	33.3
	Lin et al. (2021b)	2021	RGBD	RGBD	9DoF	rigid	estimation	source	two-stage, S+P	general	70.7	35.9
	Wen and Bekris (2021)	2021	RGBD	RGBD	9DoF	rigid	tracking	source	three-stage, S+K+P	general	-	87.4
	Peng et al. (2022)	2022	RGBD, CAD Model	RGBD	9DoF	rigid	estimation	adaptation	two-stage, S+P	general	-	33.4
	Lee et al. (2022)	2022	RGBD, CAD Model	RGBD	9DoF	rigid	estimation	adaptation	three-stage, S+N+P	general	-	34.8
	Lee et al. (2023)	2023	RGBD, CAD Model	RGBD	9DoF	rigid	estimation	generalization	three-stage, S+N+P	general	-	35.9
	Liu et al. (2023c)	2023	RGBD	RGBD	9DoF	rigid	estimation	source	two-stage, S+P	general	79.9	53.4
	Lin et al. (2023a)	2023	RGBD or Depth	RGBD or Depth	9DoF	rigid	estimation	source	two-stage, S+P	general	81.4	57.6
	Chen et al. (2024)	2024	RGBD	RGBD	9DoF	rigid	estimation	source	two-stage, S+P	general	-	63.6
	Lee et al. (2021)	2021	RGB, CAD Model	RGB	9DoF	rigid	estimation	generalization	three-stage, S+N+P	general	-	-
	Lin et al. (2024c)	2024	RGBD, Text	RGBD, Text	9DoF	rigid	estimation	source	two-stage, S+P	general	82.2	58.3
Others												

表2续

方法	RGBD 引导的 语义与几何 融合	发布年份	训练输入	推理 输入	姿势自由度	对象属性	任务	领域培训范式	推理模式	应用区域	CAMERA25 S+P 分数 (平均精度)	REAL275 S+P 分数 (平均精度)
方法	RGBD 引导的 语义与几何 融合	2019	RGBD	RGBD	RGBD	9自由度	严格的	评价	根源	一般的	40.9	10.0
		2020	RGBD	RGBD	RGBD	6自由度	严格的	跟踪	根源	一般的	-	33.3
		2021	RGBD	RGBD	RGBD	9自由度	严格的	评价	根源	一般的	70.7	35.9
		2021	RGBD	RGBD	RGBD	9自由度	严格的	跟踪	根源	一般的	-	87.4
		2022	RGBD, CAD 模型	RGBD, CAD 模型	RGBD	9自由度	严格的	评价	根源	一般的	-	33.4
		2022	RGBD, CAD 模型	RGBD, CAD 模型	RGBD	9自由度	严格的	评价	根源	一般的	-	34.8
		2023	RGBD, CAD 模型	RGBD, CAD 模型	RGBD	9自由度	严格的	评价	根源	一般的	-	35.9
		2023	RGBD	RGBD	RGBD	9自由度	严格的	评价	根源	一般的	79.9	53.4
		2023	RGBD 或 深	RGBD 或 深	RGBD	9自由度	严格的	评价	根源	一般的	81.4	57.6
		2024	RGBD	RGBD	RGBD	9自由度	严格的	评价	根源	一般的	-	63.6
		2021	RGB, CAD 模型	RGB, CAD 模型	RGB	9自由度	严格的	评价	根源	一般的	-	-
		2024	RGBD, 文本	RGBD, 文本	RGBD, 文本	9自由度	严格的	评价	根源	一般的	82.2	58.3
	其他	2021	RGB, CAD 模型	RGB, CAD 模型	RGB	9自由度	严格的	评价	根源	一般的	-	-
		2024	RGBD, 文本	RGBD, 文本	RGBD, 文本	9自由度	严格的	评价	根源	一般的	82.2	58.3

Zhou et al. (2023b) designed a two-stage pipeline consisting of deformation and registration to improve accuracy. Zou et al. (2023) introduced a graph-guided point transformer consisting of a graph-guided attention encoder and an iterative non-parametric decoder to further extract the point cloud feature. In addition, Li et al. (2023c) leveraged discrepancies in instance-category structures alongside potential geometric-semantic associations to better investigate intra-class shape information. Yu et al. (2024) further divided the NOCS shape reconstruction process into three parts: coarse deformation, fine deformation, and recurrent refinement to enhance the accuracy of NOCS shape reconstruction.

Given that the annotation of ground-truth object pose is time-consuming, He et al. (2022b) explored a self-supervised method via enforcing the geometric consistency between point cloud and category prior mesh, avoiding using the real-world pose annotation. Further, Li et al. (2023b) first extracted semantic primitives via a part segmentation network, and leveraged semantic primitives to compute SIM(3)-invariant shape descriptor to generate the optimized shape. Then, the Umeyama algorithm (Umeyama, 1991) is utilized to recover the object pose. Through this approach, they achieved domain generalization, bridging the gap between synthesis and real-world application.

Depth images may be unavailable in some challenging scenes (*e.g.*, under strong or low light conditions). Therefore, achieving monocular category-level object pose estimation is of great significance across various applications. Fan et al. (2022a) directly predicted object-level depth and NOCS shape from a monocular RGB image by deforming the shape prior, and subsequently leveraged the Umeyama algorithm (Umeyama, 1991) to solve the object pose. Unlike Fan et al. (2022a), Wei et al. (2023) estimated the 2.5D sketch and separated scale recovery using the shape prior. They then reconstructed the NOCS shape, employing the RANSAC (Fischler & Bolles, 1981) algorithm to remove outliers, before utilizing the PnP algorithm for recovering the object pose. For transparent objects, Chen et al. (2023c) proposed a new solution based on stereo vision, which defines a back-view NOCS map to tackle the problem of image content aliasing.

In general, although these NOCS shape alignment methods can recover the object pose, the alignment process is non-differentiable and is not integrated into the learning process. Thus, errors in predicting the NOCS shape/map have a significant impact on the accuracy of pose estimation.

4.1.2 Direct Regress Pose Methods

Due to the non-differentiable nature of the NOCS shape alignment process, several direct regression-based pose methods have been proposed recently to enable end-to-end training. They directly regress the object pose from the fea-

ture level, making the pose acquisition process differentiable. Irshad et al. (2022) treated object instances as spatial centers and proposed an end-to-end method that combines object detection, reconstruction, and pose estimation. Wang et al. (2023b) developed a deformable template field to decouple shape and pose deformation, improving the accuracy of shape reconstruction and pose estimation. On the other hand, Zhang et al. (2022c) proposed a symmetry-aware shape prior deformation method, which integrates shape prior into a direct pose estimation network. Further, Zhang et al. (2022b) introduced a geometry-guided residual object bounding box projection framework to address the challenge of insufficient pose-sensitive feature extraction. In order to obtain a more precise object pose, Liu et al. (2022d) designed CATRE, a pose refinement method based on the alignment of the shape prior and the object point cloud to refine the object pose estimated by the above methods. Zheng et al. (2024) extended CATRE (Liu et al., 2022d) to address the geometric variation problem by integrating hybrid scope layers and learnable affine transformations.

Due to the extensive manual effort required for annotating real-world training data, Lin et al. (2022a) explored the shape alignment of each intra-class unseen instance against its corresponding category-level shape prior, implicitly representing its 3D rotation. This approach facilitates domain generalization from synthesis to real-world scenarios. Further, Ze and Wang (2022) proposed a novel framework based on pose and shape differentiable rendering to achieve domain adaptation object pose estimation. In addition, they collected a large Wild6D dataset for category-level object pose estimation in the wild. Following Ze and Wang (2022), Zhang and Fu (2023) introduced 2D-3D and 3D-2D geometry correspondences to enhance the ability of domain adaptation. Different from the previous approaches, Remus and D'Avella (2023) leveraged instance-level methods for domain-generalized category-level object pose estimation via a single RGB image. Lin et al. (2022c) proposed a deep prior deformation-based network and leveraged a parallel learning scheme to achieve domain generalization. More recently, Liu et al. (2025a) designed a multi-hypothesis consistency learning framework. This framework addresses the uncertainty problem and reduces the domain gap between synthetic and real-world datasets by employing multiple feature extraction and fusion techniques.

Brief Discussion: Overall, while the shape prior-based methods mentioned above significantly improve pose estimation performance, obtaining the shape priors requires constructing category-level CAD model libraries and subsequently training a network, which is both cumbersome and time-consuming.

Zhou 等人 (2023b) 设计了一种包含形变处理与配准步骤的两阶段流程以提升精度。Zou 等人 (2023) 提出了一种基于图引导注意力编码器与迭代非参数解码器的图引导点变换器, 用于进一步提取点云特征。此外, Li 等人 (2023c) 通过利用实例类别结构差异及潜在几何语义关联, 更深入地探究了类内形状信息。Yu 等人 (2024) 将 NOCS 形状重建过程细分为粗粒度形变处理、精细形变处理及循环优化三个阶段, 从而显著提升了形状重建的准确性。

鉴于真实物体姿态标注耗时较长, Heetal. (2022b) 提出了一种自监督方法, 通过强制点云与类别先验网格之间的几何一致性来实现, 从而避免使用真实世界的姿态标注。此外, Li 等人 (2023b) 首先通过部件分割网络提取语义基元, 并利用这些语义基元计算 SIM (3) 不变形状描述符以生成优化后的形状, 随后采用 Umeyama 算法 (Umeyama, 1991) 恢复物体姿态。通过这种方法, 他们实现了领域泛化, 弥合了合成与实际应用之间的差距。

在某些具有挑战性的场景中 (例如强光或弱光条件下), 深度图像可能无法获取。因此, 在各类应用场景中实现单目类别级物体姿态估计具有重要意义。Fan 等人 (2022a) 通过形变形状先验, 直接从单目 RGB 图像中预测物体级深度与 NOCS 形状, 随后采用 Umeyama 算法 (Umeyama, 1991) 求解物体姿态。与 Fan 等人 (2022a) 不同, Wei 等人 (2023) 利用形状先验估计 2.5D 草图并分离尺度恢复, 继而采用 RANSAC 算法 (Fischler & Bolles, 1981) 去除离群值以重建 NOCS 形状, 最终通过 PnP 算法恢复物体姿态。针对透明物体, Chen 等人 (2023c) 提出基于立体视觉的新方案, 通过定义后视 NOCS 图来解决图像内容混叠问题。

总体而言, 尽管这些 NOCS 形状对齐方法能够恢复物体姿态, 但其对齐过程不具备可微性, 且未被整合到学习过程中。因此, NOCS 形状/地图预测误差会显著影响姿态估计的准确性。

4.1.2 直接回归姿态方法

由于 NOCS 形状对齐过程的不可微性质, 近期已提出多种基于直接回归的姿态方法以实现端到端训练。这些方法直接从特征层面回归物体姿态, 使得姿态获取

过程具有可微性。Irshad 等人 (2022) 将物体实例视为空间中心, 提出了一种结合物体检测、重建与姿态估计的端到端方法。Wang 等人 (2023b) 开发了可变形模板场以解耦形状与姿态变形, 提升了形状重建和姿态估计的准确性。另一方面, Zhang 等人 (2022c) 提出了非对称感知形状先验变形方法, 将形状先验整合到直接姿态估计网络中。此外, Zhang 等人 (2022b) 引入几何引导残差物体边界框投影框架, 以应对姿态敏感特征提取不足的挑战。为获得更精确的物体姿态, Liu 等人 (2022d) 设计了 CATRE —— 一种基于形状先验与物体点云对齐的姿态优化方法, 用于优化上述方法估计的物体姿态。Zheng 等人 (2024) 扩展了 CATRE (Liu 等人, 2022d), 通过整合混合范围层和可学习仿射变换来解决几何变化问题。

由于标注真实世界训练数据需要大量人工努力, Lin 等人 (2022a) 探索了每个类内未见实例与其对应类别级形状先验之间的形状对齐, 隐式表示其三维旋转。该方法有助于从合成场景向真实场景实现领域泛化。此外, Ze 和 Wang (2022) 提出了一种基于姿态与形状可微分渲染的创新框架, 以实现领域适应性物体姿态估计。他们还收集了大型 Wild6D 数据集用于野外环境下的类别级物体姿态估计。沿袭 Ze 和 Wang (2022) 的研究思路, Zhang 和 Fu (2023) 引入了二维-三维及三维-二维几何对应关系以增强领域适应能力。与先前方法不同, Remus 和 D'Avella (2023) 通过单张 RGB 图像采用实例级方法实现领域泛化类别的物体姿态估计。Lin 等人 (2022c) 提出基于深度先验形变的网络, 并采用并行学习方案实现领域泛化。最近, Liu 等人 (2025a) 设计了一个多假设一致性学习框架。该框架通过采用多种特征提取与融合技术, 解决了不确定性问题, 并缩小了合成数据集与真实世界数据集之间的领域差异。

简要讨论: 总体而言, 虽然上述基于形状先验的方法能显著提升姿态估计性能, 但获取形状先验需要构建类别级 CAD 模型库并进行网络训练, 这一过程既繁琐又耗时。

4.2 Shape Prior-Free Methods

Shape prior-free methods do not rely on using shape priors and thus have better generalization capabilities. These methods can be divided into three main categories: depth-guided geometry-aware (Sec. 4.2.1), RGBD-guided semantic and geometry fusion (Sec. 4.2.2), and other (Sec. 4.2.3) methods. The illustration of the first two categories is shown in Fig. 9.

4.2.1 Depth-Guided Geometry-Aware Methods

Depth-guided geometry-aware methods primarily exploit 3D geometric cues derived from depth or point cloud data to estimate object pose. By leveraging structures such as 3D Graph Convolution (3DGC), point-wise voting, or geometric reconstruction, these approaches learn pose-sensitive features directly from shape and spatial relationships, enabling robust estimation without explicit reliance on shape priors. Chen et al. (2021) leveraged 3DGC and introduced a fast shape-based method, which consists of an RGB-based network to achieve 2D object detection, a shape-based network for 3D segmentation and rotation regression, and a residual-based network for translation and size regression. Inspired by Chen et al. (2021), Liu et al. (2022a) improved the network with structure encoder and reasoning attention. Further, Di et al. (2022) proposed a geometry-guided point-wise voting method that exploits geometric insights to enhance the learning of pose-sensitive features. Specifically, they designed a symmetry-aware point cloud reconstruction network and introduced a point-wise bounding box voting mechanism during training to add additional geometric guidance. Due to the translation and scale invariant properties of the 3DGC, these methods are limited in perceiving object translation and size information. Based on this, Zheng et al. (2023) further designed a hybrid scope feature extraction layer, which can simultaneously perceive global and local geometric structures and encode size and translation information.

Besides the above 3DGC-based methods, Deng et al. (2022) combined a category-level auto-encoder with a particle filter framework to achieve object pose estimation and tracking. Wang et al. (2023c) leveraged learnable sparse queries as implicit prior to perform deformation and matching for pose estimation. In addition, Wan et al. (2023) developed a semantically-aware object coordinate space to address the semantically incoherent problem of NOCS (Wang et al., 2019b). More recently, Zhang et al. (2023) proposed a scored-based diffusion model to address the multi-hypothesis problem in symmetric objects and partial point clouds. They first leveraged the scored-based diffusion model to generate multiple pose candidates, and then utilized an energy-based diffusion model to remove abnormal poses. On the other hand, Lin et al. (2024b) first introduced

an instance-adaptive keypoints detection method and then designed a geometric-aware global and local features aggregation network based on the detected keypoints for pose and size estimation. Li et al. (2024b) leveraged category-level method to determine part object poses for assembling multi-part multi-joint 3D shape.

To perform pose estimation on articulated objects, Li et al. (2020) inspired by Wang et al. (2019b), introduced a standard representation for different articulated objects within a category by designing an articulation-aware normalized coordinate space hierarchy, which simultaneously constructs a canonical object space and a set of canonical part spaces. Weng et al. (2021) further proposed CAPTRA, a unified framework that enables 9DoF pose tracking of rigid and articulated objects simultaneously. Due to the nearly unlimited freedom of garments and extreme self-occlusion, Chi and Song (2021) introduce GarmentNets, which conceptualizes deformable object pose estimation as a shape completion problem within a canonical space. More recently, Liu et al. (2023e) developed a reinforcement learning-based pipeline to predict 9DoF articulated object pose via fitting joint states through reinforced agent training. Further, Liu and Zhang (2023) learned part-level SE(3)-equivariant features via a pose-aware equivariant point convolution operator to address the issue of self-supervised articulated object pose estimation.

To avoid using extensive real-world labeled data for training, Li et al. (2021a) used SE(3) equivariant point cloud networks for self-supervised object pose estimation. You et al. (2022) introduced a category-level point pair feature voting method to reduce the impact of synthetic to real-world domain gap, achieving generalizable object pose estimation in the wild.

In general, these methods fully extract the pose-related geometric features. However, the absence of semantic information limits their better performance. Appropriate fusion of semantic and geometric information can significantly improve the robustness of pose estimation.

4.2.2 RGBD-Guided Semantic and Geometry Fusion Methods

RGBD-guided semantic and geometry fusion methods combine the complementary strengths of RGB-based semantic features and depth-based geometric representations within a unified framework. These methods typically predict canonical object coordinates, fuse semantic and structural features, or adopt attention-based integration strategies to enhance robustness. By jointly leveraging semantic context and geometric precision, they achieve higher generalization across diverse intra-class objects and challenging real-world conditions. As a groundbreaking research, Wang et al. (2019b) designed a normalized object coordinate space to provide

4.2 先验形状无关方法

基于形状先验的无先验方法不依赖于形状先验信息, 因此具有更好的泛化能力。这些方法可分为三大类: 深度引导的几何感知方法(第4.2.1节)、RGBD引导的语义与几何融合方法(第4.2.2节)以及其他方法(第4.2.3节)。前两类方法的示意图如图9所示。

4.2.1 基于深度引导且考虑几何特性的方法

深度引导的几何感知方法主要利用从深度或点云数据中提取的三维几何线索来估计物体姿态。通过采用三维图卷积(3DGC)、逐点投票或几何重建等结构, 这些方法能够直接从形状和空间关系中学习姿态敏感特征, 从而在无需显式依赖形状先验的情况下实现稳健估计。Chen等人(2021)利用3DGC提出了一种快速形状基方法, 该方法包含基于RGB的网络用于实现二维物体检测、基于形状的网络用于三维分割与旋转回归、以及基于残差的网络用于平移与尺寸回归。受Chen等人(2021)启发, Liu等人(2022a)通过结构编码器和推理注意力机制改进了网络架构。此外, Di等人(2022)提出了一种几何引导的逐点投票方法, 通过利用几何特征洞察力来增强姿态敏感特征的学习效果。具体而言, 研究团队设计了一种具有对称性感知能力的点云重建网络, 并在训练过程中引入了逐点边界框投票机制以增强几何引导效果。由于三维几何特征(3DGC)具有平移和尺度不变特性, 这些方法在感知物体平移和尺寸信息方面存在局限性。基于此, Zheng等人(2023)进一步开发了混合范围特征提取层, 该层能够同时感知全局与局部几何结构, 并对尺寸和平移信息进行编码。

除了上述基于3DGC的方法外, Deng等人(2022)将类别级自动编码器与粒子滤波框架相结合, 实现了物体姿态估计与跟踪。Wang等人(2023c)利用可学习的稀疏查询作为隐式先验信息, 进行形变计算与匹配以实现姿态估计。此外, Wan等人(2023)开发了语义感知物体坐标空间, 以解决NOCS(Wang等人, 2019b)存在的语义不连贯问题。最近, Zhang等人(2023)提出了一种基于评分的扩散模型, 用于解决对称物体和部分点云中的多假设问题: 首先利用该评分扩散模型生成多个姿态候选解, 再通过基于能量的扩散模型剔除异常姿态。另一方面, Lin等人(2024b)首先提出了实例自适应关键点检测方法, 随后基于

检测到的关键点设计了几何感知的全局与局部特征聚合网络, 用于姿态与尺寸估计。Li等人(2024b)采用类别级方法确定多部件多关节3D形状装配中的零件对象姿态。

为对铰接物体进行姿态估计, Li等人(2020)受Wang等人(2019b)启发, 通过设计一种关节感知归一化坐标空间层次结构, 为同一类别内的不同铰接物体引入了标准表示方法, 该方法同时构建了规范物体空间和一组规范部件空间。Weng等人(2021)进一步提出CAPTRA, 这是一种能够同时实现刚性物体与铰接物体9自由度姿态跟踪的统一框架。由于服装具有近乎无限的自由度且存在极端自遮挡现象, Chi与Song(2021)提出了GarmentNets框架, 将可变形物体姿态估计概念化为规范空间内的形状补全问题。近期, Liu等人(2023e)开发了一种基于强化学习的流程, 通过强化智能体训练拟合关节状态来预测9自由度铰接物体姿态。此外, Liu和Zhang(2023)通过姿态感知等变点卷积算子学习了部分级别的SE(3)等变特征, 以解决自监督关节对象姿态估计问题。

为避免使用大量真实世界标注数据进行训练, Li等人(2021a)采用了SE(3)等变点云网络进行自监督物体姿态估计。You等人(2022)提出了一种类别级别的点对特征投票方法, 以降低合成数据与真实世界数据之间的领域差异影响, 从而在实际场景中的通用物体姿态估计。

总体而言, 这些方法能够完整提取与姿态相关的几何特征。然而, 语义信息的缺失限制了其性能提升空间。通过合理融合语义与几何信息, 可显著增强姿态估计的鲁棒性。

4.2.2 RGBD 引导的语义与几何融合方法

RGBD引导的语义与几何融合方法通过统一框架整合了基于RGB的语义特征与基于深度的几何表征的互补优势。这类方法通常采用标准物体坐标预测、语义与结构特征融合, 或运用基于注意力机制的集成策略来提升鲁棒性。通过协同利用语义上下文与几何精度, 该方法在不同类内物体及复杂现实场景中展现出更强的泛化能力。作为开创性研究, Wang等人(2019b)设计了标准化物体坐标空间以提供

a canonical representation for a category of objects. They first predicted the class label, mask, and NOCS map of the intra-class unseen object. Then, they utilized the Umeyama algorithm (Umeyama, 1991) to solve object pose by aligning the NOCS map with the object point cloud. To handle various shape changes of intra-class objects, Chen et al. (2020a) learned a canonical shape space as a unified representation. On the other hand, Lin et al. (2021a) explored the applicability of sparse steerable convolution (SSC) to object pose estimation and proposed an SSC-based pipeline. Further, Lin et al. (2021b) proposed a dual pose network, which consists of a shared pose encoder and two parallel explicit and implicit pose decoders. The implicit decoder can enforce predicted pose consistency when there are no CAD models during inference. Wang et al. (2022a) designed an attention-guided network with relation-aware and structure-aware for RGB image and point cloud features fusion. Very recently, Liu et al. (2023c) explored the necessity of shape priors for shape reconstruction of intra-class unseen objects. They demonstrated that the deformation process is more important than the shape prior and proposed a prior-free implicit space transformation network. Lin et al. (2023a) addressed the poor rotation estimation accuracy by decoupling the rotation estimation into viewpoint and in-plane rotation. In addition, they also proposed a spherical feature pyramid network based on spatial spherical convolution to process spherical signals. With the rapid development of the Large Vision Model (LVM), Chen et al. (2024) further leveraged the LVM DINOv2 (Oquab et al., 2023) to extract the SE(3)-consistent semantic features and fused them with object-specific hierarchical geometric features to encapsulate category-level information for rotation estimation.

Since the above methods still require a large amount of real-world annotated training data, their applicability in real-world scenes is limited. To this end, Peng et al. (2022) proposed a real-world self-supervised training framework based on deep implicit shape representation. They leveraged the deep signed distance function (Park et al., 2019a) as a 3D representation to achieve domain adaptation from synthesis to the real world. In addition, Lee et al. (2022) introduced a teacher-student self-supervised learning mechanism. They used supervised training in the source domain and self-supervised training in the target domain, effectively achieving domain adaptation. Recently, Lee et al. (2023) further proposed a test-time adaptation framework for domain-generalized category-level object pose estimation. Specifically, they first trained the model using labeled synthetic data and then leveraged the pre-trained model for test-time adaptation in the real world during inference.

To improve the running speed of the object pose estimation method, once the object pose of the first frame is acquired, continuous spatio-temporal information can be utilized to track the object pose. Wang et al. (2020a) proposed

an anchors-based object pose tracking method. They first detected the anchors of each frame as the keypoints, and then solved the relative object pose through the keypoints correspondence. Wen and Bekris (2021) first obtained continuous frame RGBD masks through the video segmentation network, transductive-VOS (Zhang et al., 2020a), and then leveraged LF-Net (Ono et al., 2018) for generalized keypoints detection. Next, they matched keypoints between consecutive frames and performed coarse registration to estimate the initial relative pose. Finally, a memory-augmented pose graph optimization method is proposed for continuous pose tracking.

Overall, these RGBD-guided semantic and geometry fusion methods achieve superior performance. However, if the input depth image contains errors, the accuracy of pose estimation can significantly decrease. Hence, ensuring robustness in pose estimation when dealing with erroneous or missing depth images is crucial.

4.2.3 Others

Since most mobile devices are not equipped with depth cameras Yang et al. (2025a,c); Liu et al. (2025c), Chen et al. (2020c) incorporated a neural synthesis module with a gradient-based fitting procedure to simultaneously predict object shape and pose, achieving monocular object pose estimation. Lee et al. (2021) estimated the NOCS shape and the metric scale shape of the object, and performed a similarity transformation between them to solve the object pose and size. Further, Yen-Chen et al. (2021) inverted neural radiance fields for monocular category-level pose estimation. Different from the previous methods, Lin et al. (2022e) proposed a keypoint-based single-stage pipeline via a single RGB image. Guo et al. (2022) redefined the monocular category-level object pose estimation problem from a long-horizon visual navigation perspective. On the other hand, Ma et al. (2022) enhanced the robustness of the monocular method in occlusion scenes through coarse-to-fine rendering of neural features. Given that transparent instances lack both color and depth information, Zhang et al. (2022a) proposed to utilize depth completion and surface normal estimation to achieve category-level pose estimation for transparent instances. More recently, Wei et al. (2024) decoupled 6D pose from size, leveraged a pretrained monocular estimator for inlier 2D to 3D correspondences and a separate branch for metric scale, and solved pose with RANSAC-PnP, improving rotation accuracy.

In order to improve the running efficiency of the monocular method, Lin et al. (2022d) developed a keypoint-based monocular object pose tracking approach. This approach demonstrates the significance of integrating uncertainty estimation using a tracklet-conditioned deep network and probabilistic filtering. Following Lin et al. (2022d), Yu et al.

一个对象类别的典范表示。他们首先预测了类内未见对象的类别标签、掩码及 NOCS 图。随后, 利用 Umeyama 算法 (Umeyama, 1991) 通过将 NOCS 图与对象点云对齐来求解对象姿态。为应对类内对象的多种形状变化, Chen 等人 (2020a) 学习了一个作为统一表示的典范形状空间。另一方面, Lin 等人 (2021a) 探索了稀疏可操控卷积 (SSC) 在对象姿态估计中的适用性, 并提出了一种基于 SSC 的流程。此外, Lin 等人 (2021b) 提出了一种双姿态网络, 由共享姿态编码器和两个并行显式与隐式姿态解码器组成。隐式解码器可在推理时无 CAD 模型存在时强制预测姿态一致性。Wang 等人 (2022a) 设计了一种具有关系感知与结构感知能力的注意力引导网络, 用于 RGB 图像与点云特征融合。最近, Liu 等人 (2023c) 探讨了形状先验在类内未见物体形状重建中的必要性。研究证明变形过程比形状先验更为关键, 并提出了一种无需先验的隐式空间变换网络。Lin 等人 (2023a) 通过将旋转估计分解为视角旋转和平面内旋转, 有效提升了旋转估计精度; 同时他们还提出了基于空间球面卷积的球面特征金字塔网络, 用于处理球面信号。随着大视觉模型 (LVM) 的快速发展, Chen 等人 (2024) 进一步利用 LVM DINOv2 (Oquab 等人, 2023) 提取 SE(3) 一致性语义特征, 并将其与对象特定的层次化几何特征融合, 从而封装类别级信息用于旋转估计。

由于上述方法仍需要大量真实标注的训练数据, 其在现实场景中的适用性存在局限。为此, Peng 等人 (2022) 提出了一种基于深度隐式形状表征的现实现场景自监督训练框架。他们采用深度符号距离函数 (Park 等人, 2019a) 作为三维表征手段, 实现了从合成场景到真实场景的领域适应。此外, Lee 等人 (2022) 引入了师生式自监督学习机制, 通过在源域进行监督训练、在目标域进行自监督训练的方式, 有效实现了领域适应。近期, Lee 等人 (2023) 进一步提出了用于领域泛化类别级物体姿态估计的测试时适应框架。具体而言, 他们首先使用标注合成数据训练模型, 随后在推理阶段利用预训练模型进行现实场景中的测试时适应。

为提升物体姿态估计方法的运行速度, 一旦获取第一帧的物体姿态, 即可利用连续时空信息进行物体姿态跟踪。Wang 等人 (2020a) 提出了一种基于锚点的

物体姿态跟踪方法: 首先将每帧的锚点作为关键点进行检测, 随后通过关键点对应关系求解相对物体姿态。Wen 与 Bekris (2021) 首先通过视频分割网络和转导 VOS (Zhang 等人, 2020a) 获取连续帧 RGBD 掩码, 继而利用 LF-Net (Ono 等人, 2018) 进行通用关键点检测。随后通过匹配连续帧间关键点并进行粗略配准来估计初始相对姿态, 最终提出一种基于记忆增强的姿态图优化方法实现连续姿态跟踪。

总体而言, 这些基于 RGBD 的语义与几何融合方法具有优异性能。然而, 若输入深度图像存在误差, 姿态估计精度将显著下降。因此, 在处理存在误差或缺失深度图像时, 确保姿态估计的鲁棒性至关重要。

4.2.3 其他

由于大多数移动设备并未配备深度摄像头 (Yang 等人 2025a、c; Liu 等人 2025c; Chen 等人 2020c), 研究者采用了基于梯度拟合的神经合成模块, 同时预测物体形状与姿态, 实现了单目物体姿态估计。Lee 等人 2021 估算了 NOCS 形状与物体度量尺度形状, 并通过二者间的相似性变换求解物体姿态与尺寸。此外, Yen-Chen 等人 2021 通过逆向神经辐射场实现了单目类别级姿态估计。与先前方法不同, Lin 等人 2022e 提出了一种基于关键点的单阶段流程, 仅需单张 RGB 图像即可完成。Guo 等人 2022 从长时程视觉导航视角重新定义了单目类别级物体姿态估计问题。另一方面, Ma 等人 2022 通过神经特征的粗到细渲染提升了单目方法在遮挡场景下的鲁棒性。鉴于透明物体既缺乏颜色信息又缺失深度信息, Zhang 等人 (2022a) 提出通过深度补全与表面法线估计技术, 实现透明物体的类别级姿态估计。近期, Wei 团队 (2024) 将 6D 姿态与物体尺寸解耦, 采用预训练的单目估计器进行点内 2D 到 3D 对应关系建模, 并设置独立分支处理度量尺度问题, 最终通过 RANSAC -PnP 算法实现姿态解算, 显著提升了旋转精度。

为提升单目方法的运行效率, Lin 等人 (2022d) 开发了一种基于关键点的单目物体姿态跟踪方法。该方法通过结合基于轨迹片段条件的深度网络进行不确定性估计与概率滤波, 充分展现了其重要性。沿袭 Lin 等人 (2022d) 的研究思路, Yu 等人

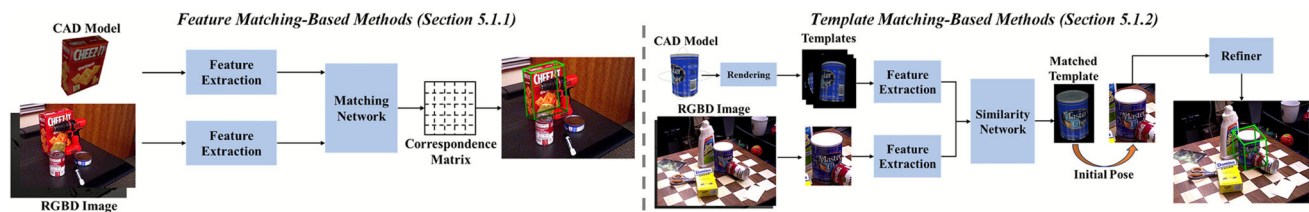


Fig. 10 Illustration of the CAD model-based methods (Sec. 5.1) for unseen object pose estimation: The feature matching-based methods (Sec. 5.1.1) focus on designing a network to match features between the CAD model and the query image, establishing correspondences (2D-3D or 3D-3D), and solving the pose using the PnP algorithm or least squares

method. The template matching-based methods (Sec. 5.1.2) utilize rendered templates from the CAD model for retrieval. The initial pose is acquired based on the most similar template, and further refinement is necessary using a refiner to obtain a more accurate pose

(2023c) further improved the pose tracking accuracy through a network that combines convolutions and transformers.

To further improve the generalization of category-level methods, Goodwin et al. (2022) introduced a reference image-based zero-shot approach, which first extracts spatial feature descriptors and builds cyclical descriptor distances. Then, they established the top-k semantic correspondences for pose estimation. Zaccaria and Manhardt (2023) proposed a self-supervised framework via optical flow consistency. Very recently, Cai et al. (2024b) developed an open-vocabulary framework that aims to generalize to unseen categories using textual prompts in unseen scene images. Di Felice et al. (2024) explored zero-shot novel view synthesis based on a diffusion model for 3D object reconstruction, and recovered the object pose through correspondences. Lin et al. (2024c) used a pre-trained vision-language model to make full use of rich semantic knowledge and align the representations of the three modalities (image, point cloud, and text) in the feature space through multi-modal contrastive learning.

Brief Discussion: On the whole, these shape prior-free methods circumvent the reliance on shape priors and further improve the generalization ability of category-level object pose estimation methods. Nevertheless, these methods are limited to generalizing within intra-class unseen objects. For objects of different categories, additional training data must be collected and the model must be retrained. This remains a significant limitation.

5 Unseen Object Pose Estimation

Unseen object pose estimation methods can further generalize to object categories not encountered during training, without retraining the model. Point Pair Features (PPF) (Drost et al., 2010) is a classical method for unseen object pose estimation that utilizes oriented point pair features to build global model description and a fast voting scheme to match locally. The final pose is solved by pose clustering

and iterative closest point (Besl & McKay, 1992) refinement. However, PPF suffers from low accuracy and slow runtime, limiting its applicability. In contrast, deep learning-based methods leverage neural networks to learn more complex features from data without specifically designed feature engineering, thus enhancing accuracy and efficiency. In this section, we review the deep learning-based unseen object pose estimation methods and classify them into CAD model-based (Sec. 5.1) and manual reference view-based (Sec. 5.2) methods. The illustration of these two categories of methods is shown in Fig. 10 and Fig. 11, respectively.

5.1 CAD Model-Based Methods

The CAD model-based methods involve utilizing the object CAD model as prior knowledge during the process of estimating the pose of an unseen object. These methods can be further categorized into feature matching-based (Sec. 5.1.1) and template matching-based (Sec. 5.1.2) methods. The illustration of these two categories of methods is shown in Fig. 10. The characteristics and performance of some representative methods are shown in Table 3.

5.1.1 Feature Matching-Based Methods

Feature matching-based methods focus on designing a network to match features between the CAD model and the query image, establishing 2D-3D or 3D-3D correspondences, and solving the pose by the PnP algorithm or least squares method. As an early exploratory work, Pitteri et al. (2019) proposed a 3DoF pose estimation approach that approximates object's geometry using only the corner points of the CAD model. Nonetheless, it only works effectively on objects having specific corners. Hence, Pitteri et al. (2020) further introduced an embedding that captures the local geometry of 3D points on the object surface. Matching these embeddings can create 2D-3D correspondences, and the pose is then determined using the PnP+RANSAC (Fischler &

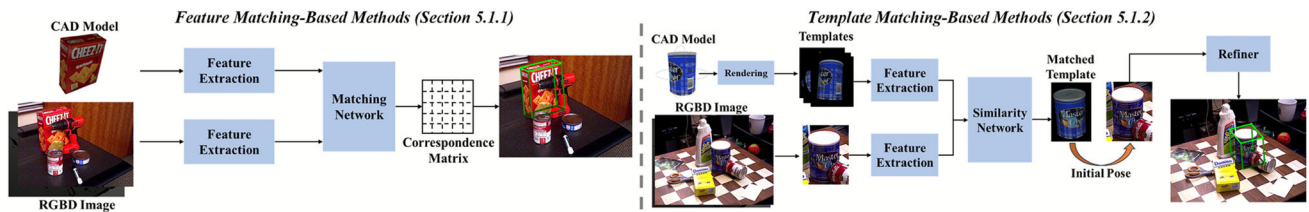


图 10基于 CAD 模型的方法（第5.1节）用于未知物体姿态估计的示意图：基于特征匹配的方法（第5.1.1节）侧重于设计一个网络，用于匹配 CAD 模型与查询图像之间的特征，建立对应关系（2D-3D 或 3D-3D），并使用 PnP 算法或最小二乘法求解姿态。

（2023c）通过结合卷积层与Transformer网络结构，进一步提升了姿态跟踪精度。

为进一步提升类别级方法的泛化能力，Goodwin 等人（2022）提出了一种基于参考图像的零样本方法，该方法首先提取空间特征描述符并构建循环描述符距离，随后建立用于姿态估计的top-k语义对应关系。Zaccaria 与 Manhardt（2023）通过光流一致性提出了自监督框架。近期，Cai 等人（2024b）开发了开放词汇框架，旨在利用未见场景图像中的文本提示实现对未见类别的泛化。Di Felice 等人（2024）基于扩散模型探索了零样本新视角合成用于3D物体重建，并通过对对应关系恢复物体姿态。Lin 等人（2024c）采用预训练视觉-语言模型充分利用丰富语义知识，并通过多模态对比学习在特征空间中对齐图像、点云和文本三种模态的表征。

简要讨论：总体而言，这些无需形状先验的方法规避了对形状先验的依赖，进一步提升了类别级物体姿态估计方法的泛化能力。然而，这些方法仅适用于类内未见物体的泛化场景。对于不同类别的物体，必须收集额外训练数据并对模型进行重新训练，这仍是其显著局限性所在。

5 未见物体姿态估计

未见过物体姿态估计方法能够进一步泛化到训练过程中未遇到的物体类别，而无需重新训练模型。点对点特征（PPF）（Drost 等人，2010）是一种经典的未见过物体姿态估计方法，它利用定向点对特征构建全局模型描述，并采用快速投票方案进行局部匹配。最终姿态通过姿态聚类 and 迭代最近点（Besl & McKay，

1992）精化得到。然而，PPF 存在精度较低和运行速度慢的问题，限制了其适用性。相比之下，基于深度学习的方法利用神经网络从数据中学习更复杂的特征，无需专门设计特征工程，从而提升精度和效率。本节将综述基于深度学习的未见过物体姿态估计方法，并将其分为基于CAD模型的方法（第5.1节）和基于人工参考视图的方法（第5.2节）。这两类方法的示意图分别如图10和图11所示。

5.1 基于CAD模型的方法

基于CAD模型的方法是在估计未见物体姿态的过程中，利用该物体的CAD模型作为先验知识。这些方法可进一步分为基于特征匹配的方法（第5.1.1节）和基于模板匹配的方法（第5.1.2节）。这两类方法的示意图如图10所示；部分代表性方法的特性与性能详见表3。

5.1.1 基于特征匹配的方法

基于特征匹配的方法主要通过设计网络来实现CAD模型与查询图像之间的特征匹配，建立二维-三维或三维-三维对应关系，并采用PnP算法或最小二乘法求解姿态。作为早期探索性研究，Pitteri 等人（2019）提出了一种仅利用CAD模型角点近似物体几何结构的三维自由度姿态估计方法。然而该方法仅对具有特定角点的物体有效。为此，Pitteri 团队（2020）进一步引入了能捕捉物体表面三维点局部几何特征的嵌入向量。通过匹配这些嵌入向量可建立二维-三维对应关系，最终采用PnP+ RANSAC（Fischler &

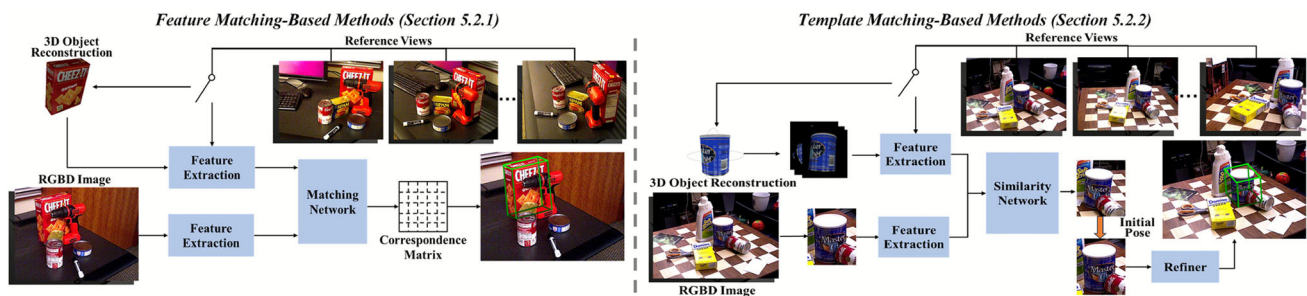


Fig. 11 Illustration of the manual reference view-based methods (Sec. 5.2) for unseen object pose estimation: There are two types of feature matching-based methods (Sec. 5.2.1). One involves extracting features from reference views and the query image, obtaining 3D-3D correspondences through a feature matching network. The other initially reconstructs the 3D object representation using the reference views and establishes the 2D-3D correspondences between the query image and the 3D representation. The object pose is solved using correspondence-

based algorithms, like PnP or the least squares method. Template matching-based methods (Sec. 5.2.2) also have two types. One reconstructs the 3D object representation using the reference views and then renders multiple templates. The initial pose is acquired by retrieving the most similar template and then refining it to get the final pose. The other directly uses the reference views as the templates for template matching

Bolles, 1981) algorithm. However, these methods only estimate the 3DoF pose.

Gou et al. (2022) defined the challenge of estimating the 6DoF pose of unseen objects, offering a baseline solution through the identification of 3D correspondences between object and scene point clouds. Similarly, Hagelskjær and Haugaard (2023) trained a network to match keypoints from the CAD model to the object point cloud. Yet, it focuses on bin picking with homogeneous bins, which only demonstrates that generalized pose estimation can achieve outstanding performance in restricted scenarios.

Inspired by point cloud registration methods on unseen objects, Zhao et al. (2023) proposed a geometry correspondence-based method using generic and object-agnostic geometry features to establish unambiguous and robust 3D-3D correspondences. Nevertheless, it still needs to get the class label and segmentation mask of unseen objects through other methods such as Mask-RCNN (He et al., 2017). To this end, Chen et al. (2023b) explored a framework named ZeroPose, which realizes joint instance segmentation and pose estimation of unseen objects. Specifically, they utilized the foundation model SAM (Kirillov et al., 2023) to generate possible object proposals and adopted a template matching method to accomplish instance segmentation. After that, they developed a hierarchical geometric feature matching network based on GeoTransformer (Qin et al., 2022b) to establish correspondences. Following ZeroPose, Lin et al. (2024a) devised a novel matching score in terms of semantics, appearance, and geometry to obtain better segmentation. As for pose estimation, they proposed a two-stage partial-to-partial point matching model to construct dense 3D-3D correspondence effectively.

Besides these methods employing geometry features, Caraffa et al. (2024) devised a method that fuses visual and

geometric features extracted from different pre-trained models to enhance pose prediction stability and accuracy. It is the first technique to estimate the unseen object pose by utilizing the synergy between geometric and vision foundation models. Additionally, Huang et al. (2024) proposed a method for object pose prediction from RGBD images by combining 2D texture and 3D geometric cues.

To sum up, feature matching-based methods aim to extract generic object-agnostic features and achieve strong correspondences by matching these features. However, these methods require not only robust feature matching models but also tailored designs to enhance the representation of object features, presenting a significant challenge.

5.1.2 Template Matching-Based Methods

Template matching has been widely used in computer vision and stands as an effective solution for tackling the pose estimation challenges posed by unseen objects. Specifically, they utilize rendered templates from the CAD model for retrieval. The initial pose is acquired based on the most similar template, and further refinement is necessary using a refiner to obtain a more accurate pose. Wohlhart and Lepetit (2015) and Balntas et al. (2017) were pioneers in using deep pose descriptors for object matching and pose retrieval. However, their descriptors are tailored to specific orientations and categories, limiting their utility to objects with similar appearances. In contrast, Sundermeyer et al. (2020) proposed a single-encoder-multi-decoder network for jointly estimating the 3D rotation of multiple objects. This approach eliminates the need to segregate views of different objects in the latent space and enables the sharing of common features in the encoder. Yet, it still requires training multiple decoders. Wen et al. (2022) addressed this problem by decoupling

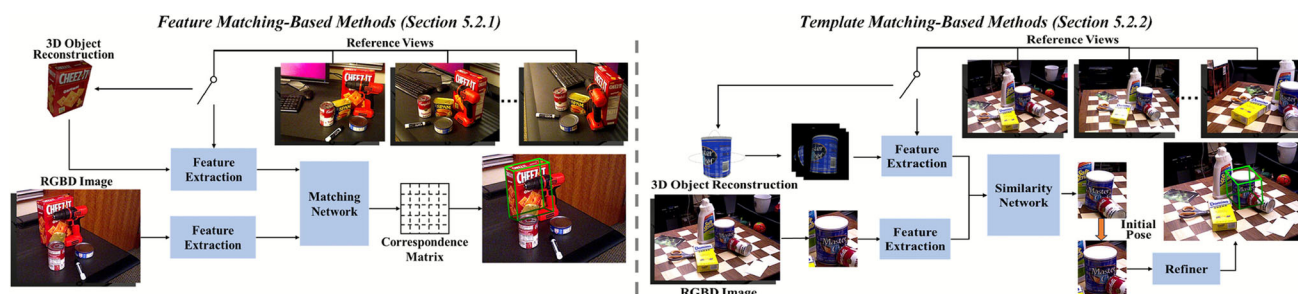


图 11 手动参考视图方法（第5.2节）用于未知物体姿态估计的示意图：基于特征匹配的方法（第5.2.1节）可分为两类。第一类通过从参考视图和查询图像中提取特征，利用特征匹配网络获得三维对应关系；第二类首先利用参考视图重建物体的三维表征，再建立查询图像与三维表征之间的二维-三维对应关系，最终采用基于对应关系的算法（如PnP或最小二乘法）求解物体姿态。基

Bolles（1981）算法。然而，这些方法仅能估计三自由度（3DoF）位姿。

Gou 等人（2022）针对未知物体六自由度姿态估计难题提出解决方案，通过建立物体与场景点云间的三维对应关系构建基准模型。Hagelskjær 与 Haugaard（2023）则训练神经网络实现CAD模型关键点与物体点云的匹配，但该方法仅针对同质化箱体的箱体拾取任务，这表明通用姿态估计技术在特定场景下仍能取得优异性能。

受未见物体点云配准方法的启发，Zhao 等人（2023）提出了一种基于几何对应的方法，利用通用且与物体无关的几何特征来建立明确且稳健的3D-3D对应关系。然而，该方法仍需通过 Mask-RCNN（Heetal., 2017）等其他方法获取未见物体的类别标签和分割掩膜。为此，Chen 等人（2023b）探索了名为 ZeroPose 的框架，实现了未见物体的联合实例分割与姿态估计。具体而言，他们利用基础模型 SAM（Kirillov 等人，2023）生成可能的物体候选，并采用模板匹配方法完成实例分割。随后，他们基于 GeoTransformer（Qin 等人，2022b）开发了分层几何特征匹配网络以建立对应关系。继 ZeroPose 之后，Lin 等人（2024a）提出了一种结合语义、外观与几何特征的新型匹配评分机制，以获得更优的分割效果。在姿态估计方面，他们提出了一种两阶段部分到部分点匹配模型，能够有效构建密集的三维空间对应关系。

除了这些基于几何特征的方法外，Caraffa 等人（2024）提出了一种创新方法，通过融合不同预训练模型提取的视

于模板匹配的方法（第5.2.2节）同样分为两类：第一类先利用参考视图重建物体三维表征，再生成多个模板；初始姿态通过检索最相似模板并优化后获得最终姿态；第二类则直接将参考视图作为模板进行模板匹配。

觉与几何特征，显著提升了姿态预测的稳定性和准确性。该技术首次利用几何模型与视觉基础模型之间的协同效应，实现了对未见物体姿态的估计。此外，Huang 等人（2024）还开发了基于 RGBD 图像的物体姿态预测方法，通过整合二维纹理特征与三维几何线索实现精准预测。

综上所述，基于特征匹配的方法旨在提取通用且与对象无关的特征，并通过特征匹配实现强对应关系。然而，这些方法不仅需要稳健的特征匹配模型，还需通过定制化设计来增强对象特征的表征能力，这构成了重大技术挑战。

5.1.2 基于模板匹配的方法

模板匹配技术在计算机视觉领域应用广泛，是应对未见物体带来的姿态估计挑战的有效解决方案。具体而言，该技术利用CAD模型生成的渲染模板进行检索。初始姿态基于最相似模板获取，随后需通过优化器进行进一步精炼以获得更精确的姿态。Wohlhart 与 Lepetit（2015）以及 Balntas 等人（2017）率先将深度姿态描述符应用于物体匹配与姿态检索。然而其描述符仅适用于特定方向和类别，导致效用受限外观相似的物体。相比之下，Sundermeyer 等人（2020）提出单编码器多解码器网络，用于联合估计多个物体的三维旋转。该方法无需在潜在空间中分离不同物体视角，且编码器可共享共同特征，但仍需训练多个解码器。Wen 等人（2022）通过解耦机制解决了这一问题。

Table 3 Representative CAD-based methods. Since the domain training paradigm of most unseen object pose estimation methods is domain generalization, we report 9 properties for each method, which have the same meanings as described in the caption of Table 1. D, S, F, T, P, R, and V denote object detection, instance segmentation, feature matching to build correspondences, template matching to retrieve pose, pose solution/regression, pose refinement, and pose voting, respectively. We report the *BOP-M* across the LM-O and YCB-V datasets (Sec. 2) for various methods. Notably, these methods use Mask-RCNN (He et al., 2017) (normal font), CNOS (Nguyen et al., 2023) or their own proposed methods (Chen et al., 2023b; Lin et al., 2024a; Shugurov et al., 2022) (**bold**), and a combination of PPF and SIFT (Okorn et al., 2021) (*italics*) for unseen object location, respectively. Moreover, Örnek et al. (2023) and Caraffa et al. (2024) don't require any task-specific training, we use "x" to denote it

Methods	Published Year	Training Input	Inference Input	Pose DoF	Object Property	Task	Inference Mode	Application Area	LM-O <i>M</i>	BOP- <i>M</i>	YCB-V BOP- <i>M</i>
CAD Model-Based Methods											
Pitteri et al. (2020)	2020	RGB, CAD Model	RGB, CAD Model	3DoF	rigid	estimation	three-stage, S+F+P	general	-	-	-
Zhao et al. (2023)	2023	Depth, CAD Model	Depth, CAD Model	6DoF	rigid	estimation	three-stage, S+F+P	general	65.2	-	-
Chen et al. (2023b)	2023	RGBD, CAD Model	RGBD, CAD Model	6DoF	rigid	estimation	four-stage, S+F+P+R	general	49.1	57.7	57.7
Lin et al. (2024a)	2024	RGBD, CAD Model	RGBD, CAD Model	6DoF	rigid	estimation	three-stage, S+F+P	general	69.9	84.5	84.5
Huang et al. (2024)	2024	RGBD, CAD Model	RGBD, CAD Model	6DoF	rigid	estimation	three-stage, S+F+P	general	56.2	60.8	60.8
Caraffa et al. (2024)	2024	x	RGBD, CAD Model	6DoF	rigid	estimation	four-stage, S+F+P+R	general	69.0	85.3	85.3
Sundermeyer et al. (2020)	2020	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	three-stage, D/S+T+R	general	-	-	-
Okorn et al. (2021)	2021	RGBD, CAD Model	RGBD, CAD Model	6DoF	rigid	estimation	two-stage, P+V	general	59.8	51.6	51.6
Shugurov et al. (2022)	2022	RGB/RGBD, CAD Model	RGB/RGBD, CAD Model	6DoF	rigid	estimation	three-stage, S+T+P	general	46.2	54.2	54.2
Nguyen et al. (2022b)	2022	RGB, CAD Model	RGB, CAD Model	3DoF	rigid	estimation	two-stage, D+T	occlusion	-	-	-
Labbé et al. (2022)	2022	RGB/RGBD, CAD Model	RGB/RGBD, CAD Model	6DoF	rigid	estimation, refinement	three-stage, D+T+R	general	58.3	63.3	63.3
Örnek et al. (2023)	2023	x	RGB, CAD Model	6DoF	rigid	estimation	four-stage, S+T+P+R	general	61.0	69.0	69.0
Nguyen et al. (2024b)	2024	RGB, CAD Model	RGB, CAD Model	6DoF	rigid	estimation	four-stage, D+T+P+R	general	63.1	65.2	65.2
Moon et al. (2024)	2024	RGB/RGBD, CAD Model	RGB/RGBD, CAD Model	6DoF	rigid	refinement	two-stage, D+R	general	62.9	82.5	82.5
Wang et al. (2024)	2024	RGB, CAD Model	RGB, CAD Model	3DoF	rigid	estimation	two-stage, D+T	general	-	-	-
Wen et al. (2024)	2024	RGBD, CAD Model	RGBD, CAD Model	6DoF	rigid	estimation, tracking	four-stage, D+T+R+V	general	78.8	88.0	88.0

方法	发布年份	训练输入	推理输入	姿势自由度	对象属性	任务	推理模式	应用区域	LM-O M	BOP	YCB-V BOP M
CAD模型 所用方法	特征匹配	Pitrii 等 (2020)	2020	RGB, CAD模型	RGB, CAD模型	3自由度	严格的	评价	三阶段S+F+P	一般的	-
	模版 匹配	Zhao等 (2023)	2023	深度, CAD 模型	深度, CAD 模型	6自由度	严格的	评价	三阶段的 S+F+P	一般的	-
		陈等人 (2023b)	2023	RGBD, CAD 模型	RGBD, CAD 模型	6自由度	严格的	评价	四级 S+F+P+R	一般的	57.7
		林 乙 等 (2024a)	2024	RGBD, CAD 模型	RGBD, CAD 模型	6自由度	严格的	评价	三阶段的 S+F+P	一般的	84.5
		Huang 等 (2024)	2024	RGBD, CAD 模型	RGBD, CAD 模型	6自由度	严格的	评价	三阶段的 S+F+P	一般的	60.8
		Caraffa等人 (2024)	2024	RGBD, CAD 模型	RGBD, CAD 模型	6自由度	严格的	评价	四级 S+F+P+R	一般的	85.3
		松德迈尔 等 (2020)	2020	RGB, CAD 模型	RGB, CAD 模型	6自由度	严格的	评价	三阶段的 D/S+T+R	一般的	-
		Okoni 等 (2021)	2021	RGBD, CAD 模型	RGBD, CAD 模型	6自由度	严格的	评价	二阶段 P+V	一般的	51.6
		Shugurov等 al. (2022)	2022	RGB/RGBD, CAD模型	RGB/RGBD, CAD模型	6自由度	严格的	评价	三阶段的 S+T+P	一般的	54.2
		Nguyen等 (2022b)	2022	RGB, CAD 模型	RGB, CAD 模型	3自由度	严格的	评价	二阶段 D+T	闭塞	-
		Labbe等 (2022)	2022	RGB/RGBD, CAD模型	RGB/RGBD, CAD模型	6自由度	严格的	评价 精炼	三阶段的 D+T+R	一般的	63.3
		Ornek 等 (2023)	2023	RGB/RGBD, CAD模型	RGB, CAD 模型	6自由度	严格的	评价	四级 S+T+P+R	一般的	69.0
		Nguyen等 (2024b)	2024	RGB, CAD 模型	RGB, CAD 模型	6自由度	严格的	评价	四级 D+T+P+R	一般的	65.2
		Moon等人 (2024)	2024	RGB/RGBD, CAD模型	RGB/RGBD, CAD模型	6自由度	严格的	精炼	二阶段 D+R	一般的	82.5
		Wang 等 (2024)	2024	RGB, CAD 模型	RGB, CAD 模型	3自由度	严格的	评价	二阶段 D+T	一般的	-
		Wen 等 (2024)	2024	RGBD, CAD 模型	RGBD, CAD 模型	6自由度	严格的	评价 跟踪	四级 D+T+R+V	一般的	88.0

object shape and pose in the latent representation, enabling auto-encoding without the necessity of multi-path decoders for different objects, thus enhancing scalability.

Instead of training a network to learn features across objects, Okorn et al. (2021) first generated candidate poses by PPF (Drost et al., 2010) and projected each into the scene. Later, they designed a scoring network to evaluate the hypothesis by comparing color and geometry differences between the projected object point cloud and RGBD image. Busam et al. (2020) reformulated 6DoF pose retrieval as an action decision process and determined the final pose by iteratively estimating probable movements. Cai et al. (2022a) retrieved various candidate viewpoints from a target object viewpoint codebook, and then conducted in-plane 2D rotational regression on each retrieved viewpoint to obtain a set of 3D rotation estimates. These estimates were evaluated using a consistency score to generate the final rotation prediction. Meanwhile, Shugurov et al. (2022) matched the detected objects with the rendering database for initial viewpoint estimation. Then, they predicted the dense 2D-2D correspondences between the template and the image via feature matching. Pose estimation was eventually performed by using PnP+RANSAC (Fischler & Bolles, 1981) or Kabsch (Kabsch, 1976)+RANSAC.

Since estimating the full 6DoF pose of an unseen object is extremely challenging, some works focus on estimating the 3D rotation to simplify it. Different from the previous works (Wohlhart and Lepetit, 2015; Balntas et al., 2017; Corona et al., 2018; Sundermeyer et al., 2018) that exploited a global image representation to measure image similarity, Nguyen et al. (2022b) used CNN-extracted local features to compare the similarity between the input image and templates, showing better property and occlusion robustness over global representation. Another noteworthy approach is an image retrieval framework based on multi-scale local similarities developed by Zhao et al. (2022). They extracted feature maps of various sizes from the input image and devised a similarity fusion module to robustly predict image similarity scores from multi-scale pairwise feature maps. Further, Thalhammer et al. (2023) and Ausserlechner et al. (2023) extended the scheme of Nguyen et al. (2022b) and demonstrated that the pre-trained Vision-Transformer (ViT) (Dosovitskiy et al., 2021) outperforms task-specific fine-tuned CNN (LeCun et al., 1989) for template matching. However, these methods still have a noticeable performance gap between seen and unseen objects. To this end, Wang et al. (2024) introduced diffusion features that show great potential in modeling unseen objects. Furthermore, they designed three aggregation networks to efficiently capture and aggregate diffusion features at different granularities, thus improving its generalizability.

In order to further improve the generalization and robustness of the 6DoF pose estimation, Labbé et al. (2022) used a render-and-compare approach and a coarse-to-fine strat-

egy. Notably, they leveraged a large-scale 3D model dataset to generate a synthetic dataset containing 2 million images and over 20,000 models. It achieved strong generalization by training the network on this dataset. Compared to the non-differentiable rendering pipeline of Labbé et al. (2022), Tremblay et al. (2023) utilized recent advancements in differentiable rendering to design a flexible refiner, allowing fine-tuning the setup without retraining. On the other hand, Moon et al. (2024) presented a shape-constraint recurrent flow framework, which predicts the optical flow between the template and query image and refines the pose iteratively. It took advantage of shape information directly to improve the accuracy and scalability. Recently, Wen et al. (2024) inherited the idea of Labbé et al. (2022) and developed a novel synthesis data generation pipeline using emerging large-scale 3D model databases, Large Language Models (LLMs), and diffusion models. It greatly expanded the amount and diversity of data, and ultimately achieved comparable results to instance-level methods in a render-and-compare manner.

It is well known that template matching methods are sensitive to occlusions and require considerable time to match numerous templates. Therefore, Nguyen et al. (2024b) achieved rapid and robust pose estimation by finding the suitable trade-off between the use of template matching and patch correspondences. In particular, the features of the query image and templates are extracted using the ViT (Dosovitskiy et al., 2021), followed by fast template matching using a sub-linear nearest neighbor search. The most similar template provides two DoFs for azimuth and elevation, while the remaining four DoFs are obtained by constructing correspondences between the query image and this template. Örnek et al. (2023) utilized DINOv2 (Oquab et al., 2023) to extract descriptors for the query image and templates. Moreover, they introduced a fast template retrieval method based on visual words constructed from DINOv2 patch descriptors, thereby decreasing the reliance on extensive data and enhancing matching speed compared to Labbé et al. (2022).

In summary, template matching-based methods make full use of the advantages provided by a multitude of templates, enabling high accuracy and strong generalization. Nonetheless, they have limitations in terms of time consumption, sensitivity to occlusions, and challenges posed by complex backgrounds and lighting variations.

Brief Discussion: The above-mentioned feature matching-based or template matching-based methods both require a CAD model of the target object to provide prior information. In practice, accurate CAD models often require specialized hardware to build, which limits the practical application of these methods to a certain extent.

在潜在表征中保留物体形状与姿态信息,使得无需为不同物体配置多路径解码器即可实现自动编码,从而提升系统可扩展性。

不同于训练网络学习跨物体特征的方法,Okorn等人(2021)首先通过PPF(Drost等人,2010)生成候选姿态,并将每个姿态投影到场景中。随后,他们设计了一个评分网络,通过比较投影物体点云与RGBD图像之间的颜色和几何差异来评估假设。Busam等人(2020)将6DoF姿态检索重新表述为动作决策过程,并通过迭代估计可能运动来确定最终姿态。Cai等人(2022a)从目标物体视角码本中检索多种候选视角,随后对每个检索视角进行平面内二维旋转回归以获得一组三维旋转估计值。这些估计值通过一致性评分进行评估以生成最终旋转预测。与此同时,Shugurov等人(2022)将检测到的物体与渲染数据库进行匹配以进行初始视角估计,随后通过特征匹配预测模板与图像之间的密集二维-二维对应关系。姿态估计最终采用PnP+RANSAC算法(Fischler & Bolles, 1981)或Kabsch算法(Kabsch, 1976)+RANSAC算法完成。

由于估计未见物体的完整6自由度姿态极具挑战性,部分研究聚焦于估计三维旋转以简化问题。与先前利用全局图像表示度量图像相似性的研究(Wohlhart和Lepetit, 2015; Balntas等人, 2017; Corona等人, 2018; Sundermeyer等人, 2018)不同,Nguyen等人(2022b)采用CNN提取的局部特征来比较输入图像与模板间的相似性,相较于全局表示展现出更优的属性鲁棒性和遮挡鲁棒性。另一值得注意的方法是Zhao等人(2022)提出的基于多尺度局部相似性的图像检索框架。他们从输入图像中提取不同尺寸的特征图,并设计了相似性融合模块以从多尺度成对特征图中稳健预测图像相似度评分。此外,Thalhammer等人(2023)与Ausserlechner等人(2023)扩展了Nguyen等人(2022b)的方案,证明了预训练视觉Transformer(ViT)(Dosovitskiy等人(2021)在模板匹配任务中的表现优于针对特定任务微调的CNN(LeCun等人, 1989)。然而,这些方法在处理已见过与未见过物体时仍存在显著性能差距。为此,Wanget等人(2024)提出了扩散特征,该方法在建模未见过物体方面展现出巨大潜力。此外,他们设计了三种聚合网络,能够高效捕获并整合不同粒度的扩散特征,从而提升了模型的泛化能力。

为了进一步提升6DoF姿态估计的泛化能力和鲁棒性,Labbe等人(2022)采用了渲染与比较相结合的方法,并采用了从粗到细的分层策略。值得注意的是,他

们利用大规模3D模型数据集生成了包含200万张图像和超过20,000个模型的合成数据集,通过在此数据集上训练网络实现了强大的泛化能力。与Labbe等人(2022)采用的不可微分渲染流程相比,Tremblay等人(2023)利用可微分渲染技术的最新进展设计了一种灵活的优化器,无需重新训练即可对模型进行微调。另一方面,Moon等人(2024)提出了基于形状约束的循环流框架,该框架能够预测模板图像与查询图像之间的光流,并通过迭代方式优化姿态估计结果,直接利用形状信息提升了精度和可扩展性。近期,Wenet等人(2024)继承了Labbe等人的研究思路(2022)并开发了一种创新的合成数据生成流程,该流程利用新兴的大规模3D模型数据库、大型语言模型(LLM)和扩散模型。该方法显著扩展了数据量与多样性,最终通过渲染与对比的方式,取得了与实例级方法相当的效果。

众所周知,模板匹配方法对遮挡敏感且需要大量时间来匹配大量模板。因此,Nguyen等人(2024b)通过在模板匹配与图像块对应关系之间找到合适的平衡点,实现了快速且稳健的姿态估计。具体而言,首先利用ViT(Dosovitskiy等人, 2021)提取查询图像与模板的特征,随后采用次线性最近邻搜索进行快速模板匹配。最相似的模板可提供方位角和仰角两个自由度,其余四个自由度则通过构建查询图像与该模板之间的对应关系获得。Ornek等人(2023)采用DINOv2(Oquab等人, 2023)提取查询图像与模板的描述符,并引入基于DINOv2图像块描述符构建视觉词汇的快速模板检索方法,相较于Labbe等人(2022)显著降低了对海量数据的依赖性并提升了匹配速度。

综上所述,基于模板匹配的方法充分利用了多种模板的优势,能够实现高精度和强泛化能力。然而,该方法在时间消耗、遮挡敏感性以及复杂背景与光照变化带来的挑战方面仍存在局限性。

简要讨论:上述基于特征匹配或模板匹配的方法均需目标物体的CAD模型作为先验信息。然而实际应用中,精确的CAD模型往往需要专用硬件支持,这在一定程度上制约了这些方法的实际应用范围。

5.2 Manual Reference View-Based Methods

Aside from these CAD model-based approaches, there are some manual reference view-based methods that do not require the unseen object CAD model as a prior condition but instead require providing some manual labeled reference views with the target object. Similar to CAD model-based methods, these methods are also categorized into two types: feature matching-based (Sec. 5.2.1) and template matching-based (Sec. 5.2.2) methods. These two categories of methods are illustrated in Fig. 11. The attributes and performance of some representative methods are shown in Table 4.

5.2.1 Feature Matching-Based Methods

Different from CAD model-based feature matching methods, manual reference view-based feature matching methods primarily establish 3D-3D correspondences between the RGBD query image and RGBD reference images, or 2D-3D correspondences between the query image and sparse point cloud reconstructed by reference views. Subsequently, the object pose is solved according to the different correspondences. He et al. (2022c) proposed the first few-shot 6DoF object pose estimation method, which can estimate the pose of an unseen object by a few support views without extra training. Specifically, they designed a dense RGBD prototype matching framework based on transformers to fully explore the semantic and geometric relationship between the query image and reference views. Corsetti et al. (2024) used a textual prompt for object segmentation and reformulated the problem as a relative pose estimation between two scenes. The relative pose was obtained via point cloud registration.

Some methods took an alternative route from the perspective of matching after reconstruction. Wu et al. (2021b) developed a global registration-based method that used reference and query images to reconstruct full-view and single-view models, and then searched for point matches between the two models. Sun et al. (2022b) drew inspiration from visual localization and revised the pipeline to adapt it for pose estimation. More precisely, they reconstructed a Structure from Motion (SfM) model of the unseen object using RGB sequences from all reference viewpoints. Then, they matched 2D keypoints in the query image with the 3D points in the SfM model by a graph attention network. Nevertheless, it performed poorly on low-textured objects because of its reliance on repeatably detected keypoints. To deal with this problem, He et al. (2022a) designed a new keypoint-free SfM method to reconstruct semi-dense point cloud models of low-textured objects based on the detector-free feature matching method LoFTR (Sun et al., 2021). Castro and Kim (2023) pointed out that these pre-trained feature matching models (Sarlin et al., 2020; Sun et al., 2021) fail to capture the optimal descriptions for pose estimation. Based on this,

they redesigned the training pipeline based on a three-view system for one-shot object-to-image matching.

The aforementioned works still require dense support views (*i.e.*, ≥ 32 views). To address this problem, Fan et al. (2024a) turned the 6DoF object pose estimation task into relative pose estimation between the retrieved object in the target view and the reference view. Given only one reference view, they achieved it by using the DINOv2 model (Oquab et al., 2023) for global matching and the LoFTR model (Sun et al., 2021) for local matching. Note that this method cannot estimate absolute translation (or object scale), as this is an ill-posed problem when only considering two views. Beyond that, Lee et al. (2024) applied a powerful pre-trained technique tailored for 3D vision (Weinzaepfel et al., 2022) and demonstrated geometry-oriented visual pre-training can get better generalization capability with fewer reference views.

Generally, due to the lack of prior geometric information from CAD models, manual reference view-based feature matching methods often require special designs to extract the geometric features of unseen objects. The number of reference views also constrains the actual application of such approaches to a certain extent.

5.2.2 Template Matching-Based Methods

Template matching-based methods mainly adopt the strategy of retrieval and refinement. There are two types: one reconstructs a 3D object representation using reference views, renders multiple templates based on this 3D representation, and employs a similarity network to compare the query image with each template for the initial pose. A refiner is then used to refine this initial pose for increased accuracy. The other directly uses reference views as templates, requiring plenty of views for retrieval and greater reliance on a refiner for accuracy. Park et al. (2020a) introduced a novel framework for pose estimation of unseen objects without the CAD model. They reconstructed 3D object representations from a few reference views, followed by estimating translation using mask bounding boxes and corresponding depth values. The initial rotation was determined by sampling angles and refined using gradient updates via a render-and-compare approach. By training the network to render and reconstruct diverse 3D shapes, it achieved excellent generalization performance on unseen objects.

Unlike Park et al. (2020a) that used the strategy of render-and-compare after reconstruction, Liu et al. (2022f) designed a pipeline for detection, retrieval, and refinement. They first designed a detector to identify object bounding boxes in the target view. Next, they compared the query and reference images at the pixel level to acquire the initial pose based on the similarity score. The pose was then refined using feature volume and multiple 3D convolution layers. However, object-centered reference images from cluttered scenes are

5.2 基于参考文献的手动方法

除了这些基于CAD模型的方法外, 还存在一些基于手动参考视图的方法。这些方法不需要未见物体的CAD模型作为先验条件, 而是要求提供带有目标物体的手动标注参考视图。与基于CAD模型的方法类似, 这些方法同样可分为两类: 基于特征匹配的方法(第5.2.1节)和基于模板匹配的方法(第5.2.2节)。这两类方法如图11所示; 部分代表性方法的属性与性能详见表4。

5.2.1 基于特征匹配的方法

与基于CAD模型的特征匹配方法不同, 基于手动参考视图的特征匹配方法主要建立 RGBD 查询图像与 RGBD 参考图像之间的三维对应关系, 或查询图像与参考视图重建的稀疏点云之间的二维-三维对应关系。随后, 根据不同的对应关系求解物体的姿态。He 等人(2022c)提出了首个少样本6DoF物体姿态估计方法, 无需额外训练即可通过少量支撑视图估计未见过物体的姿态。具体而言, 他们设计了一个基于Transformer的密集 RGBD 原型匹配框架, 以全面探索查询图像与参考视图之间的语义和几何关系。Corsetti 等人(2024)采用文本提示进行物体分割, 并将问题重新表述为两个场景之间的相对姿态估计, 该相对姿态通过点云配准获得。

部分方法从重建后的匹配角度采取了替代路径。Wu 等人(2021b)开发了一种基于全局配准的方法, 利用参考图像和查询图像重建全视角与单视角模型, 随后在两个模型间搜索点匹配。Sun 等人(2022b)从视觉定位技术中汲取灵感, 对流程进行改进以适配姿态估计任务。具体而言, 他们利用所有参考视角的RGB序列重建未见物体的运动学结构(SfM)模型, 再通过图注意力网络将查询图像中的二维关键点与SfM模型中的三维点进行匹配。但由于依赖重复检测的关键点, 该方法在低纹理物体上的表现欠佳。为解决此问题, He 等人(2022a)基于无检测器特征匹配方法LoFTR(Sun 等人, 2021), 设计了一种无关键点SfM方法, 用于重建低纹理物体的半密集点云模型。Castro 和 Kim(2023)指出, 这些预训练特征匹配模型(Sarlin 等人, 2020; Sun 等人, 2021)无法捕捉姿态估计的最佳描述。基于此, 他们基于三视角系统重新设计了一

次性物体到图像匹配的训练流程。

上述研究仍需要密集的支持视图(≥ 32 个视图)。为解决这一问题, Fan 等人(2024a)将6DoF物体姿态估计任务转化为目标视图中检索到的物体与参考视图之间的相对姿态估计。仅给定一个参考视图时, 他们通过使用DINOv2模型(Oquab 等人, 2023)进行全局匹配以及LoFTR模型(Sun et al. 等人(2021)提出用于局部匹配的方法。需注意该方法无法估计绝对平移量(或物体尺度), 因为仅考虑两个视图时这属于不适定问题。此外, Lee 等人(2024)应用了一种专为三维视觉设计的高效预训练技术(Weinzaepfel 等人, 2022), 并证明基于几何特征的视觉预训练可在较少参考视图条件下获得更强的泛化能力。

通常情况下, 由于缺乏来自CAD模型的先验几何信息, 基于参考视图的手动特征匹配方法往往需要特殊设计来提取未见物体的几何特征。参考视图的数量也在一定程度上限制了此类方法的实际应用范围。

5.2.2 基于模板匹配的方法

基于模板匹配的方法主要采用检索与优化相结合的策略。该方法可分为两种类型: 第一种通过参考视角重建三维物体表征, 基于此三维表征生成多个模板, 利用相似性网络对查询图像与各模板进行初始姿态比对, 随后通过优化器对初始姿态进行精度提升; 第二种直接以参考视角作为模板, 需要大量视角进行检索, 且对优化器的依赖程度更高。Park 等人(2020a)提出了一种无需CAD模型即可对未见物体进行姿态估计的新框架。该方法通过少量参考视角重建三维物体表征, 继而利用掩码边界框及对应深度值估算平移参数。初始旋转角度通过角度采样确定, 并采用渲染-比较方法通过梯度更新进行优化。通过训练网络渲染并重构多样化三维形状, 该方法在未见物体姿态估计任务中展现出优异的泛化性能。

与 Park 等人(2020a)采用重建后渲染与比较策略不同, Liu 等人(2022f)设计了一套包含检测、检索与优化的完整流程。他们首先开发检测器用于识别目标视图中的物体边界框, 随后通过像素级比对查询图像与参考图像, 基于相似度评分获取初始姿态。接着利用特征体积和多层三维卷积层对姿态进行优化。然而, 来自杂乱场景的以物体为中心的参考图像存在以下局限性:

Table 4 Representative manual reference view-based methods. For each method, we report its 9 properties, which have the same meanings as described in the caption of Table 1. D, S, F, T, P, R, and V have the same meanings as Table. We report the average recall of ADD(S) within 10% of the object diameter, termed as ADD(S)-0.1d (Sec. 2). Notably, "YOLOv5" and "GT" denote the use of YOLOv5 (Ultralytics, 2022) and ground-truth bounding box/segmentation mask for object localization, respectively. For a fair comparison, we also report the number of used reference views. "Full" represents all views

Methods		Published Year	Training Input	Inference Input	Pose DoF	Object Property	Task	Inference Mode	Application Area	LM ADD(S)-0.1d	
Manual Reference View-Based Methods	Feature matching	He et al. (2022c)	2022	RGBD	RGBD	6DoF	rigid	estimation	three-stage, D+F+P	general	83.4
		Sun et al. (2022b)	2022	RGB	RGB	6DoF	rigid	estimation	three-stage, D+F+P	general	(GT+16) 63.6 (YOLOv5+Full)
		He et al. (2022a)	2022	RGB	RGB	6DoF	rigid	estimation	three-stage, D+F+P	general	76.9 (YOLOv5+Full)
		Castro and Kim (2023)	2023	RGB	RGB	6DoF	rigid	estimation	three-stage, D+F+P	general	87.5 (GT+Full)
		Lee et al. (2024)	2024	RGB	RGB	6DoF	rigid	estimation	three-stage, D+F+P	general	78.4 (YOLOv5+64)
	Template matching	Park et al. (2020a)	2020	RGBD	RGBD	6DoF	rigid	estimation	three-stage, S+T+R	general	87.1 (GT+16)
		Nguyen et al. (2022a)	2022	RGB	RGB	6DoF	rigid	tracking	three-stage, D+S+P	general	-
		Liu et al. (2022f)	2022	RGB	RGB	6DoF	rigid	estimation	three-stage, D+T+R	general	-
		Gao et al. (2023)	2023	RGB	RGB	6DoF	rigid	estimation	four-stage, S+D+P+R	occlusion	-
		Cai et al. (2024a)	2024	RGB	RGB	6DoF	rigid	estimation	two-stage, P+R	general	81.7 (Full)
	Wen et al. (2024)	2024	RGBD	RGBD	6DoF	rigid	estimation, tracking	four-stage, D+T+R+V	general	99.9 (GT+16)	

表4代表性基于人工参考视图的方法。针对每种方法, 我们报告其9个属性, 这些属性的含义与表1标题中所述一致。D、S、F、T、P、R和V的含义与表中定义相同。
 3. 我们报告了ADD(S)在物体直径10%范围内的平均召回率, 称为ADD(S)-0.1d (详见第2节)。值得注意的是, “YOLOv5”和“GT”分别表示使用YOLOv5 (Ultralytics, 2022) 以及用于物体定位的真实边界框/分割掩码。为确保公平比较, 我们还报告了所使用的参考视角数量: “Full”代表所有视角。

方法	特征匹配	发布年份	训练输入	推理输入	姿势自由度	对象属性	任务	推理模式	应用区域	LM ADD(S)-0.1d
基于参考文献的自动方法	模板匹配	He 等人 (2022c)	RGBD	RGBD	6自由度	严格的	评价	三阶段D+F+P法	一般的	83.4
		Sun 等人 (2022b)	RGB	RGB	6自由度	严格的	评价	三阶段D+F+P法	一般的	(GT+16) 63.6 (YOLOv 5+Full) 76.9
		He 等人 (2022a)	RGB	RGB	6自由度	严格的	评价	三阶段, D+F+P	一般的	(YOLOv5+ Full) 87.5
		Castro 与 Kim (2023)	RGB	RGB	6自由度	严格的	评价	三阶段, D+F+P	一般的	(GT+完整版) 78.4
		Lee 等人 (2024)	RGB	RGB	6自由度	严格的	评价	三阶段D+F+P法	一般的	(YOLOv5+64) 87.1
		Park 等人 (2020a)	RGBD	RGBD	6自由度	严格的	评价	三阶段, S+T+R	一般的	(GT+16)
		Nguyen 等人 (2022)	RGB	RGB	6自由度	严格的	跟踪估算	三阶段D+S+P	一般的	-
		a) Liu 等人 (2022f)	RGB	RGB	6自由度	严格的	跟踪估算	三阶段, D+T+R 四阶段, S+D+P+R 两阶段, P+R	全身性	-
		Gao 等人 (2023)	RGB	RGB	6自由度	严格的	评价		全身性	-
		Cai 等人 (2024a)	RGB	RGB	6自由度	严格的	评价		全身性	81.7 (完整版) 99.9
		Wen 等人 (2024)	RGBD	RGBD	6自由度	严格的	估计, 跟踪	四阶段: D+T+R+V	一般的	(GT+16)

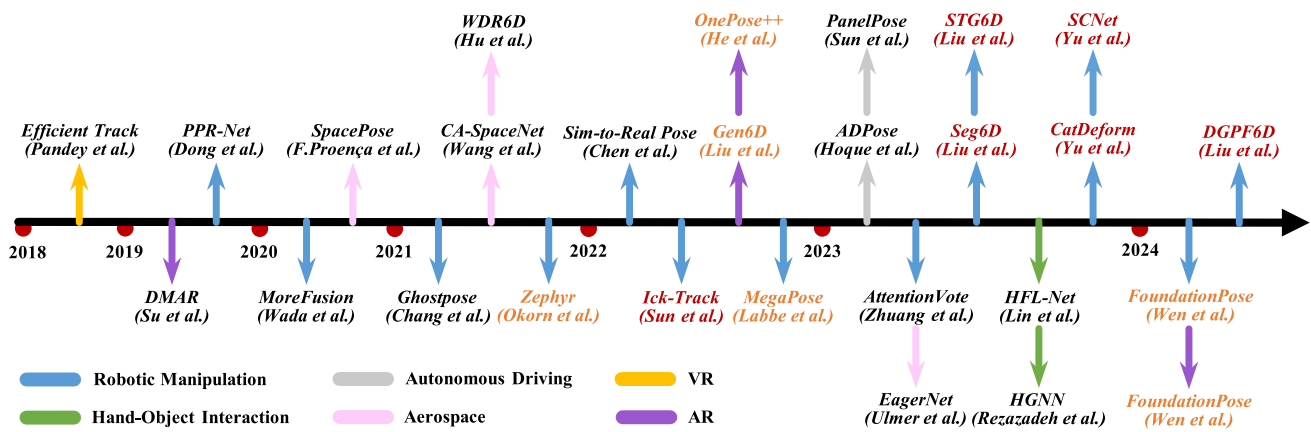


Fig. 12 Chronological overview of some representative applications of object pose estimation methods. The black references, red references, and orange references represent the application of instance-level, category-level, and unseen object pose estimation methods, respectively.

constrained by actual segmentation or bounding box cropping, limiting its real-world applicability. To overcome this limitation, Gao et al. (2023) proposed adaptive segmentation modules to learn distinguishable representations of unseen objects, and Zhao et al. (2024) leveraged distributed reference kernels and translation estimator to achieve multi-scale correlation computation and object translation parameter prediction, thus robustly learning the prior translation of unseen objects.

To further enhance the robustness of the translation estimation for object detection, Pan et al. (2024) modified the framework of Liu et al. (2022f). Precisely, they utilized pre-trained ViT (Dosovitskiy et al., 2021) to learn robust feature representations and adopted a top-K pose proposal scheme for pose initialization. Additionally, they applied a coarse-to-fine cascaded refinement process, incorporating feature pyramids and adaptive discrete pose hypotheses. Besides Pan et al. (2024), Cai et al. (2024a) revisited the pipeline of Liu et al. (2022f). They proposed a generic joint segmentation method and an efficient 3D Gaussian Splatting-based refiner, improving the performance and robustness of object localization and pose estimation.

In unseen object tracking, Nguyen et al. (2022a) proposed the first method that extended to invisible categories without requiring 3D information and extra reference images, given the ground-truth object pose in the first frame. Their transformer-based architecture outputs continuous relative object poses between consecutive frames, combined with the initial object pose, to provide the object pose for each frame. Wen et al. (2023) used the collaborative design of concurrent tracking and neural object fields to perform 6DoF tracking from RGBD sequences. Key aspects of it include online pose graph optimization, concurrent neural object fields for 3D

shape and appearance reconstruction, and a memory pool facilitating communication between the two processes.

More recently, Nguyen et al. (2024a) reconsidered template matching from the perspective of generating new views. Given a single reference view, they trained a model to directly predict the discriminative embeddings of the novel view-points of the object. In contrast, Wen et al. (2024) applied an object-centric neural field representation for object modeling and RGBD rendering.

Brief Discussion: In summary, similar to template matching-based methods using CAD models, manual reference view-based methods also rely on massive templates. Moreover, due to limited reference views, these methods need to generate new templates or employ additional strategies to optimize the initial pose obtained through template matching.

6 Applications

With the advancement of object pose estimation technology, several applications leveraging this progress have been deployed. In this section, we elaborate on the development trends of these applications. Specifically, these applications include robotic manipulation (Sec. 6.1), Augmented Reality (AR)/Virtual Reality (VR) (Sec. 6.2), aerospace (Sec. 6.3), hand-object interaction (Sec. 6.4), and autonomous driving (Sec. 6.5). The chronological overview is shown in Fig. 12. The illustration of application scenarios is shown in Fig. 13.

6.1 Robotic Manipulation

We categorize the robotic manipulation application into instance-level, category-level, and unseen objects. This clas-

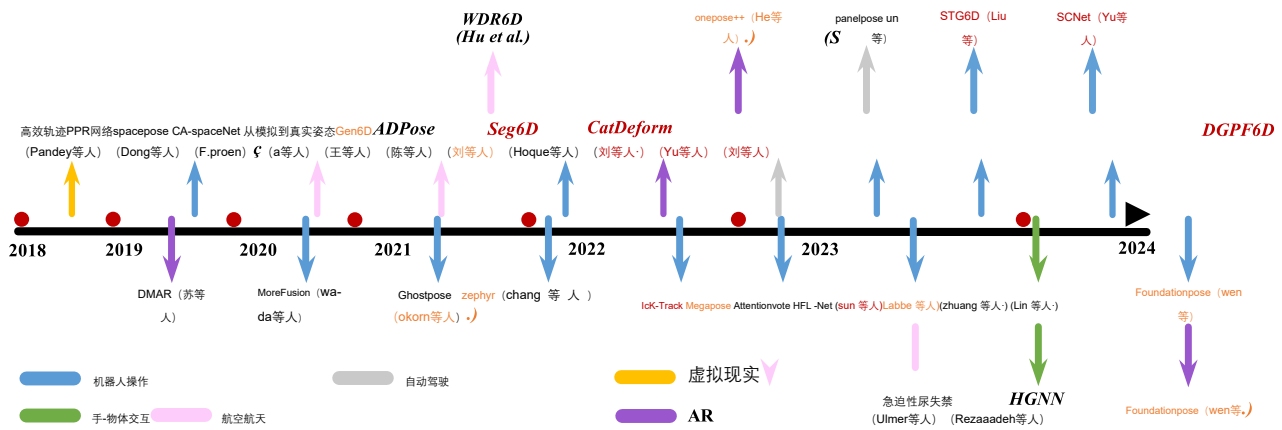


图12物体姿态估计方法部分代表性应用的时间线概述。黑色参考图、红色参考图与橙色参考图分别对应实例级、类别级及未见物体姿态估计方法的应用场景。由此我们还可观察到发展趋势，即从实例级方法向类别级及未见物体姿态估计方法的演进路

径。

受限于实际分割或边界框裁剪的约束，其实际应用价值受到限制。为突破这一局限，Gao 等人（2023）提出自适应分割模块，通过学习可区分的未见物体表征；Zhao 等人（2024）则采用分布式参考核与平移估计器，实现多尺度相关性计算及物体平移参数预测，从而稳健地学习未见物体的先验平移信息。

为进一步提升目标检测中翻译估计的鲁棒性，Pan 等人（2024）改进了 Liu 等人（2022f）提出的框架。具体而言，他们利用预训练的ViT模型（Dosovitskiy 等人，2021）学习鲁棒特征表示，并采用top-K姿态提议方案进行姿态初始化。此外，他们应用了粗到细的级联精炼过程，整合特征金字塔与自适应离散姿态假设。除 Pan 等人（2024）外，Cai 等人（2024a）重新审视了 Liu 等人（2022f）的流程框架，提出通用联合分割方法及高效的基于3D高斯点阵的精炼器，显著提升了目标定位与姿态估计的性能与鲁棒性。

在不可见物体跟踪领域，Nguyen 等人（2022a）提出了首个无需三维信息和额外参考图像即可扩展至不可见类别的方法，该方法仅需第一帧中的真实物体姿态作为输入。其基于Transformer架构的模型通过输出连续帧间的相对物体姿态，并结合初始物体姿态，为每帧提供物体姿态信息。Wen 等人（2023）采用并行跟踪与神经物体场协同设计方法，实现了 RGBD 序列中的六自由度跟踪。该方法的关键特征包括在线姿态图优化、用于三维形态与外观重建的并行神经物体

场，以及促进两过程间通信的内存池机制。

最近，Nguyen 等人（2024a）从生成新视图的角度重新审视了模板匹配问题。他们以单一参考视图为基础，训练模型直接预测物体新视角的判别性嵌入向量。相比之下，Wen 等人（2024）采用以物体为中心的神经场表征方法进行物体建模与 RGBD 渲染。

简要讨论：总之，与基于CAD模型的模板匹配方法类似，基于手动参考视图的方法同样依赖于大量模板。此外，由于参考视图有限，这些方法需要生成新模板或采用其他策略来优化通过模板匹配获得的初始姿态。

6 应用

随着物体姿态估计技术的进步，利用这一进展的多种应用已得到部署。本节将详细阐述这些应用的发展趋势。具体而言，这些应用包括机器人操作（第6.1节）、增强现实（AR）/虚拟现实（VR）（第6.2节）、航空航天（第6.3节）、手-物体交互（第6.4节）以及自动驾驶（第6.5节）。时间线概述见图12，应用场景示意图见图13。

6.1 机器人操作

我们将机器人操作应用分为实例级、类别级和未见物体三类。该分类

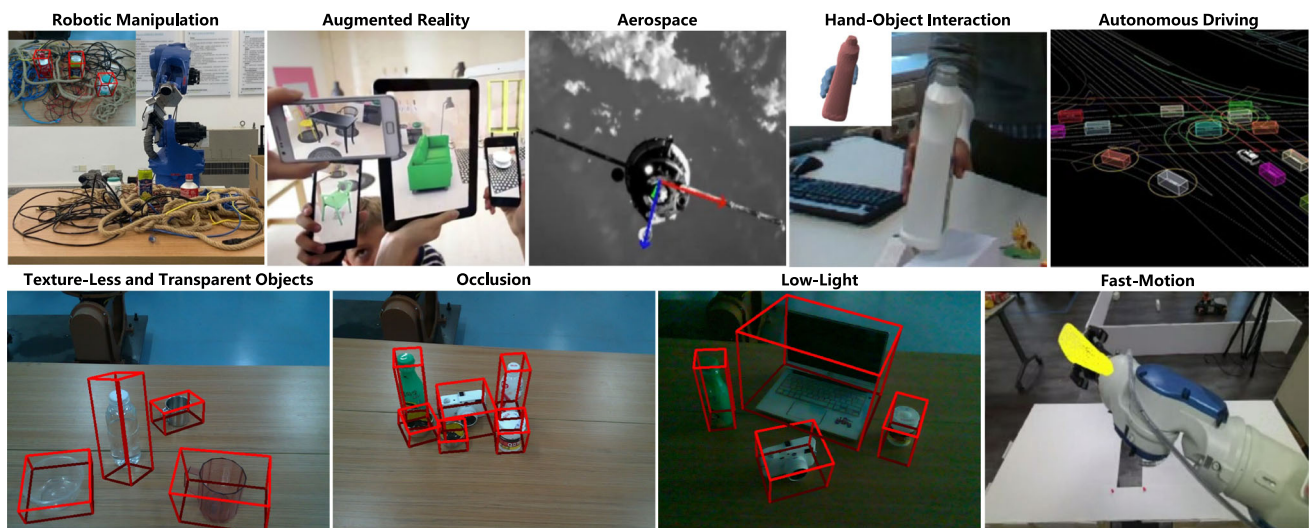


Fig. 13 Illustration of application scenarios of object pose estimation. The first row shows its applications in robotic manipulation, augmented reality, aerospace, hand–object interaction, and autonomous driving. The second row presents examples involving challenging objects and scenes

sification helps in better understanding the challenges and requirements across different levels.

6.1.1 Instance-Level Manipulation

To tackle the challenge of annotating real data during training, many works utilize synthetic data for training as it is easy to acquire and annotate (Park et al., 2020b; Chen et al., 2023a). At the same time, synthetic data can simulate various scenes and environmental changes, thus helping to improve the adaptability of robotic manipulation. Li et al. (2018b) used a large-scale synthetic dataset and a small-scale weakly labeled real-world dataset to reduce the difficulty of system deployment. Additionally, Chen et al. (2022b) proposed an iterative self-training framework, using a teacher network trained on synthetic data to generate pseudo-labels for real data. Meanwhile, Fu et al. (2022) trained only on synthetic images based on physical rendering. One of the critical challenges of synthetic data is bridging the gap with reality, which Tremblay et al. (2018) addressed by combining domain randomization with real data.

Handling stacked occlusion scenes is another significant challenge, especially in industrial automation and logistics. In these scenarios, robots must accurately identify and localize objects stacked on each other, which requires an effective process of occluded objects and accurate pose estimation. Dong et al. (2019) argued that the regression poses of points from the same object should tightly reside in the pose space. Therefore, these points can be clustered into different instances, and their corresponding object poses can be estimated simultaneously. This method can handle severe object occlusion. Moreover, Zhuang et al. (2024) established an end-to-end pipeline to synchronously regress all

potential object poses from an unsegmented point cloud. Most recently, Wada et al. (2020) proposed a system that fully utilized identified accurate object CAD models and non-parametric reconstruction of unrecognized structures to estimate the occluded objects pose in real-time.

Low-textured objects lack object surface texture information, making robotic manipulation challenging. Therefore, Zhang and Cao (2019) proposed a pose estimation method for texture-less industrial parts. Poor surface texture and brightness make it challenging to compute discriminative local appearance descriptors. This method achieves more accurate results by optimizing the pose in the edge image. In addition, Chang et al. (2021) carried out transparent object grasping by estimating the object pose using a proposed model-free method that relies on multiview geometry. In agricultural scenes, Kim et al. (2022) constructed an automated data collection scheme based on a 3D simulator environment to achieve three-level ripeness classification and pose estimation of target fruits.

6.1.2 Category-Level Manipulation

To investigate the application of category-level object pose estimation for robotic manipulation, Liu et al. (2023a) introduced a fine segmentation-guided category-level method with difference-aware shape deformation for robotic grasping. Yu et al. (2023a) proposed a shape prior-based approach and explored its application for robotic grasping. Further, Liu et al. (2023d) developed a robotic continuous grasping system with a pre-defined vector orientation-based grasping strategy, based on shape transformer-guided object pose estimation. To improve efficiency and enable the pose estimation method to be applied to tasks with higher real-time

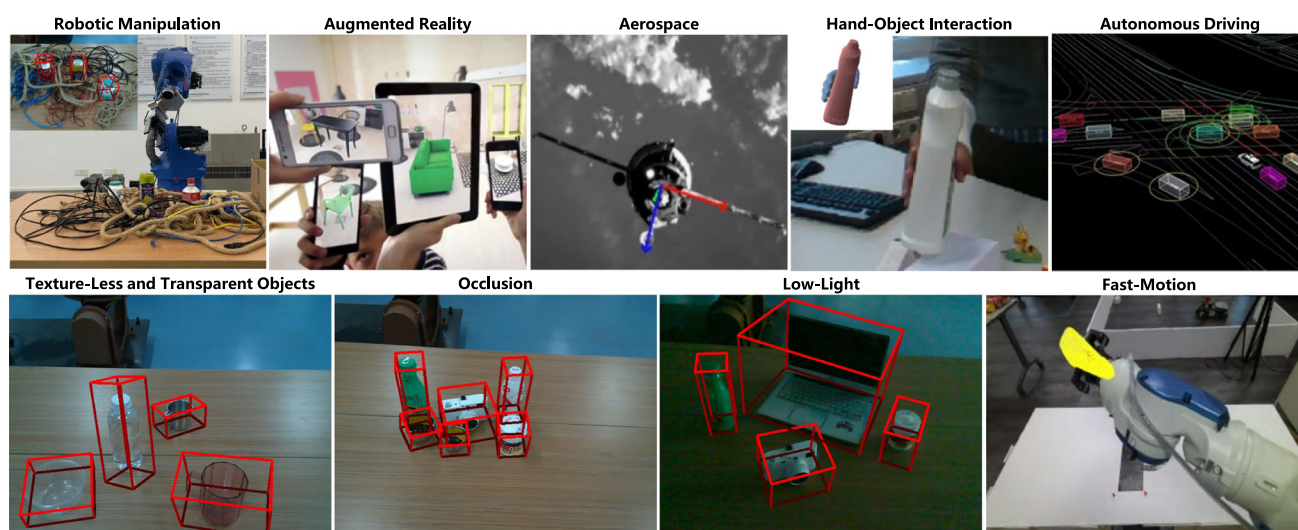


图13物体姿态估计应用场景示意图。第一行展示了其在机器人操作、增强现实、航空航天、人机交互及自动驾驶等领域的应用实例。第二行则呈现了涉及复杂物体与场景的典型应用场景。

系统化分析有助于更深入地理解不同层级面临的挑战与需求。

6.1.1 实例级别操作

为应对训练过程中标注真实数据的挑战,许多研究采用合成数据进行训练,因其易于获取和标注(Park 等人, 2020b; Chen 等人, 2023a)。同时,合成数据能够模拟多种场景及环境变化,从而提升机器人操作的适应性。Li 等人(2018b)采用大规模合成数据集与小规模弱标注真实数据集,降低了系统部署难度。此外,Chen 等人(2022b)提出迭代自训练框架,利用基于合成数据训练的教師网络为真实数据生成伪标签。与此同时,Fu 等人(2022)仅基于物理渲染的合成图像进行训练。合成数据的关键挑战在于弥合与现实的差距,Tremblay 等人(2018)通过结合领域随机化与真实数据解决了这一问题。

处理堆叠遮挡场景是工业自动化与物流领域面临的另一重大挑战。在这些应用场景中,机器人需要精准识别并定位相互堆叠的物体,这要求具备有效的遮挡物体处理流程及精确的姿态估计能力。Dong 等人(2019)提出,同一物体上各点的回归姿态应紧密分布于姿态空间内,因此可将这些点聚类为不同实例,并同步估算对应物体姿态。该方法能有效应对严重物体遮挡问题。此外,Zhuang 团队(2024)构建了端到端处理流程,可同步从未分割点云中回归所有潜在物体姿态。最新研究中,Wada 团队(2020)提出了一套

系统,通过充分利用已识别的精确物体CAD模型及未识别结构的非参数化重建技术,实现实时估算遮挡物体姿态。

低纹理物体缺乏表面纹理信息,导致机器人抓取操作面临挑战。为此,Zhang 与 Cao(2019)提出了一种适用于无纹理工业零件的姿态估计方法。由于表面纹理和亮度较差,难以计算具有区分度的局部外观描述符。该方法通过优化边缘图像中的物体姿态来提升识别精度。此外,Chang 等人(2021)采用基于多视角几何的无模型方法估计物体姿态,实现了透明物体抓取。在农业场景中,Kim 团队(2022)通过三维模拟器环境构建自动化数据采集方案,实现了目标水果的三级成熟度分类与姿态估计。

6.1.2 类别-层级操作

为研究类别级物体姿态估计在机器人操作中的应用,Liu 等人(2023a)提出了一种基于精细分割引导、具备差异感知形状变形特性的类别级方法,用于机器人抓取任务。Yu 等人(2023a)提出了一种基于形状先验的方法,并探索了其在机器人抓取中的应用。此外,Liu 等人(2023d)开发了一种基于形状变换器引导的物体姿态估计的机器人连续抓取系统,该系统采用预定义的基于矢量方向的抓取策略。为提高效率并使姿态估计方法能够应用于更高实时性的任务.....

requirements, Sun et al. (2022a) utilized the inter-frame consistent keypoints to perform object pose tracking for aerial manipulation. To further avoid manual data annotation in the real-world scene, Yu et al. (2023b) built a robotic grasping platform and designed a self-supervised-based method for category-level robotic grasping. More recently, Liu et al. (2024a) explored a contrastive learning-guided prior-free object pose estimation method for domain-generalized robotic picking. Zhou et al. (2025) proposed a single-shot SSM-based framework that predicts the full 3D information (pose, size, and shape) of multiple object for end-to-end robotic scene understanding.

6.1.3 Unseen Object Manipulation

Since unseen object pose estimation belongs to an emerging research, there is currently a lack of specialized designs for robotic. Here, we report several methods that validate the effectiveness of unseen object pose estimation through robotic manipulation. Okorn et al. (2021) introduced a method for zero-shot object pose estimation in clutter. By scoring pose hypotheses and choosing the highest-scoring pose, they successfully grasped a novel drill object using a robotic arm. Labbé et al. (2022) and Wen et al. (2024) adopted the render-and-compare strategy and trained the network on a large-scale synthetic dataset, resulting in an outstanding generalization. They further verified the effectiveness of their methods through robotic grasping experiments.

6.2 Augmented Reality/Virtual Reality

Object pose estimation has various specific applications in AR and VR fields. In AR, accurate pose estimation allows for a precise overlay of virtual objects onto the real world. The key to VR technology lies in tracking the head-mounted display pose and controller in 3D space.

Su et al. (2019) combined two CNN architectures (LeCun et al., 1989) into a network, consisting of a state estimation branch and a pose estimation branch explicitly trained on synthetic images, to achieve AR assembly applications. Pandey et al. (2018) introduced a method for automatically annotating the handheld objects pose in camera space, addressing the efficient 6Dof pose tracking problem for handheld controllers from the perspective of egocentric cameras. Liu et al. (2022f) presented a generalizable model-free 6DoF object pose estimator that has realized the complete object detection and pose estimation process. By simply capturing reference images of an unseen object and retrieving the poses of reference images, this method can predict the object pose on arbitrary query images and be easily applied to daily objects for AR/VR applications. He et al. (2022a) adopted matching after the reconstruction strategy, which establishes the correspondences between the query image and the reconstructed

point cloud from reference views. This method does not rely on keypoint matching and allows for AR applications even on low-texture objects. Wen et al. (2024) achieved strong generality by employing large-scale comprehensive training and innovative transformer-based architecture. This method has been successfully applied in various domains, including AR and robotic manipulation.

6.3 Aerospace

Estimating the object pose in space presents unique challenges not commonly encountered in terrestrial environments. One of the most significant differences is the lack of atmospheric scattering, which complicates lighting conditions and makes objects invisible over long distances. In-orbit proximity operations for space rendezvous, docking, and debris removal require precise pose estimation under diverse lighting conditions and on high-texture backgrounds. Proença and Gao (2020) proposed URSO, a simulator developed on Unreal Engine 4 (UnrealEngine4, 2020) for generating annotated images of spacecraft orbiting Earth, which can be used as valuable data for aerospace application. Hu et al. (2021) proposed an encoder-decoder architecture that reliably handles large-scale changes under challenging conditions, enhancing robustness. Wang et al. (2022c) introduced a counterfactual analysis framework to achieve robust pose estimation of spaceborne targets in complex backgrounds. Ulmer et al. (2023) generated multiple pose hypotheses for objects and introduced a pixel-level posterior formula to estimate the probability of each hypothesis. This approach can handle extreme visual conditions, including overexposure, high contrast, and low signal-to-noise ratio.

6.4 Hand-Object Interaction

When humans/robots interact with the physical world, they primarily do so through their hands. Therefore, accurately understanding how hands interact with objects is crucial.

Hand-object interaction methods often rely on the object CAD model, and obtaining the object CAD model from daily life scenes is challenging. Patten et al. (2021) reconstructed high-quality object CAD model to mitigate the reliance on object CAD model in hand-object interaction. To further enhance hand-object interaction, Lin et al. (2023b) utilized an effective attention model to improve the representation capability of hand and object features, thereby improving the accuracy of hand and object pose estimation. However, this method has limited utilization of the underlying geometric structures, leading to an increased reliance on visual features. Performance may degrade when objects lack visual features or when these features are occluded. Therefore, Rezazadeh et al. (2023) introduced a hierarchical graph neural network architecture combined with multimodal (visual

在需求研究方面, Sun 等人 (2022a) 利用帧间一致关键点实现了空中操作中的物体姿态跟踪。为避免在真实场景中进行人工数据标注, Yu 等人 (2023b) 构建了机器人抓取平台, 并设计了基于自监督学习的类别级机器人抓取方法。近期, Liu 等人 (2024a) 探索了对比学习引导的无先验物体姿态估计方法, 用于领域泛化机器人抓取任务。Zhou 等人 (2025) 提出了一种基于单次 SSM 的框架, 可预测多个物体的完整三维信息 (姿态、尺寸和形状), 实现端到端机器人场景理解。

6.1.3 不可见物体操作

由于未见物体姿态估计属于新兴研究领域, 目前缺乏针对机器人领域的专用设计方案。本文通过机器人操作验证了多种未见物体姿态估计方法的有效性。Okorn 等人 (2021) 提出了一种在杂乱环境中进行零样本物体姿态估计的方法, 通过评估姿态假设并选择得分最高的姿态, 成功实现了机械臂对新型钻头物体的抓取。Labbe 等人 (2022) 与 Wenet 等人 (2024) 采用渲染-比较策略, 并在大规模合成数据集上训练网络, 实现了卓越的泛化性能。他们进一步通过机器人抓取实验验证了方法的有效性。

6.2 增强现实/虚拟现实

物体姿态估计在增强现实 (AR) 和虚拟现实 (VR) 领域具有多种具体应用。在 AR 中, 精确的姿态估计可实现虚拟物体与现实世界的精准叠加。VR 技术的核心在于对头戴式显示器的姿态及控制器在三维空间中的实时追踪。

Su 等人 (2019) 将两种 CNN 架构 (LeCun 等人, 1989) 整合为一个网络, 该网络包含状态估计分支和姿态估计分支, 并通过合成图像进行显式训练, 以实现 AR 装配应用。Pandey 等人 (2018) 提出了一种在相机空间中自动标注手持物体姿态的方法, 从第一人称视角解决了手持控制器高效 6 自由度姿态跟踪问题。Liu 等人 (2022f) 提出了一种通用的无模型 6DoF 物体姿态估计器, 实现了完整的物体检测与姿态估计流程。该方法仅需捕获未见物体的参考图像并提取其姿态信息, 即可预测任意查询图像上的物体姿态, 可轻松应用于 AR/VR 场景中的日常物体识别。He 等人 (2022a) 采用重建后匹配策略, 通过参考视角重建点云与查询图像建立对应关系。该方法无需依赖关键点匹配,

即使在低纹理物体上也能实现增强现实 (AR) 应用。

Wen 等人 (2024) 通过采用大规模综合训练和创新的 Transformer 架构, 实现了高度通用性。该方法已成功应用于 AR 和机器人操作等多个领域。

6.3 航空航天

在空间环境中估计物体姿态面临独特的挑战, 这些挑战在地面环境中并不常见。最显著的区别在于缺乏大气散射效应, 这使得光照条件复杂化, 并导致物体在远距离下不可见。针对空间交会、对接及碎片清除等在轨近距离操作, 需要在多样化光照条件和高纹理背景下进行精确的姿态估计。Proenca 与 Gao (2020) 提出了 URSO —— 基于 Unreal Engine 4 (UnrealEngine 4, 2020) 开发的模拟器, 用于生成环绕地球的航天器标注图像, 这些图像可作为航空航天应用的重要数据来源。Hu 等人 (2021) 提出了一种编码器-解码器架构, 能够在严苛条件下可靠处理大规模变化, 从而提升鲁棒性。Wang 等人 (2022c) 引入了反事实分析框架, 实现了复杂背景中空间目标的鲁棒姿态估计。Ulmer 等人 (2023) 为物体生成了多个姿态假设, 并引入了像素级后验公式来估计每个假设的概率。该方法能够应对极端视觉条件, 包括过曝、高对比度和低信噪比。

6.4 人机交互

当人类/机器人与物理世界互动时, 其主要通过手部进行。因此, 准确理解手部与物体的交互方式至关重要。

手-物体交互方法通常依赖于物体 CAD 模型, 而从日常场景中获取物体 CAD 模型具有挑战性。Patten 等人 (2021) 通过重建高质量物体 CAD 模型来降低手-物体交互中对 CAD 模型的依赖性。为进一步提升手-物体交互效果, Lin 等人 (2023b) 采用有效注意力模型增强手部与物体特征的表征能力, 从而提高手部与物体姿态估计的准确性。然而该方法对底层几何结构的利用有限, 导致对视觉特征的依赖性增强。当物体缺乏视觉特征或这些特征被遮挡时, 性能可能下降。因此 Rezazadeh 等人 (2023) 提出了一种结合多模态 (视觉

and tactile) data to compensate for visual deficiencies and improve robustness. Moreover, Qi et al. (2024) introduced a hand-object pose estimation network guided by Signed Distance Fields (SDF), which jointly leverages the SDFs of both the hand and the object to provide a complete global implicit representation. This method aids in guiding the pose estimation of hands and objects in occlusion scenarios. Yang et al. (2025b) proposed an occlusion-aware framework based on masked autoencoders, where a target-focused masking strategy and multi-scale SDF features are leveraged to enhance structural reasoning under occlusions.

6.5 Autonomous Driving

Object pose estimation can be used to perceive surrounding objects such as vehicles, pedestrians, and obstacles, aiding the autonomous driving system in making timely decisions.

In order to address the pose estimation problem in autonomous driving, Hoque et al. (2023) proposed a 6DoF pose hypothesis based on a deep hybrid structure composed of CNNs (LeCun et al., 1989) and RNNs (Elman, 1990). More recently, Sun et al. (2023) designed an effective keypoint selection algorithm, which takes into account the shape information of panel objects within the scene of robot cabin inspection, addressing the challenge of 6DoF pose estimation of highly variable panel objects.

6.6 Challenging Objects and Scenes

In practical deployment, object pose estimation models frequently encounter challenging objects and scenes that amplify the limitations of existing methods. To better guide practitioners, we provide targeted recommendations for method selection under these conditions.

Texture-less objects pose a major difficulty for correspondence-based methods, which depend heavily on distinctive local descriptors (Drost et al., 2010; Birdal & Ilic, 2015). In such cases, template-based (Dang et al., 2024) or regression-based (Wang et al., 2021c) approaches often provide more reliable estimates by leveraging global image evidence rather than sparse correspondences. **Transparent** and **reflective** objects present a different challenge due to noisy or missing depth. Approaches that integrate photometric and geometric reasoning (Chen et al., 2022d), or that incorporate physics-aware rendering constraints (Zhang et al., 2022a), have shown greater promise for handling these cases by compensating for unreliable sensor inputs.

Under severe **occlusions**, correspondence-based (Lian & Ling, 2023; Xu et al., 2024b) and voting-based (Peng et al., 2019; He et al., 2020) methods are typically more effective. Correspondence methods, especially dense correspondence models with strong local descriptors, can maintain stability when only partial object views are visible. Voting strate-

gies are also advantageous in these cases, as aggregated hypotheses from visible parts can recover accurate global poses despite missing regions. Moreover, **low-light** conditions often degrade RGB-based pipelines, as appearance cues become unreliable. Since most RGBD sensors are active (using LEDs or lasers) and can provide depth and point cloud data that remain robust under challenging lighting conditions, RGBD fusion or point cloud-centric approaches are therefore recommended (Wang et al., 2019a; He et al., 2021). By explicitly integrating geometric depth information, these methods remain resilient to illumination variations and sensor noise. For improved robustness, fusion architectures that exploit cross-modality relationships rather than naive concatenation are preferable (Wang et al., 2022a; Lin et al., 2023a). Furthermore, for **fast-motion** scenarios, temporal consistency becomes critical. Methods that incorporate spatio-temporal constraints, such as keypoint tracking (Wang et al., 2020a) or pose graph optimization (Wen & Bekris, 2021), offer better stability. End-to-end regression pipelines with temporal feature integration also show promise by learning motion-aware pose embeddings that reduce prediction jitter during rapid dynamics. Finally, in tracking tasks across continuous frames, anchor-based or memory-augmented methods are particularly effective (Wang et al., 2020a; Wen & Bekris, 2021). By exploiting inter-frame coherence, these approaches reduce the reliance on per-frame detection and provide smoother, real-time performance.

Overall, practitioners should select methods by aligning the strengths of each paradigm with the specific challenges of the target application, balancing accuracy, robustness, and efficiency according to object characteristics and environmental conditions.

7 Emerging Trends and Future Directions

In this survey, we have provided a systematic overview of the latest deep learning-based object pose estimation methods, covering a comprehensive classification, a comparison of their strengths and weaknesses, and an exploration of their applications. Despite the great success, many challenges still exist, as discussed in Sec. 3, Sec. 4, and Sec. 5. Building on these challenges, we further highlight emerging trends and outline promising future directions to advance research in object pose estimation.

7.1 Emerging Trends

From the perspective of **label-efficient learning**, prevailing methodologies predominantly rely on the utilization of real-world labeled datasets for training purposes. Nevertheless, the labor-intensive nature of manually collecting and annotating training data is widely acknowledged. Hence,

通过整合触觉数据来弥补视觉缺陷并提升系统鲁棒性。此外, Qi 等人 (2024) 提出了一种基于符号距离场 (SDF) 的手-物体姿态估计网络, 该网络同时利用手部和物体的SDF特征, 构建完整的全局隐式表征, 有效指导遮挡场景下的手部与物体姿态估计。Yang 等人 (2025b) 则提出了一种基于掩码自编码器的遮挡感知框架, 通过采用目标聚焦式掩码策略和多尺度SDF特征, 显著提升了遮挡条件下的结构推理能力。

6.5 自动驾驶

物体姿态估计可用于感知周围环境中的物体 (如车辆、行人及障碍物), 从而帮助自动驾驶系统及时做出决策。

为解决自动驾驶中的姿态估计问题, Hoque 等人 (2023) 提出了一种基于深度混合结构的6自由度姿态假设模型, 该结构由卷积神经网络 (CNN, LeCun 等人, 1989) 和循环神经网络 (RNN, Elman, 1990) 构成。近期, Sun 等人 (2023) 设计了一种有效的关键点选择算法, 该算法充分考虑了机器人舱室检测场景中面板物体的形状信息, 有效解决了高度可变面板物体的6自由度姿态估计难题。

6.6 具有挑战性的物体与场景

在实际部署中, 物体姿态估计模型常会遇到具有挑战性的物体和场景, 这些情况会加剧现有方法的局限性。为更好地指导实践者, 我们针对此类场景提供了针对性的方法选择建议。

无纹理物体对基于对应关系的方法构成重大挑战, 这类方法高度依赖独特的局部描述符 (Drost 等人, 2010; Birdal & Ilic, 2015)。在此类情况下, 基于模板的方法 (Dang 等人, 2024) 或基于回归的方法 (Wang 等人, 2021c) 通常能通过利用全局图像证据而非稀疏对应关系提供更可靠的估计结果。透明与反射性物体因存在噪声或缺失深度而带来不同挑战。整合光度学与几何推理的方法 (Chen 等人, 2022d), 或结合物理感知渲染约束的方法 (Zhang 等人, 2022a) 通过补偿不可靠传感器输入, 在处理此类场景时展现出更大潜力。

在严重遮挡情况下, 基于对应关系的方法 (Lian & Ling, 2023; Xu 等人, 2024b) 和基于投票的方法 (Peng 等人, 2019; He 等人, 2020) 通常更为有效。对应关系方法, 尤其是具有强局部描述符的密集对应模型, 在仅可见部分物体视图时仍能保持稳定。投票策略在此类场景中同样具有优势, 因为来自

可见区域的聚合假设可恢复精确的全局姿态, 即使存在缺失区域。此外, 弱光条件会削弱基于RGB的流程性能, 因外观线索变得不可靠。由于大多数 RGBD 传感器为有源式 (使用LED或激光器), 且能提供在严苛光照条件下仍具鲁棒性的深度和点云数据, 因此推荐采用 RGBD 融合或以点云为中心的方法 (Wang 等人, 2019a; He 等人, 2021)。通过显式整合几何深度信息, 这些方法对光照变化和传感器噪声仍具有抗干扰能力。为提升鲁棒性, 相较于简单的拼接方式, 利用跨模态关系的融合架构更为理想 (Wang 等人, 2022a; Lin 等人, 2023a)。此外, 在快速运动场景中, 时间一致性变得至关重要。结合时空约束的方法 (如关键点追踪 (Wang 等人, 2020a) 或姿态图优化 (Wen & Bekris, 2021)) 能提供更好的稳定性。采用时间特征整合的端到端回归流程也展现出潜力, 通过学习运动感知姿态嵌入来减少快速动态过程中的预测抖动。最后, 在连续帧间的跟踪任务中, 基于锚点或记忆增强的方法尤为有效 (Wang 等人, 2020a; Wen & Bekris, 2021)。这些方法通过利用帧间一致性, 降低了对逐帧检测的依赖, 从而实现更平滑的实时性能。

总体而言, 实践者应通过将各范式的优势与目标应用的具体挑战相匹配来选择方法, 并根据对象特性和环境条件平衡准确性、稳健性与效率。

7 新兴趋势与未来发展方向

在本综述中, 我们系统性地概述了最新的基于深度学习的物体姿态估计方法, 内容涵盖全面的分类体系、各方法优缺点的对比分析以及其应用探索。尽管取得了显著进展, 但正如第3节、第4节和第5节所讨论的, 仍存在诸多挑战。基于这些挑战, 我们进一步指出了新兴趋势, 并勾勒出推动物体姿态估计研究发展的前景广阔的方向。

7.1 新兴趋势

从标签高效学习的角度来看, 主流方法主要依赖于使用真实世界标注数据集进行训练。然而, 人工收集和标注训练数据的劳动密集型特性已得到广泛认可。因此,

we advocate for the exploration of label-efficient learning techniques for object pose estimation, which can be pursued through the following avenues:

1) LLMs/LVMs-guided weak/self-supervised learning methods. With the rapid advancements in pre-trained LLMs/LVMs, their versatile application in various scenarios through an unsupervised manner has become feasible. Leveraging LLMs/LVMs as prior knowledge holds promise for exploring weak or self-supervised learning techniques in object pose estimation. **2) Synthesis to real-world domain adaptation and generalization methods.** Due to the high costs associated with acquiring real-world training data through manual efforts, synthetic data generation offers a cost-effective alternative. We believe that by exploring domain adaptation and generalization techniques from synthetic to real-world domains, we can mitigate domain gaps and achieve the capability to generalize synthetic data-trained models for real-world applications. **3) Novel view reconstruction for pose estimation.** Recent advances in neural rendering, including 3D Gaussians, differentiable rendering, and volumetric rendering, have enabled the synthesis of object-consistent novel views from only a few input images (Di Felice et al., 2024; Wen et al., 2024; Nguyen et al., 2024a; Legrand et al., 2024). These approaches offer a CAD-free route to densify viewpoints, regularize correspondences, and support render-and-compare estimation under sparse supervision. In category-level and unseen-category scenarios, such models can serve as priors to promote object-centric canonicalization and metric scaling, or provide uncertainty-aware multi-hypothesis images and embeddings that enhance retrieval and RANSAC-based PnP. Open challenges remain in preserving geometric fidelity under intra-class variation, mitigating generative artifacts and view inconsistency, and achieving real-time performance on robotic platforms.

In terms of **applications**, facilitating the deployment of object pose estimation methods on mobile devices and robots is crucial. We argue that enhancing the deployability of existing methods can be achieved through the following approaches:

1) End-to-end methods integrating detection or segmentation. Current SOTA approaches typically require initial object detection or segmentation using a pre-trained model before inputting the image into a pose estimation model (indirect pose estimation models even need to use non-differentiable PnP or Umeyama algorithms to solve pose), which complicates deployment. Future research can enhance the deployability on mobile devices and robots by exploring end-to-end object pose estimation methods that seamlessly integrate detection or segmentation.

2) Single RGB image-based methods. Given that most mobile devices (such as smartphones and tablets) lack depth cameras, achieving high-precision estimation of unseen object poses using a single RGB image is crucial. Due to

inherent geometric limitations in 2D images, future research can explore LVMs-based monocular depth estimation methods to enhance the accuracy of monocular object pose estimation by incorporating scene-level depth information.

3) Model lightweighting. Existing SOTA models often have large parameter sizes and inefficient running performance, which presents challenges for deployment on mobile devices and robots with limited computational resources. Future work can explore effective lightweight methods, such as teacher-student models, to research reducing model parameter count (GPU memory) and improving model running efficiency.

Existing methods are predominantly designed for common objects and scenes, rendering them ineffective for **challenging objects and scenes**. We believe that the applicability can be enhanced through the following avenues:

1) Articulated object pose estimation. Articulated objects (such as clothing and drawers) exhibit multiple DoF and significant self-occlusion compared to rigid objects, making pose estimation challenging. Achieving high-precision pose estimation for articulated objects is an important research problem that remains to be addressed in the future.

2) Transparent object pose estimation. The simultaneous absence of texture, color, and depth information poses a significant challenge for estimating the pose of transparent objects. Future research endeavors could focus on enhancing the geometric information of transparent objects through depth augmentation or completion techniques, thereby improving the accuracy of pose estimation.

3) Robust methods for handling occlusion. Occlusion is the most common challenge. Currently, there exists no object pose estimation method that can effectively handle severe occlusion. Severe occlusion leads to an incomplete representation of texture and geometric features in objects, introducing uncertainty into the pose estimation model. Hence, improving the model's ability to perceive severe occlusion is crucial for enhancing its robustness.

From the aspect of **problem formulation**, recent instance-level methods have achieved high precision but exhibit poor generalization. Category-level methods demonstrate good generalization for intra-class unseen objects but fail to generalize to unseen object categories. Unseen object pose estimation methods have the potential to generalize to any unseen object, yet they still rely on object CAD models or reference views. The following paths can be explored from the problem formulation to further enhance the generalization of object pose estimation:

1) Few-shot learning-based category-level methods for unseen categories. Since category-level methods need to re-obtain a large amount of annotated training data for unseen object categories, their generalization is severely limited. Therefore, future research could focus on exploring how to

我们倡导探索适用于物体姿态估计的标签高效学习技术, 具体可通过以下途径实现:

1) 基于LLM/LVM的弱监督/自监督学习方法。随着预训练LLM/LVM的快速发展, 通过无监督方式在多种场景中灵活应用已成为可能。将LLM/LVM作为先验知识加以利用, 为探索物体姿态估计中的弱监督或自监督学习技术提供了前景。**2) 合成数据在现实领域适应与泛化方法。**由于手动获取现实训练数据成本高昂, 合成数据生成提供了一种经济高效的替代方案。我们认为, 通过探索从合成数据到现实领域的领域适应与泛化技术, 可以弥合领域差距, 使基于合成数据训练的模型具备在现实场景中泛化的能力。**3) 面向姿态估计的新型视角重建。**神经渲染技术的最新进展(包括3D高斯渲染、可微分渲染和体绘制)使得仅需少量输入图像即可生成与物体一致的新型视角(Di Felice等人, 2024; Wen等人)。2024; Nguyen等人, 2024a; Legrand等人, 2024)。这些方法提供了一种无需计算机辅助设计(CAD)的解决方案, 可用于提升视图密度、规范对应关系, 并支持在稀疏监督下进行渲染与对比估计。在类别级及未见类别场景中, 此类模型可作为先验知识, 促进以物体为中心的规范处理与度量缩放; 或生成具备不确定性感知的多假设图像及嵌入表示, 从而提升检索性能及基于RANSAC的点对点(PnP)匹配效果。当前仍面临若干挑战: 如何在类内变异中保持几何保真度、减轻生成伪影与视图不一致性, 以及在机器人平台上实现实时性能。

在应用方面, 推动物体姿态估计方法在移动设备和机器人上的部署至关重要。我们认为, 可通过以下方法提升现有方法的可部署性:

1) 集成检测与分割的端到端方法。当前SOTA方法通常需要先使用预训练模型进行初始目标检测或分割, 再将图像输入姿态估计模型(间接姿态估计模型甚至需要采用不可微分的PnP或Umeyama算法来求解姿态), 这导致部署过程复杂化。未来研究可通过探索能无缝集成检测或分割的端到端目标姿态估计方法, 提升移动设备与机器人系统的可部署性。

2) 基于单张RGB图像的方法。鉴于智能手机和平板电脑等大多数移动设备缺乏深度摄像头, 如何利用单张RGB图像实现对未知物体姿态的高精度估计至关重要。鉴于二维图像固有的几何限制, 未来研究可探索基于线

性向量机(LVM)的单目深度估计方法, 通过整合场景级深度信息来提升单目物体姿态估计的准确性。

3) 模型轻量化研究。现有SOTA模型普遍存在参数量过大、运行效率低下等问题, 这给计算资源有限的移动设备和机器人部署带来了挑战。未来研究可探索教师-学生模型等高效轻量化方法, 以降低模型参数量(GPU内存占用)并提升运行效率。

现有方法主要针对常见物体和场景设计, 导致其在处理复杂物体与场景时效果欠佳。我们认为可通过以下途径提升适用性:

1) 关节式物体姿态估计。与刚性物体相比, 关节式物体(如服装和抽屉)具有更多自由度(DoF)且存在显著的自遮挡现象, 这使得姿态估计面临巨大挑战。如何实现关节式物体的高精度姿态估计, 仍是未来亟待解决的重要研究课题。

2) 透明物体的姿态估计。由于缺乏纹理、颜色和深度信息, 对透明物体的姿态估计构成了重大挑战。未来的研究可重点通过深度增强或信息补全技术来丰富透明物体的几何信息, 从而提升姿态估计的准确性。

3) 应对遮挡的鲁棒方法。遮挡是当前最普遍的技术挑战。目前尚无物体姿态估计方法能有效处理严重遮挡场景。严重遮挡会导致物体纹理与几何特征的表征不完整, 从而给姿态估计模型引入不确定性。因此, 提升模型对严重遮挡的感知能力对增强其鲁棒性至关重要。

从问题建模的角度来看, 近期实例级方法虽然实现了高精度, 但泛化能力较弱。类别级方法在处理类内未知物体时表现出良好的泛化性能, 但在面对未知物体类别时却难以推广。未见过物体姿态估计方法虽具备向任意未知物体泛化的潜力, 但仍依赖于物体CAD模型或参考视图。针对这一问题, 我们可从问题建模层面探索以下改进路径以提升物体姿态估计的泛化能力:

1) 基于少样本学习的未见类别分类方法。由于这类方法需要为未见对象类别重新获取大量标注训练数据, 其泛化能力受到严重限制。因此, 未来研究可重点探索如何

leverage few-shot learning to enable the rapid generalization of category-level methods to unseen object categories.

2) CAD model-free and sparse manual reference view-based unseen object pose estimation. While current unseen object pose estimation methods do not require retraining for unseen objects, they still rely on either the CAD models or extensive annotated reference views of unseen objects, both of which still require manual acquisition. To this end, exploring CAD model-free and sparse manual reference view-based unseen object pose estimation methods is crucial.

3) Open-vocabulary strong generalization methods. Given the broad applicability of object pose estimation in human-machine interaction scenes, future research could leverage open vocabulary provided by humans as prompts to enhance generalization to unseen objects and scenes.

7.2 Future Directions

While emerging trends primarily reflect ongoing developments, future work should focus on open-ended directions and broader applications that remain underexplored. Building on the discussions in the preceding sections, we summarize the following research directions that can substantially advance the field.

1) Improving generalization to intra-class unknown and unseen objects: While instance- and category-level methods have shown impressive accuracy on trained objects, their performance often degrades when facing intra-class unknown or category-unseen objects. Future research could explore disentangled geometric representations that encode both category-level priors and fine-grained instance variations. Leveraging generative shape modeling, compositional learning, or implicit neural representations may enable networks to infer pose from limited supervision and adapt to new object geometries without retraining.

2) Foundation model-based pose representation and adaptation: The rapid progress of large-scale vision and vision-language foundation models offers new opportunities for transferable 3D understanding. Integrating these models into pose estimation pipelines could allow pretraining on massive visual data while adapting to the geometry of novel objects via lightweight finetuning or prompt-based adaptation. Developing geometry-aware adapters or SE(3)-equivariant heads for such models may bridge the gap between semantic representation learning and geometric reasoning.

3) Label-efficient and self-supervised learning: A major bottleneck in current methods is the dependence on dense 6D pose annotations. Future work could focus on label-efficient learning paradigms, including weakly-supervised, self-supervised, and active learning approaches that exploit geometric constraints, photometric consistency, or multi-view correspondences to learn pose without explicit labels.

This will be crucial for scaling pose estimation to large and diverse object collections.

4) Uncertainty modeling and temporal consistency: Current models typically output a single deterministic pose, which limits their robustness in ambiguous or occluded conditions. Future methods could explicitly model pose uncertainty to better handle symmetric, transparent, or reflective objects. In dynamic environments, temporal consistency constraints through tracking or sequential optimization can also help stabilize predictions and improve overall reliability.

5) Cross-modal and physically grounded estimation: Beyond visual input, tactile and inertial modalities provide complementary cues for robust pose inference. Future work may explore multimodal fusion strategies that incorporate physical constraints, such as contact geometry or force sensing, to estimate pose in manipulation or interaction scenarios. The combination of visual and physical feedback could lead to more accurate and physically consistent pose estimation.

6) Scalable benchmarks and realistic simulation: As illustrated in Sec. 2, most benchmarks remain limited in scale, diversity, or realism. Future efforts should aim to develop large-scale, physics-aware simulation environments and real-world datasets with fine-grained annotations for both rigid and articulated objects. Such resources will support more comprehensive evaluation and training of generalizable models.

Together, these research avenues represent critical steps toward achieving robust, data-efficient, and generalizable object pose estimation across diverse domains and real-world conditions.

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Data Availability The data that support the research and a more readable version of this paper are available at [Awesome-Object-Pose-Estimation](#).

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利用少样本学习技术, 实现类别级方法对未见物体类别快速泛化的能力。

2) 基于CAD模型无关且稀疏式手动参考视图的未见物体姿态估计。尽管当前未见物体姿态估计方法无需针对新物体进行再训练, 但仍依赖CAD模型或大量标注过的未见物体参考视图——这两者仍需人工获取。因此, 探索基于CAD模型无关且稀疏式手动参考视图的未见物体姿态估计方法具有关键意义。

3) 开放词汇强泛化方法。鉴于物体姿态估计在人机交互场景中的广泛应用性, 未来研究可利用人类提供的开放词汇作为提示, 以增强对未见物体和场景的泛化能力。

7.2 未来方向

新兴趋势主要反映当前研究进展, 未来工作应聚焦于尚未充分探索的开放性方向及更广泛的应用领域。基于前文讨论, 我们总结以下研究方向, 这些方向将显著推动该领域发展。

1) 提高对类内未知及未见对象的泛化能力:尽管实例级和类别级方法在训练数据上展现出优异的准确率, 但当面对类内未知对象或类别未见对象时, 其性能往往会下降。未来研究可探索解耦几何表征方法, 这种表征方式能同时编码类别级先验信息与细粒度实例差异。通过运用生成式形状建模、组合学习或隐式神经表征技术, 网络可在有限监督数据下推断物体姿态, 并无需重新训练即可适应新几何形态。

2) 基于基础模型的态度表示与适应技术:大规模视觉模型与视觉语言基础模型的快速发展, 为可迁移的三维理解能力提供了新机遇。将这些模型整合到姿态估计流程中, 既可在海量视觉数据上进行预训练, 又能通过轻量级微调或基于提示的适应机制, 灵活适配新物体的几何特征。开发具备几何感知能力的适配器或SE(3)-等变头, 或将有效弥合语义表征学习与几何推理之间的技术鸿沟。

3) 标签高效与自监督学习:当前方法面临的主要瓶颈在于对密集六维姿态标注的依赖。未来研究可聚焦于标签高效学习范式, 包括弱监督学习、自监督学习及主动学习等方法——这些方法通过利用几何约束条件、光度一致性或多视角对应关系, 在无需显式标注的情况下实现姿态学习。这对于将姿态估计扩展至大

规模多样化物体集合具有关键意义。

4) 不确定性建模与时间一致性:

当前模型通常仅输出单一确定性的姿态, 这限制了其在模糊或被遮挡条件下的鲁棒性。未来的方法可明确建模姿态不确定性, 从而更好地处理对称、透明或反光物体。在动态环境中, 通过跟踪或顺序优化实现的时间一致性约束也有助于稳定预测结果并提升整体可靠性。

5) 跨模态与物理基础的估计:

除视觉输入外, 触觉与惯性模态可为稳健的姿态推断提供互补性线索。未来研究可探索整合物理约束条件(如接触几何结构或力传感)的多模态融合策略, 以实现机械操作或交互场景中的姿态估计。视觉反馈与物理反馈的结合有望实现更精确且符合物理规律的姿态估计。

6) 可扩展基准测试与真实模拟:如第2节所示, 现有基准测试在规模、多样性及真实感方面仍存在局限。未来研究应致力于构建大规模、物理感知型模拟环境, 并为刚性与关节运动物体提供细粒度标注的真实数据集。此类资源将为通用模型的全面评估与训练提供有力支持。

这些研究方向共同构成了实现跨领域及真实场景中稳健、数据高效且可泛化的物体姿态估计的关键步骤。

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